Request for Proposal (RFP)

for

Design, Supply, Implementation and Maintenance of CCTV based Surveillance System for Pune City

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Volume 2: Scope of Work

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1. Scope of Work

The broad scope of work can be categorized into two tracks which include maintenance of entire existing surveillance system and design, supply, installation, operations and maintenance of newly proposed surveillance system.

1.1 TRACK 1 – O&M of Existing Project

TRACK 1 - Operations & Maintenance of existing CCTV system: SI shall also take over the O&M from the incumbent (ADSL) and manage the upkeep of existing cameras.

TRACK 1 - Operations & Maintenance of existing CCTV system (Pune City only)	
Locations	**Pune: 367 existing** locations (470 Poles) **PCMC: 88 existing** locations (123 Poles)
Cameras	Total of **1043 existing** cameras (**790** Fixed, **197** PTZ, **56** ANPR) in Pune & 298 existing cameras (**243** Fixed, **33** PTZ, **22** ANPR) in PCMC
Command Control / Viewing Centres	**Pune Area :** ▪ CP Office Command Centre ▪ Addl. CP East office ▪ Addl. CP West office (Traffic Control Room at DCP Traffic office in Yerwada) ▪ 3 Zonal DCP Offices (Zone 1, 2 & 4) ▪ PMC and District Collector's Office ▪ 30 Police Station Viewing Centers **PCMC Area**: ▪ 1 Zonal DCP Office (PCPC) ▪ Pimpri Chinchwad Municipal Corporation (PCMC) ▪ 10 Police Station Viewing Centers (9 Police Stations & 1 DCP Traffic office PCPC)
Network Connectivity	MPLS redundant ring supporting multicast on BSNL network
Data Center	Co-location facility at Nxtra Data Center in Kharadi, Pune
Application Portfolio	▪ Video Management System : Nextiva Review 6.4.2022.0 ▪ Incident Management : Verint SMC 7.3.5400.02 ▪ ANPR : Kritikal 7.0 ▪ Video Analytics : Savvi Agent VI 5.3.0.222 (100 licenses) ▪ Helpdesk & Ticketing : BMC Remedy (Version 8.1.02) For detailed application portfolio, refer Annexure II
External Integrations	VAHAN CCTNS
Go-Live Date for project	Go-Live on 27th October 2015
Maintenance & Support Duration	**6 years** (1 year + 5 years) from issuance of work order

From the perspective of project implementation and support, the scope has been categorized as follows:

System Integrator (ADSL in this case) has secured the network infrastructure for this project from BSNL. While several components from the existing project will be utilised, few components such as bandwidth and storage require augmentation as per additional requirement.

1.1.1 O&M of Component #1: Existing Surveillance Equipment Infrastructure

Scope:

The existing project includes surveillance of 367 locations across Pune area and 88 locations across PCMC area. These locations are covered through different types of surveillance cameras including Fixed Box, PTZ and ANPR cameras. There are currently 790 Fixed, 197 PTZ and 56 ANPR cameras deployed in the Pune city & 243 Fixed, 33 PTZ and 22 ANPR in PCMC. Video feed from all these locations is transmitted to the data center and then the feed is made available to the respective Command Centre, Viewing Centers and Police Stations based on their jurisdiction. For details of specifications refer **Annexure II**.

CCTV Cameras

Total 1077 Cameras in Pune

- About 811 Fixed Cameras
- About 202 PTZ Cameras
- About 64 ANPR Cameras

Total 264 Cameras in PCMC

- About 221 Fixed Cameras
- About 29 PTZ Cameras
- About 14 ANPR Cameras

Operation and Maintenance Services

The implementation component for edge level equipment shall primarily cover assessing & maintenance of the existing edge devices at different locations already available in Pune City Surveillance Project.

A. **Operate and Maintain**

A1. Maintenance of the entire existing Surveillance Equipment as per SLAs

A1. Maintaining the Uptime of Surveillance Equipment as per SLA

As mentioned in scope section above, the 1040 cameras are currently deployed in Pune City under this project. These are currently maintained by incumbent SI which is the property of GoM. Selected SI will take over the maintenance of these surveillance equipment and related components such as Junction Box, UPS, Batteries, Pole Box, etc.

Major portion of SLA is based on upkeep and uptime of edge level equipment. As SLA is directly linked to uptime of these edge level equipment, selected SI will ensure the uptime of these equipment and strictly adhere to the defined SLA (as described in this RFP).

The selected SI may require to change the field of view of the camera, adjust focal length of lens, adjust height of the camera, etc. as required by Pune Police. Maintenance of the edge level equipment shall include the services like:

- **Repair** - In case of damage, failover, etc.
- **Replacement** - In case of irreparable loss. Selected SI must factor in spares to carry out replacement.
- **Cleansing** - Routine checkup, proactive and reactive cleansing of lens

Selected SI will undertake above activities till the end of project O&M phase. Selected SI will create SOP to ensure the maintenance of existing equipment.

1.1.2 O&M of Component #2: Existing Network Connectivity Infrastructure

Scope:

The BSNL MPLS network connectivity is provided throughout the city for this project. The points of connection include cameras, Data Center, CP Office Command Center (including Helpdesk), and Viewing Centers at Addl. CP East and West region, DCP Traffic office in Yerwada, Zonal DCP offices, PMC, PCMC, Collectorate and 40 Police Stations. All viewing centers at various locations are connected through MPLS backbone and the details are available in **Annexure II.**

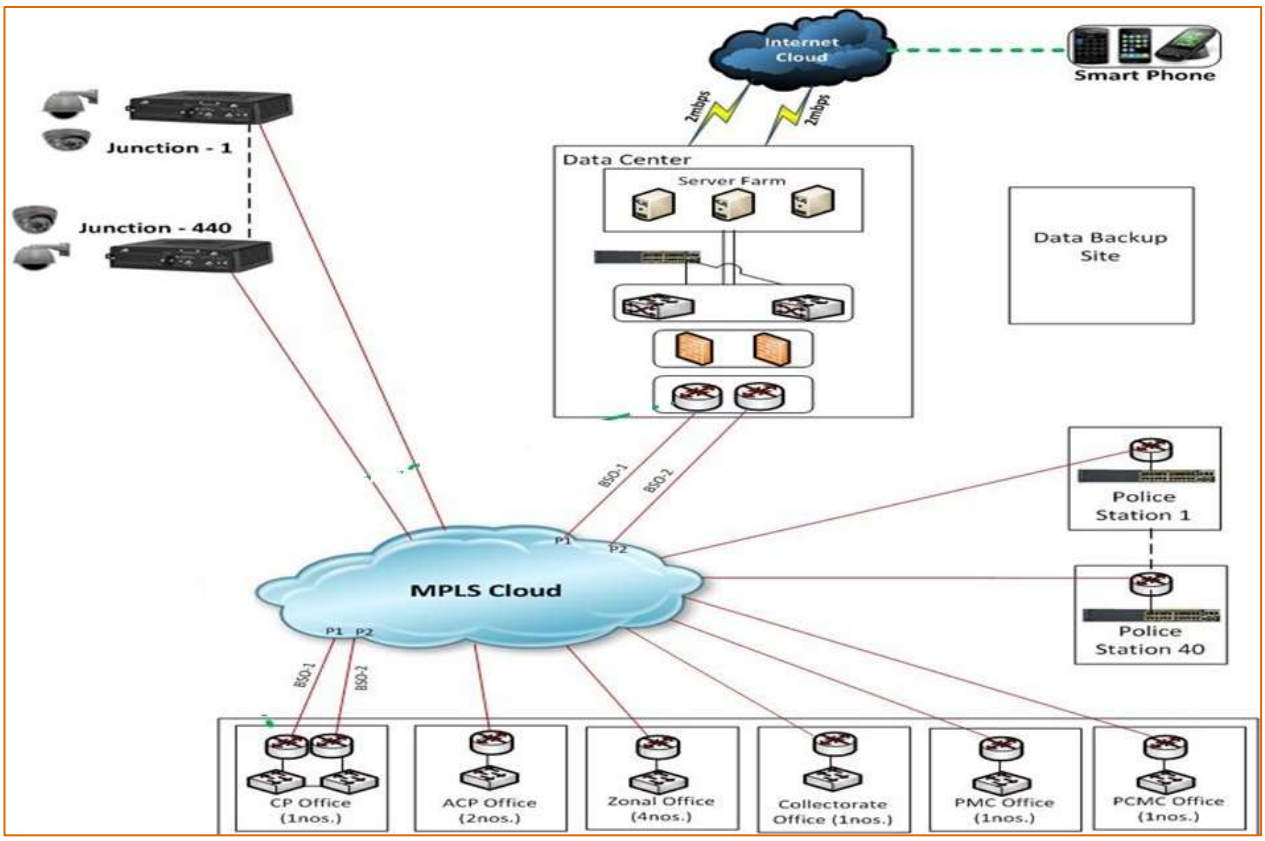
Establish connectivity between following -

- Between Cameras & Aggregation Point at edge level
- Between Aggregation Point & Data Center
- Between Data Center & Command/ Viewing Centers
- Between Data Centers & the Mobile Vans

Operation and Maintenance Services

Network Connectivity is backbone of this project. All the surveillance equipment function on the basis of network layout that has been established currently. Existing network connectivity vendor **BSNL** has set up this backbone in

form of MPLS ring. The current network architecture comprises of various fiber rings which are interconnected to



provide redundancy. These rings are connected to junction level components via BSNL Point of Presence (POP).

Bandwidth of 4 MBPS is required for each camera. Selected SI to use the services of existing ISP i.e. BSNL to provide required bandwidth as below:

SN	Location	Minimum Cameras to be viewed simultaneously
1	CP Office Command Center	150
2	PMC	60
3	PCMC	25
4	Collector office	10
5	Additional CP office East	10% of camera of their Jurisdiction
6	Additional CP office West (DCP traffic Yerwada)	10% of camera of their Jurisdiction
7	Zonal DCP office (Zone 1, 2,4 & Zone-PCPC)	10% of camera of their Jurisdiction
8	Police Stations (39) + PCPC traffic department	50% of camera of their Jurisdiction

Existing Network Architecture

The selected SI should use existing network infrastructure & ISP (BSNL) for providing network connectivity for existing locations as well as new locations i.e. Track I & Track II. .

A. **Operate and Maintain**

A1. Maintain uptime and ensure the SLA

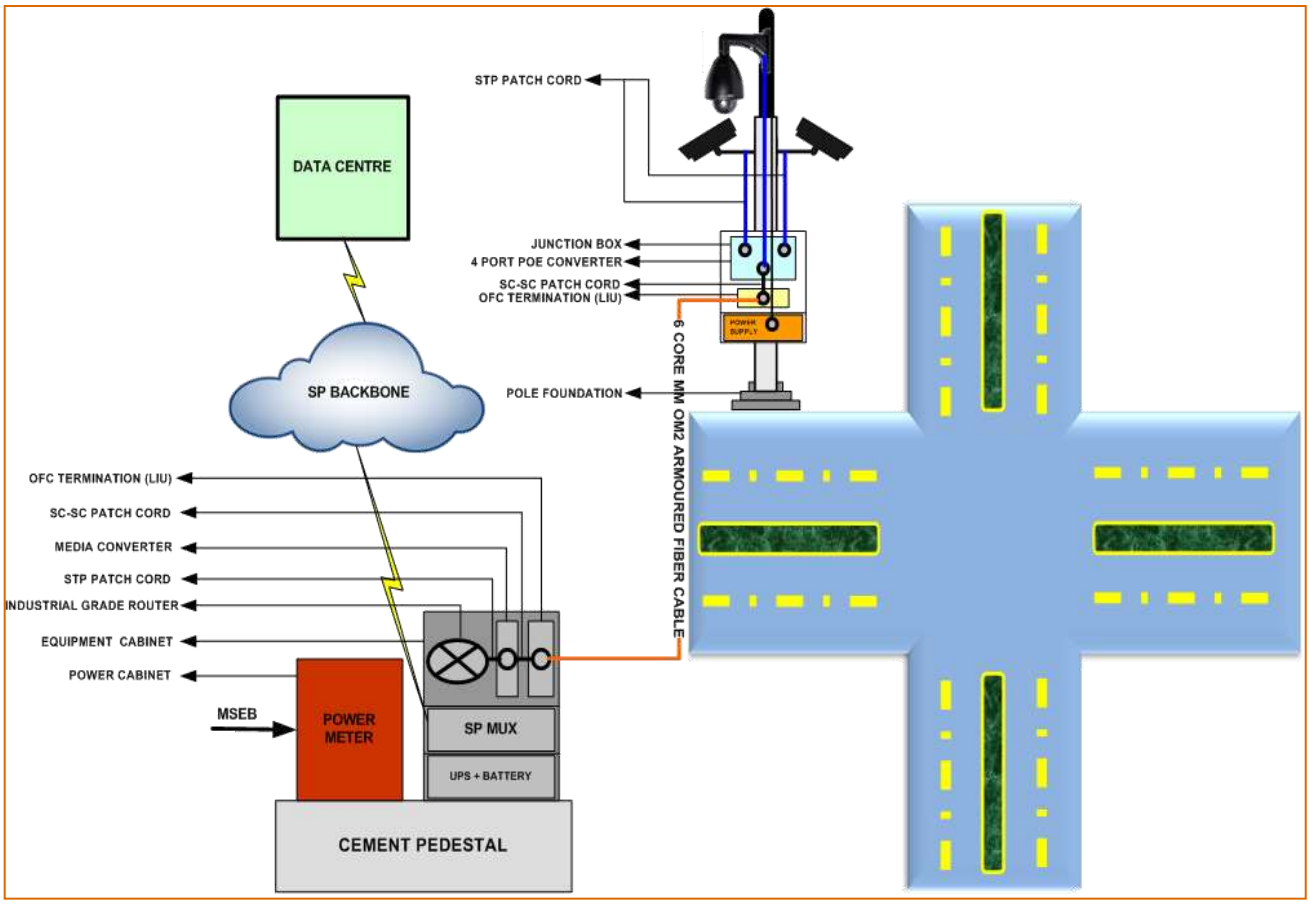
A2. Upkeep of the equipment by ensuring Repair/ Replacement/ Cleaning Services

A1. Maintain uptime and ensure the SLA

Irrespective of any option chosen by SI for setup of network connectivity, it should be able to ensure the uptime and adhere to SLA which is described in this RFP.

A2. Upkeep of the equipment by ensuring Repair/ Replacement/ Cleansing Services

Maintenance of the network equipment shall include the services like:



- **Repair** - In case of damage, failover, etc.
- **Replacement** - In case of irreparable loss. Selected SI must factor in spares to carry out replacement.
- **Cleaning** - Routine checkup and cleaning of devices

Selected SI will undertake above activities till the end of project O&M phase. Selected SI will create SOP to ensure the maintenance of network equipment. The details of the equipment are available in **Annexure II**.

The layout of current field junction is shown in below diagram:

1.1.3 O&M of Component #3: Data Center

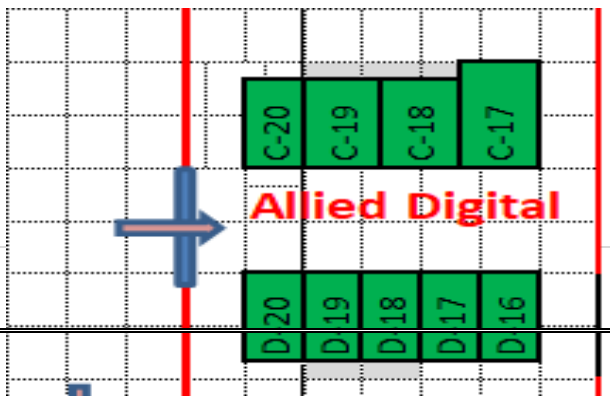
Scope:

The Data Center equipment for this project is co-located at Nxtra Data Ltd. Tier III data center at Kharadi, Pune (**2nd Floor, Airtel E Park Center, Chandan Nagar, Thite Nagar, Kharadi, Pune 411014**). The secondary Data Backup site is kept 10 kms away from the Data Centre at CP Office Pune. The incumbent SI is into lease agreement with Nxtra Data Ltd. till end of project for the requisite space in data center, details of which are mentioned as below:

- No. of Racks = 9 racks
- Total rented space = 252 Sq. ft.
- Used Space = 70 Sq. ft.
- Primary storage available & used - 2.2PB & 1.85PB respectively

Indicative layout of data center showcasing position of different racks is as mentioned below:

Rack No.	Rack details
D-16	NAS Storage rack
D-17	NAS Storage rack
D-18	NAS Storage rack
D-19	NAS /SAN Storage rack



D-20	IBM Server rack
C-20	IBM Server rack
C-19	Network rack
C-18	Network rack
C-17	TSM Library

Selected SI needs to make a provision for hosting DC and may continue using the same location for data center (with certain augmentation) or select a new location for data center. In this case, bidder must consider reuse of existing servers hardware procured for existing project. Details of data center infrastructure is covered in **Annexure II**.

Operation and Maintenance Services

A. **Operate and Maintain**

A1. Provisioning of colocation space for Data center infrastructure

A2. Maintaining the uptime of Server Infrastructure and ensure the SLA

A1. Provisioning of colocation space for Data center infrastructure

Selected SI will be responsible for provisioning of colocation space for data center infrastructure as per the benchmark requirements which should act as guiding factors for the bidders to propose the locations for the Data center and the Data backup center.

- Data Center should provide a dedicated caged rack space controlled by access control for the City Surveillance Project Infrastructure.
- Data Center should be at least a Tier III Data Center as per Telecommunications Infrastructure Standard for Data Centers and should be certified as ISO 27001.
- Access to the Data Center Space where the City Surveillance Project Infrastructure is hosted should be demarcated and physical access to the place would be given only to the authorized personnel through access control.
- Indoor CCTV cameras would be required to be installed to monitor the physical access of the system from remote location.
- Data Center should be in the jurisdiction of Pune Police Commissionerate. Data Backup Center site should also be in the Pune region preferably at an aerial distance of 10 Kms from the data center.
- Physical Access to the building hosting Data Center and the Data Backup Center should be armed and it must be possible to even depute police personnel for physical security of the premises.

A2. Maintaining the Uptime of Server Infrastructure and ensure the SLA

The various components are installed at the Data Center which act as brain of this project. Server infrastructure consisting of Application Server, Recording Server, Analytics Server, Database Server, Management Server, Enterprise Backup Server, etc. are installed in Data Center. Detailed list of components with capacities, make/ model is available in **Annexure II.**

Selected SI shall take handover of these components from incumbent SI for operations & maintenance. It will be sole responsibility of SI to undertake the maintenance of existing Server Infrastructure. The upkeep and consistency of the existing system should be maintained by selected SI to avoid the penalties due to shortfall of SLA.

1.1.4 O&M of Component #4: Existing Application Portfolio

Scope

The applications deployed in the Data Center are central part of the system. Keeping in view the needs of interoperability and integration, especially the possibility that the solution would become the focal point for maintaining law and order in the city by the Police in future, the solution is built on Open Standards and adaptive to Service Oriented Architecture (SOA). Details are available in **Annexure II.**

The key components of the Application Portfolio are Security Management System (SMS), Recording System, Video Analytics System, ANPR Analytics, etc. Several of these applications are COTS whose licenses are perpetual in nature, but annual maintenance charges are to be paid by SI to OEM's. Custom software applications which have been developed by the incumbent SI are available with the source code. Details of existing application are described in **Annexure II**. Reusability of existing applications can be done by selected SI to avoid duplication of applications as previous applications are property of GoM. O&M of applications will include renewals wherever required, upgrades, etc.

A. **Operate and Maintain**

A1. Maintenance of existing Application Portfolio

A2. Assess and Finalize the requirements for integration of Surveillance System with other Systems like CCTNS system, VAHAN system, etc

A3. GIS Integration

A4. SMS Gateway integration

A5. Maintain uptime and ensure the SLA

The key components of the existing Application Portfolio are:

- Video Management System
- Command Center Application
- Enterprise Management System
- Video Analytics and ANPR Software
- GIS

A1. Maintenance of existing Application Portfolio

Selected SI shall ensure that the existing applications are functioning as per technical and functional specifications mentioned in this RFP catering to the requirement of Pune Police. The selected SI must also ensure that the applications are upgraded to latest firmware versions for the period of contract.

The list of applications along with make/model and current status provided in **Annexure II** are part of the Application Stack. These applications play vital role in this project as they ensure various activities like alarms, alerts, monitoring, analytics, ticketing, etc.

In case bidder is proposing new OEMs for VMS, ANPR, etc. Bidder needs to ensure seamless integration between existing applications & new proposed applications to comply with all required functionalities as defined in the RFP.

A2. Integration of Surveillance System with other Systems like CCTNS, VAHAN system, etc.

Surveillance system should use information of some other operational systems to generate intelligent reporting. In this regard, Home Department has identified integration with at least 2 systems to start with - CCTNS system (being implemented for Police Station Computerization) and the VAHAN systems (operationalized by transport department). Currently CCTV System is integrated with CCTNS for Missing persons & stolen vehicles data. However, Selected SI during the Assess and Prepare stage of the project shall carry out detail requirement analysis and propose a full-fledge integration (with dynamic data sharing) to meet the end objectives of the project.

A3. GIS Integration

Currently map is available from MapmyIndia. The selected SI must provide updated GIS map with added additional assets (such as new locations, cameras, etc.)

Selected SI shall also undertake detailed assessment of the existing GIS solution with Police Dept and undertake its integration with the proposed additional cameras (in Track II) so that physical location of cameras, GPS fitted police vehicles are brought out on the GIS map. SI is required to carry out the seamless integration to ensure ease of use of GIS in Command Control Centers, Police Stations & by higher management. SI is required to update GIS maps at least once every year. Different layers to be covered under GIS are as follows:

- Crime hotspots
- Accident hotspots
- CCTV cameras setup by private establishments
- Cameras
- Buildings / Structures
- Police Stations
- Important Govt / Public Buildings
- Public Places like Gardens, Maidans, etc.
- Roads
- Bridges, Flyovers
- Footpaths
- Any other layer as an when required by Dept.

Please note that data of hotspots and private CCTV cameras shall be provided as lat-long geo coordinates by Pune Police. Geographical coverage of the project is the jurisdiction of Pune Police Commissionerate. Selected SI shall ensure that GIS application is integrated with VMS to support the Command Center / Police Station personnel to navigate on the map and use it for better spatial understanding. It should also help higher management of Police Department to analyse the events on a spatial perspective. This map should be made available to police personnel via web access.

A4. SMS Gateway Integration

SI shall implement an SMS Gateway, integrate it with the Surveillance System and develop necessary applications to send Mass SMSes or SMSes to the group/individuals, which can be either manual or system generated.

A5. Maintain uptime and ensure the SLA

Selected SI must ensure the uptime of all applications and adherence to SLA parameters as described in this RFP.

1.1.5 O&M of Component #5: Existing Command Center and Viewing Centers Infrastructure

Scope:

All the feeds from cameras are transmitted to Data Center and then the feed is distributed to viewing centers. Command Center is located at Pune Police Commissionerate Office. Apart from Command Center, there are viewing centers in various establishments like PMC, Collectorate Office, 3 Zonal DCP Offices, 1 Add CP office & 1 DCP Traffic officer, Yerwada, 1 PCMC Zone and 30 Viewing centers at Police Stations in Pune City and 10 Police Stations & 1 viewing center at PCMC.

Operation and Maintenance Services

A. **Operate and Maintain**

A1. Maintenance of Command Center equipment as per SLA

A2. Maintenance of Viewing Centers at zonal offices and other institutions as per SLA

A3. Maintenance of 40 Viewing Centers at Police Stations as per SLA

A1. Maintenance of Command Center equipment as per SLA

CCTV Command Center is located at Pune Police Commissionerate Office. CP Office Command Center is having simultaneous viewing capability for about 150 cameras. For the purpose of common viewing of the cameras, video wall is installed in the CP Office Command Center. Along with video wall, there is a setup of 10 workstations at command center for simultaneous viewing for CCTV operators. The Command Center is equipped with IP Phones for **24x7** support. The detailed list of Command Center infrastructure which also includes video wall is described in **Annexure II**.

Selected SI shall maintain the existing Command Center infrastructure to meet the SLA requirement as described in this RFP.

A2. Maintenance of Viewing Centers at major establishments as per SLA

Apart from Command Center, there are viewing centers (VCs) at major establishments.

- 1 Viewing Centers of Additional CP Office West, currently deployed at DCP Traffic at Yerwada office
- 1 Viewing Center at Pune Municipal Corporation building.
- 1 Viewing Center at PCMC building.
- 1 Viewing Center at Collectorate Office.
- 4 Viewing Centers at DCP Offices of Zone 1, Zone 2, Zone 4 and Zone (PCMC)
- 1 Viewing Center of Additional CP East Office

PMC, PCMC Office & Collectorate Office Viewing Centers are having simultaneous viewing capability for about 60, 25 & 10 cameras respectively. The Addl. CP and Zonal DCP offices are have viewing capability for about 10% of the cameras of their regions. LED TV sets are these locations for viewing of camera feeds. These viewing centers are also equipped with IP Phones for 24x7 support. There is setup of workstations installed at all these viewing centers. The details of infrastructure at these viewing centers are available in **Annexure II.**

Selected SI shall maintain the existing Viewing Center infrastructure to meet the SLA requirement as described in this RFP**.**

A3. Maintenance of 40 Viewing Centers at Police Stations as per SLA

Apart from above mentioned viewing centers, there are 40 Viewing Centers deployed at 40 Police Stations. Police stations are having simultaneous viewing capability for about 50% of cameras of the respective jurisdiction. LED TV set along with a workstation is installed in every Police Station. These viewing centers are also equipped with IP Phones for **24x7 support**.

Selected SI shall maintain the existing Viewing Center infrastructure to meet the SLA requirement as described in this RFP. Details of components in these viewing centers is available in **Annexure II.**

1.1.6 O&M of Component #6: Existing Collaborative Monitoring Infrastructure

Scope:

Certain surveillance systems have been deployed by many public & private establishments. The incumbent SI has added one location - Dagadusheth Halwai Ganpati Mandir for Collaborative Monitoring with the project.

Implementation Services

A. **Operate and Maintain**

A1. Ensuring maintenance of collaborative monitoring with existing establishments

A1. Ensuring Maintenance of Existing Integrations

It will be responsibility of selected SI to maintain the existing integrations. Currently, only 1 external establishment's CCTVs are integrated with the Project. Selected SI shall facilitate with the private establishment to ensure that connectivity of cameras is maintained.

1.1.7 O&M of Component #7: Helpdesk and Facility Management

Scope:

Facility Management Service (FMS) at Command Center with dedicated manpower is provided by the incumbent SI to troubleshoot & solve issues / problems. The FMS is supported by a centralized helpdesk setup at CCTV Command Center. This Helpdesk is single point of contact for complaint management and resolution for all the users of the surveillance system. This helpdesk is integrated with the Enterprise Management System. The helpdesk is designed to meet the SLA response & resolution timelines.

Operation and Maintenance Services

A. **Operate and Maintain**

A1. Uptime of Existing Components of Helpdesk and EMS as per SLA

A2. Provision of manpower for FMS as per SLA

A1. Uptime of existing components of Helpdesk and EMS as per SLA

The incumbent SI has setup a central helpdesk which is supported by their field units across various locations. Specifications of existing Helpdesk, which is fully integrated with the enterprise monitoring and network management system (EMS), is provided in **Annexure II**. The system can be accessed by the Police officials for raising their incidents and logging calls for support. The detailed service levels and response time which the selected SI is required to maintain for provisioning of the FMS services as per SLA described in this RFP.

A2. Provision of manpower for FMS as per SLA

The incumbent SI has deployed a dedicated, centralized project management & technical team for the overall project management and interaction with Sr. Police Dept / Home Dept personnel. The details and profiles of existing manpower deployed for FMS are described in **Annexure II**.

1.2 TRACK 2 – Implementation of new initiatives for Pune Police

Implementation of new initiatives such as additional CCTV cameras, drones, mobile vans, etc. which will include Site Survey, Detailed Design, Supply, Installation and Maintenance of the entire city-wide surveillance system (including the existing and new initiatives),

The detailed work to be undertaken by the Selected SI for setting up & operationalization of the new initiatives is given in subsequent sections and is to be performed as per the specifications and conditions mentioned in this RFP and as per any further amendments issued and the contract to be signed with the Selected SI subsequently. From the perspective of project implementation, the scope has been categorized as follows:

1.2.1 Finalization and submission of a detailed technical architecture for integration with additional components

Within 3 weeks of the work order, the Systems Integrator needs to deploy the team proposed for the project and ensure that the Project Inception Report is submitted to GoM, which should cover following aspects:

- Names of the Project Team members, their roles & responsibilities
- Approach & methodology to be adopted to implement the Project (which should be in line with what has been promised during bidding stage but may have value additions / learning in the interest of the project).
- Responsibility matrix for all stakeholders
- Risks the bidder anticipates and the plans they have towards their mitigation.
- **Detailed Project Plan**, specifying dependencies between various project activities / sub-activities and their timelines.

Thereafter, within 3 weeks, SI shall submit the detailed Technical Architecture which should take into consideration following guiding principles:

- **Scalability** - Important technical components of the architecture must support scalability to provide continuous growth to meet the growing demand of Pune Police. The system should also support vertical and horizontal scalability so that depending on changing requirements from time to time, the system may be scaled upwards. There must not be any system-imposed restrictions on the upward scalability in no. of cameras. Main technological components requiring scalability are Storage, Bandwidth, Computing Performance (IT Infrastructure), Software / Application performance and advancement in camera features. In quantitative terms, there may not be major change in number of Command Centers & Police Stations.

- **Availability** - Components of the architecture must provide redundancy and ensure that are no single point of failures in the key project components. Considering the high sensitivity of the system, design should be in such a way as to be resilient to technological sabotage. To take care of remote failure, the systems need to be configured to mask and recover with minimum outage. The bidder shall make the provision for high availability for all the services of the system.

- **Security** - The architecture must adopt an end-to-end security model that protects data and the infrastructure from malicious attacks, theft, natural disasters etc. Selected SI must make provisions for security of field equipment as well as protection of the software system from hackers and other threats. Using Firewalls and Intrusion detection systems such attacks and theft should be controlled and well supported (and implemented) with the security policy. The virus and worms' attacks should be well defended with Gateway level Anti-virus system, along with workstation level Anti-virus mechanism. There should also be an endeavor to make use of the SSL/VPN technologies to have secured communication between Applications and its end users. Furthermore, all the system logs should be properly stored & archived for future analysis and forensics whenever desired. Home Department may carry out the Security Audit of the entire system in approx. 3 months of Acceptance / operationalization through a Third-Party Auditor (TPA). The following guidelines need to be observed for security

- Build a complete audit trail of all activities and operations using log reports, so that errors in system – intentional or otherwise – can be traced and corrected.
- The most appropriate level of security commensurate with the value to that function for which it is deployed must be chosen
- Access Controls must be provided to ensure that the system is not tampered or modified by the system operators.

- Implement data security to allow for changes in technology and business needs.
  - **Manageability** - Ease of configuration, ongoing health monitoring, and failure detection are vital to the goals of scalability, availability, and security and must be able to match the growth of the environment.
  - **Interoperability** - The system should have capability to take feed from edge devices installed by private / Govt. at public places, digitize (if required) & compress (if required) this feed & store as per requirements.
  - **Open Standards** - System should use open standards and protocols to the extent possible.
- The bidder would not be allowed to sub-contract / outsource work, except for following:
  - Passive networking & Civil Work during implementation
  - Viewing Manpower at Command / Viewing Centers during post-implementation

Stage Deliverables:
1. Project Inception Report
2. Detailed Project Plan
3. Technical Architecture

1.2.2 Implementation & Maintenance of Component #1: Additional Edge Devices

Scope:

The scope of work for Selected SI with regards to the edge devices consists of implementation and maintenance of edge devices not limited to CCTV, drones, etc.

The additional edge devices will be covering the jurisdiction on Pune Police Commissionerate. These locations would get covered through different types of surveillance cameras including IP based Fixed cameras & PTZ cameras, IP based PA speakers and Drones for aerial surveillance. The successful bidder will also be commissioning the surveillance cameras required in the Mobile Vans.

Edge level equipment

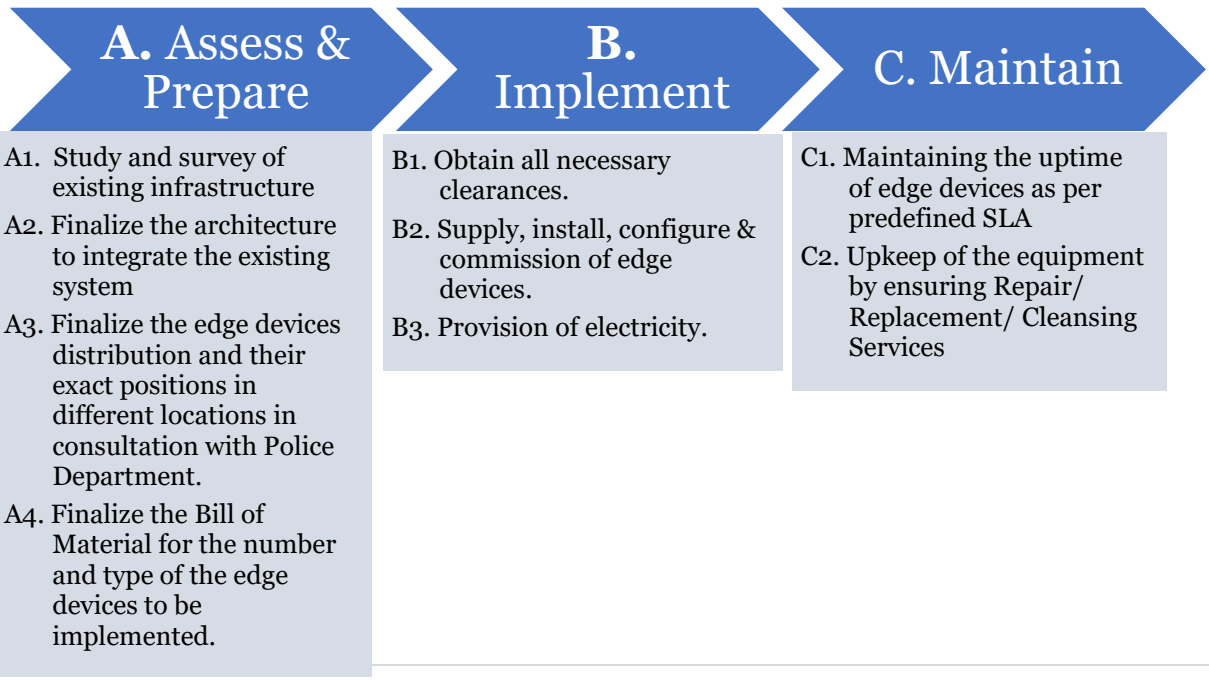
Cameras for Surveillance

- About 2240 Fixed Cameras
- About 328 PTZ Cameras
- About 318 ANPR Cameras
- 5 Mobile Vans
- 5 Drones
- 500 IP based PA speakers

Electrical online UPS has to provide minimum 60 minutes power backup to all locations and viewing centers. SI should ensure that proper protection is taken against power surges and ensure power stabilization to the surveillance equipment. In addition, rack design should be modular, and each component should be well organized and clamped inside to ensure components do not heat up or fall out on opening.

Implementation Services

The implementation component for edge level equipment shall primarily cover planning & implementation of the edge devices at different locations identified by Police Department and SI.



A1. Study and survey of existing infrastructure

Selected SI will do detailed site survey to understand the existing infrastructure and components deployed in existing project. The exhaustive list is given in **Annexure II** of this RFP**.** It will be sole responsibility of selected SI to assess the existing components to create the final architecture that will integrate existing edge devices with additional components.
Certain locations may entail sharing of poles/ junction cabinet/ electricity meter of existing locations with new cameras. The list of such locations is available in **Annexure II** of this RFP. In such cases, selected SI must carry detailed study of existing infrastructure and ensure no damage to existing infrastructure during installation.

A2. Finalize the technical architecture to integrate the existing system

After the detailed assessment of existing infrastructure, the selected SI will create a technical feasibility report detailing the approach to integrate existing infrastructure with the new initiatives.

A3. Finalize the edge devices distribution and their exact positions in different locations in consultation with Police Department:

Bidders are required to note that while executing the project, the selected SI shall prepare the final camera & other edge devices distribution plan at all the locations across Pune city. Actual place for placement of number of cameras at each location, type of cameras, height & angle for the cameras would be done carefully to ensure optimum impact. Due to this, there could be some variation in type of the cameras at certain locations as compared to the list given in **Annexure III** of this RFP. Payments to be made to the Systems Integrator shall be based on actual number of cameras installed and unit rates quoted by the Selected SI shall be used to arrive at the same. The locations must be carefully selected to ensure appropriate functioning of advanced video analytics, facial recognition, etc. For better field of view, it is suggested that SI to use cantilever of minimum 2 meter length and gantry where ever required.

A4. Finalize the Bill of Material for the number and type of edge devices to be implemented:

Selected SI shall prepare the detailed report on edge level requirements – Cameras (types & numbers), Camera Mounting requirements, Power requirements and connectivity requirements. Indicative list of the edge level hardware / services is as follows:

- Field Cameras (Fixed Box Cameras with Internal/External IR, PTZ Cameras with Internal/External IR, ANPR Cameras with External IR/FR Cameras)
- Managed Switches
- Routers
- Junction boxes
- Pole / Cantilever
- Digging & Trenching
- Networking cables and other related infrastructure
- Provisioning of Electrical Power
- IP based PA speakers

B1. Obtain all necessary clearances:

Selected SI will have to identify and obtain necessary clearances for installing cameras, for provisioning of the required power, network etc. It is important to mention that a timely communication and follow-up will be required by the Selected SI for the clearances. SI will have to then supply & erect poles at these locations well in advance to meet the camera installation timelines.

During post-implementation period, in case the Pole is damaged by a vehicular accident (or due to any other reason outside the control of SI) and needs repair, then the corresponding cameras won't be part of the SLA monitoring for max. 15 days. That means, SI will need to repair / have the new pole within 15 days of the incidence. Post 15 days, the corresponding cameras would be again considered for SLAs. Damages to be borne by SIs in such cases through proper insurance.

B2. Supply, install, configure & commission of edge devices:

The Selected SI will be required to supply, install, configure and integrate the edge devices at the identified locations and then undertake necessary work towards their commissioning. The successful bidder will also be commissioning the surveillance cameras required in the Mobile Vans.

It is a must that the poles erected to house the edge devices are good, both qualitatively and aesthetically. Benchmark specifications for the poles are given in **Annexure I**. Benchmark specifications for various types of edge devices to be supplied & operationalized as part of this project are given in **Annexure I**. Bidders are required to ensure that edge devices proposed are capable to meet these benchmark specifications and are also able to adhere to the functional requirements specified in this RFP.

SI should use the industry best practice while positioning and mounting the cameras (like using of cantilever/gantry). Some of the check-points which need to be adhered by the SI while installing / commissioning cameras are as follows: Ensure Surveillance objective is met while positioning the camera, create the required field of view ensure appropriate housing is provided to protect camera from the on field challenges carry out proper adjustments to have the best possible image ensure that the pole / mast implementation is vibration resistant.

It's the SI responsibility to take approvals of pole design from Government authorized structural engineer in design phase and structural stability and safety certificate of all poles to be submitted by Government authorized structural engineer after installation of poles.

B3. Provision of electricity

For the successful commissioning & operationalization of the edge devices and to provide the video feeds to Command Center, Viewing Centers and Police Stations, the Selected SI will be required to provide electricity to the edge devices through the aggregation point. Since this component has dependency on approval from other agencies, it is recommended that SI plans this requirement well in advance & submits the application to the concerned electricity distribution agency. Registration of Electrical connections (existing as well as new locations) to be done in the name of SI. For ensuring uptime as per SLA, bidder shall provide power back-up for min. 1 hour. Solar based power back-up would be accepted only if it is in addition to the alternate back-up mechanism.

B4. Procure & configure the required Vans

Mobile Vans would be used as and when the situation demands to capture the real-time video feed of an incident. These will be built on rugged vehicles and will house communications equipment that may be required to stream video feeds to command centres.

Mobile Vans shall have the appropriate wireless connectivity to connect to the nearest command centre through the data centre, local storage for storing at least 24-hour video feed for 4 cameras, local computing and processing capabilities & a seating capability for minimum 4 people.

Indicative list of the Bill of Material for each Pre-Fabricated Mobile Vans is as follows:

- Laptop for viewing min. 14" Screen (Rugged)
- Outdoor PTZ Camera (Vehicle mounted Rugged)
- Outdoor Fixed Box Cameras (Vehicle mounted Rugged)
- Body worn cameras
- Local Storage for 24 hours
- Router, modem (Industrial grade)
- LAN for the Van
- Invertor with Battery (Industrial grade)

SI shall bear the cost of Vehicle Insurance, PUC, Maintenance. SLA considerations for the Vehicles shall be HelpDesk call based, with benchmark resolution period for normal calls being 1 day. If the vehicle is damaged in mob rage or terrorist attack, it would be qualified as Force Majeure and SI shall be given due period to get the vehicle serviced and make it operational. Decision of Commissioner of Police, Mumbai, would be final regarding qualification of any event as Force Majeure or not.

Please note that the Mobile Van component (including all the related components) is optional, and Government shall decide upon it's inclusion or exclusion at a later date. Government decision in this regard shall be final and binding upon the successful bidder.

A Rapid Re-deployable system, to be setup in 5 Police Vans, which would involve setup of a temporary Viewing center where video feeds from 4 wireless CCTV cameras can be captured. These cameras shall transmit video feeds over Wi-Fi / 4G / 5G connectivity to the temporary viewing center in the Police vans. Viewing in these Police vans will be done over Laptops. The video feeds (from each of the cameras) may be stored locally for 24 hours in the Police van and if necessary, archived for investigation purposes.

C1. Maintaining the Uptime of edge level equipment as per predefined SLA

As mentioned in scope section above the existing cameras and new cameras will be part of this project. SI will take over these existing surveillance equipment and related components such as Junction Box, UPS, Batteries, Pole Box, etc. SI will do the maintenance of these old and new edge level equipment.

As SLA is directly linked to uptime of these edge level equipment, SI will ensure the uptime of these equipment and strictly adhere to the defined SLA.

C2. Upkeep of the equipment by ensuring Repair/Replacement/Cleansing Services

Maintenance of the edge level equipment will include the services like:

- **Repair** – In case of damage, failover, etc.
- **Replacement** – In case of irreparable loss
- **Cleansing** – Routine checkup and cleansing of the edge devices

SI will undertake above activities till the end of project O&M phase. SI will create SOP to ensure the maintenance of existing equipment

Stage Deliverables:

1. Detailed Site Survey Report with Coverage Study
2. Detailed design of poles with cantilever/gantry (including civil).
3. Detailed electricity plan for each location
4. Identification of video analytics per camera
5. SLA plan/report for edge devices

1.2.3 Implementation & Maintenance of Component #2: Network Connectivity for additional components

Scope:

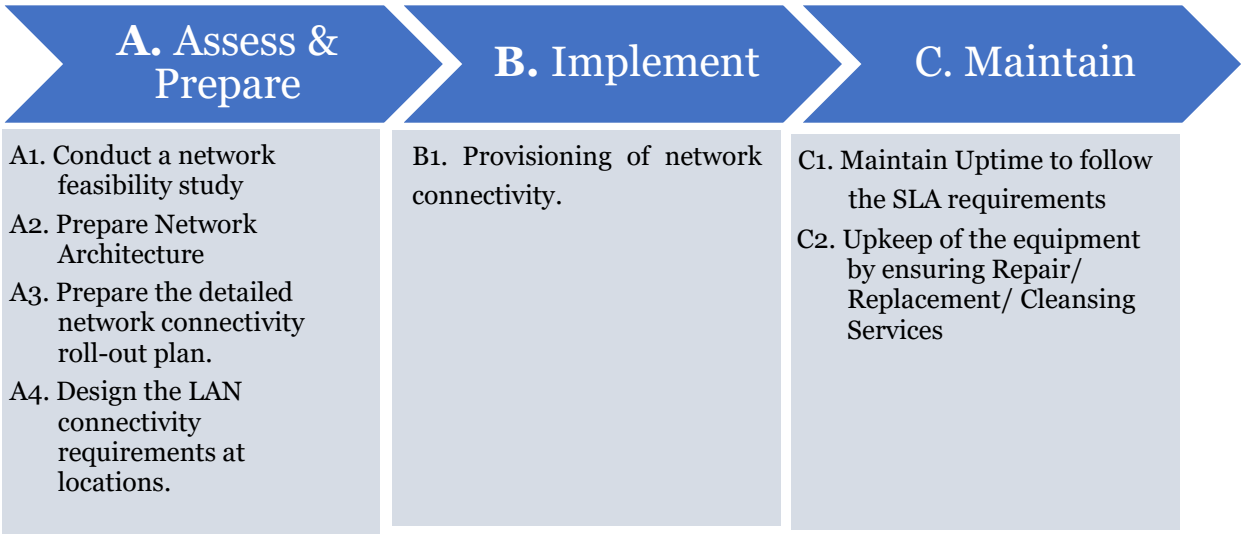
The Selected SI shall deploy robust, reliable and scalable network to enable converged communication. The bidder will be required to provide all the required equipment (active and passive both) for establishing such connectivity and the service levels specified in the RFP will need to be deployed as a part of the overall networking solution.

Implementation Services

Network Connectivity is one of the most important components of the project and needs very careful attention in assessment, planning and implementation. It is important not only to ensure that the required connectivity is provisioned within the required timelines but also ensure that it is reliable, secure and supports the required SLA parameters of Latency, Jitter, Packet Loss and Performance.

Establish connectivity between following -

- Between Cameras/edge devices & Aggregation Point at edge level
- Between Command/ Viewing Centres & Aggregation Point/Data Centre
- Between Aggregation Point & Data Centre



A1. Conduct a network feasibility study

Based on the finalized locations of edge devices, selected SI must conduct a feasibility study along with existing ISP i.e BSNL.

CAPEX & OPEX cost involved in setting up and operating the network connectivity will be in scope of SI. SI must consider cost of (Digging, fiber laying, cable/conduit laying, refilling, active and passive network components, operational cost, Switch, Node etc. required to terminate the network from edge level to aggregation point/data center as part of their proposal.

Bidder will not be given any extra timeline to setup new network. Bidder must adhere to the timelines mentioned in the tender document.

A2. Prepare Network Architecture

The selected SI will be required to prepare detailed network architecture of the overall system, incorporating findings of site survey exercise. The network so envisaged should be able to provide real time video stream/data to the command centers and police stations through data center. All the components of the technical network architecture should be of industry best standard and assist bidder in ensuring that all the connectivity SLAs are adhered to during the operational phase. The network architecture must ensure convergence with the existing Pune CCTV data network while maintaining security standards ad provisions.

A3. Prepare the detailed network connectivity roll-out plan

The Selected SI shall prepare the overall network connectivity establishment plan for this project. The plan shall comprise of deployment of network equipment at the locations to be connected over network, any clearances required from other State government departments, timelines for setting up of the entire network. The detailed plan shall be also comprising of the scalability, expandability, and security that such architecture will implement under this project.

It is necessary that at least 80% of the proposed last mile connectivity should be wired. Bidder shall also agree to increase the wired connectivity to 90% within 1 year of Go Live. Last Mile to be defined as “the access link from the service provider’s PoP – (as per Telco Standards) to the Camera”.

A4. Design the LAN connectivity requirements at locations

The Selected SI shall be responsible for deploying the LAN connectivity requirements at the Police locations such as Police stations, and other offices. The LAN connectivity may involve setting up the structured cabling, commissioning of active and passive components for patronization the surveillance system. At present the LAN establishment would be done at the proposed Police Stations and other viewing centers proposed in this RFP as indicated in **Annexure III** of this RFP.

B1. Provisioning of connectivity between Cameras, Command Center, Police Stations

Aggregation points and Data Centers

A combination of network technology including leased lines, MPLS and Mobile Network technologies is expected to be used to provide seamless connectivity to all cameras. Connectivity to Data Centers and control rooms shall be provided with scalable capacities to allow for expansion in the future.

Bidders are required to note that cameras already installed by GoM as a part of the existing project at various junctions, and these are connected to the network so that their feed is made available at Control Center. Bidder would be required to provide connectivity from the existing cameras and additional edge devices to the proposed CCTV network. Feeds are required to be stored at DC & DR. Bidders are required to include the last mile connectivity costs for the edge devices.

Bidders are required to do the estimation of bandwidth & storage requirements considering following benchmark parameters:

Minimum benchmark parameters for existing cameras (Track 1): -

Minimum benchmark parameter		**PTZ Camera**	**Fixed Camera**	**ANPR Camera**
Resolution		1920 X 1080 (Full HD)	1920 X 1080 (Full HD)	1920 X 1080 (Full HD)
Bitrate		4 Mbps	4 Mbps	4 Mbps
Frame per second (for Viewing & Storage)	Normal day Time (6 am to 12 am)	18 FPS	18 FPS	25 FPS
	Night Time (12 am to 6 am)	10 FPS	10 FPS	25 FPS

Minimum benchmark parameters for newly proposed cameras (Track 2): -

Minimum benchmark parameter		**PTZ Camera**	**Fixed Camera**	**ANPR Camera**
Resolution		1920 X 1080 (Full HD)	1920 X 1080 (Full HD)	1920 X 1080 (Full HD)
Bitrate		4 Mbps	2.4 Mbps	4 Mbps
Frame per second (for Viewing & Storage)	Normal day Time (6 am to 12 am)	18 FPS	18 FPS	25 FPS
	Night Time (12 am to 6 am)	10 FPS	10 FPS	25 FPS

The actual bandwidth requirement and network solution to cater to above mentioned bandwidth & storage parameters and to meet SLAs would be calculated by the selected SI and the same shall be clearly proposed in the technical proposal with detail calculations. GoM also requires the bidder to meet the parameters of video feed quality; security & performance and bidders should factor the same while designing the solution.

The acceptance of the solution would be subject to the proposed networking solution achieving the service level requirements of GoM with respect to the Video based Surveillance System. The solution needs to adhere to the following minimum benchmark requirements –

- Latency should be less than 20 ms

- Jitter should be less than 10% of one-way latency
- Packet loss should be less than 0.5%

It is expected that SI shall design the networking solution in such a manner that there is no single point of failures for connectivity at every pole and have back-to-back agreement with Telecom Service Providers to meet all the uptime & and quality related SLAs.

In case the Telecommunication guidelines of Government of India require the purchaser to place Purchase Order to the Service Provider for bandwidth, Home Department shall do so. However, Systems Integrator shall sign a contract with Telecom Service Provider(s) and ensure the performance. Home Department shall make payments to the Systems Integrator.

C1. Maintain Uptime to follow the SLA requirements

The entire project's backbone is network. SI will ensure that the all the network components are maintained with requisite uptime. Majorly, network component defines the uptime/downtime of the surveillance equipment. SI should note that SLA will be same for existing and new surveillance equipment.

C2. Upkeep of the network components by ensuring Repair/Replacement/Cleansing Services

Performance of SI will be assessed on basis of SLA. Hence, upkeep of all network components from Go Live till end of project by providing Repair/ Replacement/ Cleansing Services will be sole responsibility of SI.

Stage Deliverables:
1. Detailed Network feasibility report with Coverage Study
2. Detailed trenching plan
3. Proposed network architecture integrating existing and new systems
4. Network SLA plan

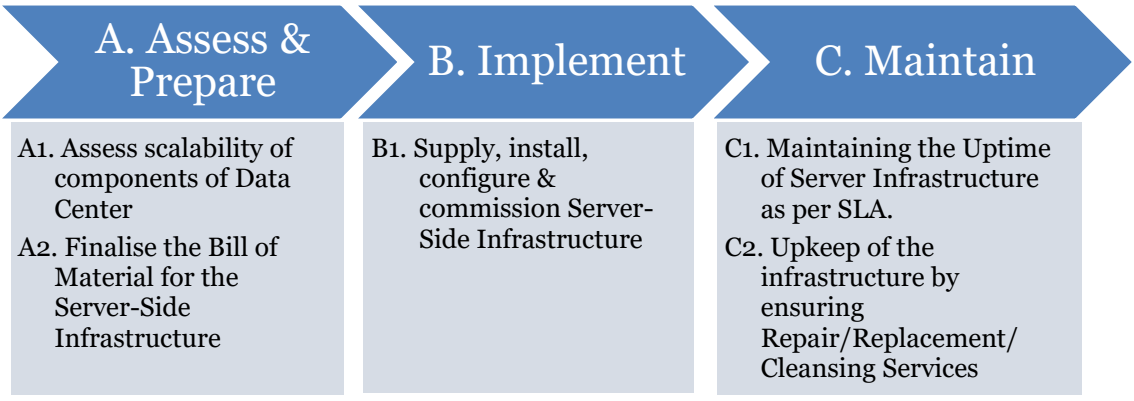
1.2.4 Implementation & Maintenance of Component #3: Augmentation of existing IT infrastructure in Data Center

Scope:

Selected SI has to augment the infrastructure in identified DC for additional components which are to be deployed as a part of Track II of this project. New computational requirement needs to be assessed for augmenting the infrastructure of DC.

Implementation Services:

This component includes assessment and procurement of Server-Side Infrastructure and operationalization of the same at the Data Center.



A1. Assess scalability of components of DC

Selected SI will do detailed site survey of current infrastructure at the co-located Data Center to assess scalability since new components are being added in this Track II of the project. The Data Center equipment for this project is co-located at Nxtra Data Ltd. Tier III data center at Kharadi, Pune (**2nd Floor, Airtel E Park Center, Chandan Nagar, Thite Nagar, Kharadi, Pune 411014**). Details of existing data center infrastructure is mentioned in **Annexure II.**

A2. Finalise the Bill of Material for the Server-Side Infrastructure

Bidders are required to carry out the detailed technical assessment including solution sizing and propose detailed Bill of material and bill of quantity (for compute, storage, enterprise management system, security system, etc.). Bidder to submit the certificate from OEM about the compatibility of new hardware/software with existing system and share the same in the Technical Bid Proposal. Pune Police reserves the right to dis-qualify the bidder if the BoQ & corresponding sizing is not sufficient to cater to the requirements of this tender and doesn't meet the following key parameters:

- High Availability of Network.
- Should provision failover/ redundancy for servers at DC.
- High Performance to ensure the end users get best of the UI/ UX Experience
- Scalable to meet the increased client-side requirements. The proposed IT infrastructure (Servers, storage and network) must cater to (from day 0):
    - Increase in number of CCTV cameras by 10 % of total camera count (existing + new)

B1. Supply, install, configure & commission Server-Side Infrastructure

The selected SI shall provide system integration services to procure and commission the required software and hardware infrastructure at the Data Center and deploy the complete surveillance management applications. The SI shall be completely responsible for the sourcing, installation, commissioning, testing and certification of the necessary software licenses and infrastructure required to deploy the solution at the Data Center. The selected SI shall be responsible for the provisioning of connectivity with adequate bandwidth from cameras to Data Center and from data center to all the command centers.

The selected SI shall be required to submit a detailed installation report post installation of all the equipment at approved locations. The report shall be utilized during the acceptance testing period of the project to verify the actual quantity of the equipment supplied and commissioned under the project.

C1. Maintain Uptime to follow the SLA requirements

Selected SI will ensure that the data center components are maintained with requisite uptime. Same SLA shall be applicable for existing equipment as well as the newly procured equipment.

C2. Upkeep of the network components by ensuring Repair / Replacement / Cleaning Services

Performance of SI will be assessed on basis of SLA. Hence, upkeep of all component from Go Live till end of project by providing Repair/ Replacement/ Cleaning Services shall be sole responsibility of the selected SI.

Stage Deliverables:

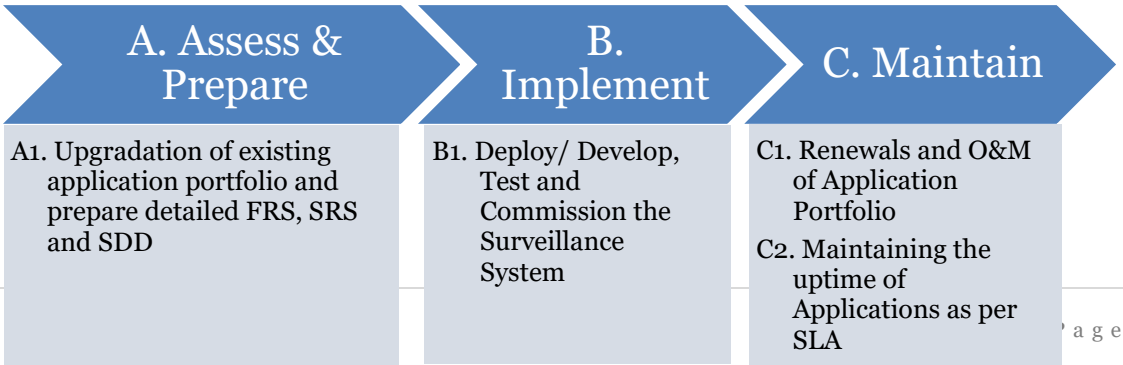
1. Detailed Data Center layout
2. Data Center architecture (including reusability)
3. Data Center Scalability plan

1.2.5 Implementation & Maintenance of Component #4: Application Portfolio

Scope:

Selected SI needs to deploy enhanced applications which can effectively address the requirement as described in this RFP.

Implementation Services



A1. Upgradation of existing application portfolio and prepare detailed FRS, SRS and SDD

Indicative application portfolio to be commissioned as part of the project is as follow:

Application & System Software

- Video Management System
- Command control center application
- Analytics Software
- Enterprise Management System including SLA Management, Helpdesk Mint, Network Mgmt. & BMS
- Anti-virus Software
- Drone application
- Facial recognition
- Traffic violation detection
- Advanced video analytics
- Video Summarization
- GIS
- Vehicle tracking
- Any other application

The exact quantity and requirement would emerge after the Project Design Document, prepared by the Selected SI and is approved by GoM. Benchmark specifications for various items mentioned above are given in the **Annexure I** to this RFP document. The present RFP document covers the key expectations from the project. However, it is required that the selected SI documents the requirements in detail before the work on execution begins. Following documents are expected to be delivered as part of this documentation:

Functional Requirements Documentation, giving complete details of the functional aspects of the project. Some of the key functional requirements for designing the system are given in **Section 2** of this RFP.

System Requirements documentation, giving complete details of the entire system components and their inter relationships to execute the project once operational.

System Design Documentation, detailing out the design of the Surveillance System, Command Control Center in particular, including integration with various systems like Transport Department Systems, CCTNS, etc. This will also include detailed use cases for advanced video analytics.

The selected SI should prepare above mentioned documents in discussion with all key stakeholders (Home Dept, Police Dept, IT Dept, Consultants, etc.). It is expected that the Selected SI brings on table some International Experience in this field and ensure that a progressive system is implemented for the Pune Surveillance.

Integration requirement for the application portfolio is listed below (non-exhaustive):

a) Addition of new cameras and edge devices in SMC/ Command Center application
b) Addition of new cameras and edge devices in VMS
c) Integration of existing VTS solution with the SMC/ Command Center application
d) All alerts generated from video analytics to be integrated with SMC/ Command Center application
e) All assets to be plotted on single GIS map
f) All collaborative monitoring feeds to be integrated with SMC/ Command Center Application
g) Configuration of advanced video analytics and integration with SMC/ Command Center application
h) Integration of Drones with SMC/Command Center Application

B1. Deploy/ Develop, Test and Commission the Surveillance System

The Selected SI will be responsible for the solution deployment / customization for implementing end-to-end surveillance management system including its integration with other systems as mentioned above. The application or COTS will be customized to meet the project objectives and the requirements of Pune Police. The bidder will ensure that the Best Practices for Software Development and Customization are used during the software development / customization and implementation exercise. This would at a minimum include:

(a) Software development / customization based on the functional requirement specifications, system requirement specifications, software requirement specifications and solution designs finalized after the approval of Pune Police. Wherever necessary, the Selected SI shall develop additional functionality/modules in order to meet the project requirements.

(b) Delivering the surveillance management system, along with all of the necessary modules and additional functionalities/ integrated COTS products, utilities, system drivers and documentation consistent with proven standards, including product updates, technology upgrades and patches to run on the selected operating system(s) and hardware according to the solution.

(c) Deployment and commissioning of surveillance management system with all the necessary solution elements at the data center. It is pertinent to mention that application hosted at the data center shall be accessible by the intended users as desired under this project.

(d) Provision for Police officers to login into the system remotely from any location via a secure private network.

Note: Over the period of the contract, even after the Go-Live of the system, department may require certain modifications or additions in the application or the development of new modules. Also, the selected Bidder can suggest certain improvements in the software to make the operations more effective at no additional cost. In such a situation, the selected Bidder shall be responsible for carrying out software enhancement / development activities, as requested by the Department. Any software development / modification will need to pass through the following envisaged phase:

- Feasibility study / proposal for change
- Conceptualization of solution
- Requirement Study
- Design
- Development
- Unit and Integration testing
- Regression Testing
- User acceptance testing
- Roll out

At each of the above phases, the Selected SI would have the deliverables (including documentation) reviewed and approved by the Pune Police or its nominated agencies/ representatives. Pune Police will approve all the deliverables; only then the Selected SI shall commence with the next course of action. Software modifications / development will be considered completed only after formal acceptance provided by Pune Police.

C1. Licenses for new components

Selected SI shall assess the additional license requirement catering to the additional edge level components to be deployed as a part of Track 2 of this project. Selected SI to factor in adequate number of licenses to ensure complete functionality of video analytics as described in Functional Requirement section of this RFP

C2. Maintain Uptime to follow the SLA requirements

Selected SI will ensure that the data center components are maintained with requisite uptime. Same SLA shall be applicable for existing applications as well as the newly installed applications.

Stage Deliverables:
1. Detailed application integration architecture
2. Detailed list of customizations required
3. System Requirements Document
4. Functional requirement document
5. System design document

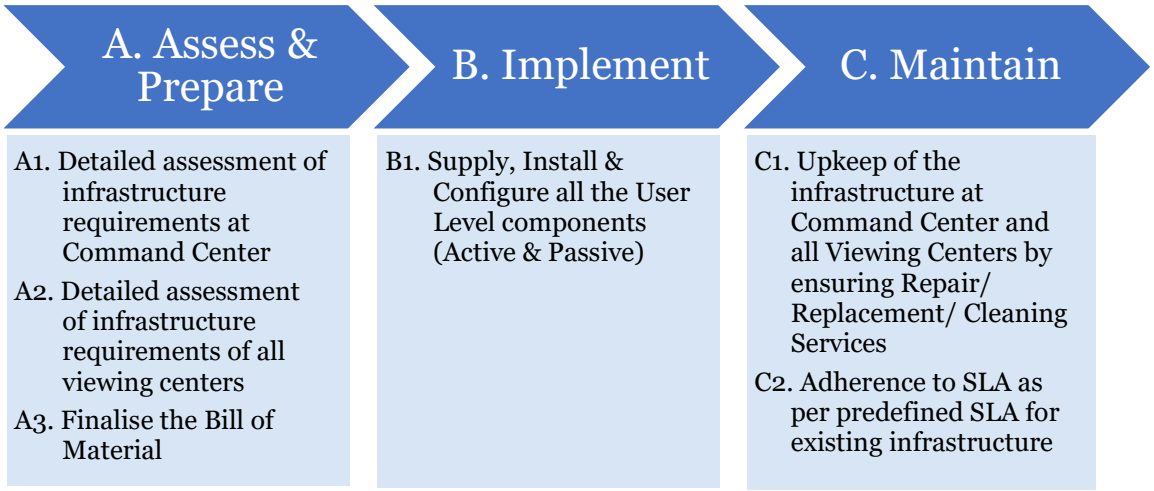
1.2.6 Implementation & Maintenance of Component #5: Command Center and setup of additional Viewing Centers

Scope:

As a part of the existing project, full-fledged Command Center is already functional at CP Office and the details of infrastructure and components deployed are provided in **Annexure II**. Additional viewing centers are required in the locations mentioned in the Annexure III- Annexure C: List of proposed Viewing Centers.

Selected SI shall integrate existing Command Center applications, fine-tune existing 30+ SOPs and create new SOPs, if required; and creation of updated dashboards to incorporate additional components proposed as a part of Track 2. Deployment of additional edge devices shall entail upgrades in Command Center applications and the selected SI should ensure that existing infrastructure is scaled up to incorporate additional edge devices.

Implementation Services



A1. Detailed assessment of infrastructure requirements at Command Center

SI will do detailed study of existing command center and scale it up as per the new components that will be added due to installation of cameras at new locations. The details of current infrastructure, application portfolio is mentioned in **Annexure II.** SI must commission new LED Video wall with controller & workstations in existing CCC and buyback the existing Video wall ,controller & workstations. Bidder to augment the graphic card by minimum 4GB, in each workstation installed in CCC for viewing of video feeds. SI shall also assess the Command Center ergonomics and implement modification where necessary. It may require shifting of existing command center to suitable location in same building where current command center exits. It is the responsibility of SI to reuse the existing IT infrastructure & Non-IT infrastructure and propose suitable solution. To assess the actual condition of command center, bidder may visit the current command and control center.

A2. Detailed assessment of infrastructure requirements for viewing centers

Apart from command center the infrastructural components are available at viewing centers. The list of viewing centers wise infrastructure is available in **Annexure II.** SI will implement setup of new viewing centers keeping in mind the architecture followed for the existing viewing centers. Locations for new viewing centers is listed in Annexure III.

A3. Finalize the Bill of Material

Indicative Bill of Material for Client-Side IT Infrastructure at viewing centers is given below:

Viewing Center at DCP office/ Police Station:

IT Components

- 40" LED TV (1 no. at each)
- Workstations with 1 monitor (1 no. at each)
- Adequate LAN Components
- IP Phone (1 no. at each)

Non-IT Components

- Chairs & Computer Tables (1 no. at each)
- Electrical Work

- UPS (adequate capacity to cover all above IT Components with 30 min. Backup)

B1. Supply, Install & Configure all the User Level components (Active & Passive)
The System Integrator shall develop a plan to procure, install and configure all the necessary items for the Command Center/ Viewing Center in a timely fashion. There should be a tracker created and shared with Pune Police that would track all the commissioning of the equipment, the timelines adhered to and the compliance to the requirements.

C1. SLA Management for the Command Center and additional Viewing Centers

Existing Command Center is located at Pune Police Commissionerate Office. The existing setup at command center is given in **Annexure II.** SI will maintain the command center by providing services including complete maintenance over the period of project O&M phase.

SI will provide the support for all these above said Viewing Centers and ensure availability of them throughout the O&M phase to ensure adherence to SLA.

C2. Upkeep of the infrastructure at Command Center and all Viewing Centers by ensuring Repair/ Replacement/ Cleaning Services

Number of components are deployed in each of above-mentioned viewing centers and in command center. These will require maintenance till the project O&M phase. Selected SI will provide the all the services to:

- Repair
- Replacement
- Cleaning

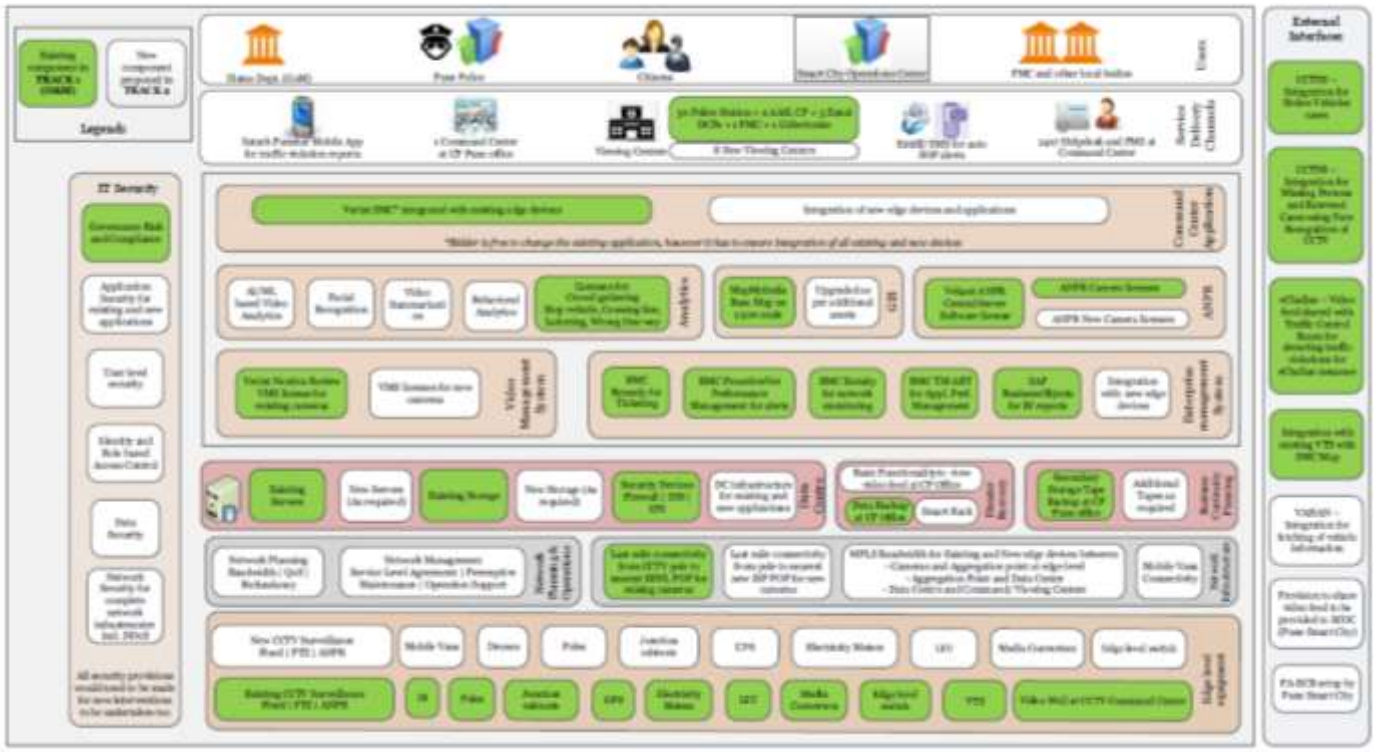
of the equipment as and when required to ensure the uptime of these equipment. This will help SI to maintain the SLA requirements. Details of quantity, make, model, specifications, etc. about existing components is available in **Annexure II.**

Stage Deliverables:
1. Command Center modified layout
2. Command Center revised SOP's after considering additional components

1.2.7 Integration of new system with existing system

The project is a combination of O&M of existing infrastructure as well as setup of new infrastructure. It will sole responsibility of SI to integrate the existing system with new components. All the components of existing infrastructure are to be studied by SI and accordingly augment the existing technical architecture of project comprising of both existing and new systems. Same SLA should be applicable to existing and new components of project.

A technology blueprint comprising of Track 1 and Track 2 has been depicted below, the components highlighted in green are existing and need to be integrated with new initiatives.



1.2.8 Preparation and implementation of the Information security policy, including policies on backup

The selected SI shall prepare the Information Security Policy for the overall project and the same would be reviewed by the Home Department & it's Consultant & then finalized by the Home Department. The Security policy needs to be submitted by the Systems Integrator within 1st quarter of the successful Final Acceptance Tests.

The Systems Integrator should then obtain ISO 27001 certification for CP Office Control Center within 1 quarter of final acceptance test. Payment from 2nd Quarter onwards shall be withheld till this certification is obtained by the Selected SI.

1.2.9 Training to the Police Personnel for operationalization of the system

Training is an important aspect of this project, and Government expects the Selected SI to undertake it in a very professional manner. Bidder has to conduct a proper Training Needs Analysis of all the concerned staff and draw up a systematic training plan in line with the overall project plan. For all these training programs the bidder has to provide necessary course material and reference manuals (user/maintenance/ administration).

Trainings would be of three types:

Functional Training: This training would focus on the use of the surveillance software at Command & Control Rooms, so that the users are aware of all the operations of the surveillance systems and are able to implement the overall process defined by the Police Department for optimum use of the system. Broad training requirement defined for the purpose of calculation of effort is as follows –

- Training program of 2 batch (of 15 personnel) of 1 day - Twice for 1st year
- Training program of 1 batch (of 15 personnel) of 1 day – Once in every subsequent year

Administrative Training: This training would focus on the administration of Surveillance System and Server Infrastructure and would be imparted to about 8 – 10 staff identified by the Government for Administration of the System from Government side. SI shall also provide additional training program of 1 batch (of 8 - 10 personnel) every year. Expected training time would be 4 hrs. for each batch.

Senior Management Training: This training would focus on how to use the surveillance system for day-to-day monitoring by the Sr. Management and access various exception reports. Broad training requirement defined for the purpose of calculation of effort is as follows –

- Initial Training of approx. 20 police officers (i.e. about 2 batches of 10 officers each). Expected training time of 4 hrs.
- Additional training program of 1 batch (of 10 personnel) every 2.5 years.

Other requirements to be fulfilled by the Selected SI with respect to Training are as follows:

- Prepare the training material in consultation with Home / Police Department & it's Consultant. Detailed Training manuals would be prepared by the Systems Integrator prior to the start of the training.
- One Hard Copy & One Soft Copy of the training material shall be given by the Systems Integrator to all the trainees.
- Functional training shall be delivered at each of the Zonal offices, while location for the Administrative & Sr. Management Training would be decided subsequently
- Selected SI should ensure that the knowledge transfer to the Pune Police staff happens effectively post training.

1.2.10 Partial Acceptance Testing & Final Acceptance Testing of IT & Non-IT Equipment

The acceptance test for the project shall be carried out in two phases by Home Department or duly appointed third party agency by Home Department. The Selected SI should cooperate with the third-party agency to ensure successful completion of Acceptance tests.

The acceptance test shall consist of a Partial Acceptance Test (PAT) and Final acceptance test (FAT). The Selected SI shall submit a detailed acceptance testing document at the stage of planning (as mentioned in section 1.1 of this Volume) and Home Department & the Selected SI shall mutually agree upon the same.

1.2.10.1 Partial Acceptance Test

Partial Acceptance Test shall involve scrutiny of documents for various IT / Non-IT components to verify if the specifications conform to the requirements mentioned in the RFP and subsequent corrigendum. Police Department reserves right to conduct physical inspection of the equipment delivered to ensure that they arrive at the sites in good condition and are free from physical damage and incomplete shipments and shall return the products to the Supplier at the supplier's expenses if required quality is not maintained. Physical inspection of hardware will also include physical checking and counting of the delivered equipment in presence of the Selected SI. This equipment will only be acceptable as correct when each received item corresponds with the checklist that will be prepared by the Selected SI prior to shipment. Any shortfalls in terms of number of items received may render the delivered equipment incomplete. SI shall submit third party test reports on performance for the critical components like Cameras, Active Network Equipment, Servers, Video Wall, etc. Physical verification of the individual items would be undertaken as part of FAT for Server-Side equipment & Mobile Vans.

1.2.10.2 Final acceptance test

After successful installation of equipment in accordance with the requirements in the RFP, the Selected SI would need to carry out Final Acceptance Testing in 2 different phases - **(a) Unit Testing** and **(b) Integration Testing**. These tests would be carried out based on the test cases developed and validated by Home Department and Pune Police. Apart from the functional testing of the entire system components, the testing would also verify following aspects –

- Configuration Testing (to ensure that all the components are configured properly)
- Security Testing (to review & evaluate security controls)

Home Department will issue final acceptance certificate to the selected SI after successful testing in a real time condition for at least 15 days of trouble-free operation. The date on which final acceptance certificate is issued shall be deemed date of the successful commissioning of the project (either whole or partial, depending upon the implementation strategy adopted). Home Department shall consider implementation of 95% cameras as a sufficient condition for the overall project Go-Live. Any delay by the Selected SI in the performance of its contracted obligations shall render the Selected SI liable to the imposition of appropriate liquidated damages, unless agreed otherwise by tenderer.

1.2.11 System Documents, User Documents

The Selected SI will provide documentation, which should follow the ITIL (Information Technology Infrastructure Library) standards. This documentation should be submitted as the project undergoes various stages of implementation. Indicative list of documents includes:

- Project Commencement Documentation: Project Plan in giving out micro level activities with milestones & deadlines.
- Cabling Layout: Systems Integrator shall submit the detail cabling Layout including cable routing, telecommunication closets and telecommunication outlet/ connector designations. The layout shall detail locations of all equipment and indicate all wiring pathways.
- Equipment Manuals: Original Manuals from OEMs.
- Installation Manual: For all the Application Systems
- Training Material: Training Material will include the presentations used for trainings and also the required relevant documents for the topics being covered.
- User Manuals: For all the Application Software Modules, required for operationalisation of the system.
- System Manual: For all the Application Software Modules, covering detail information required for its administration.
- Operational Manual: The bidder shall be responsible for preparing Operational Manual relating to operation and maintenance of each and every service as mentioned in this RFP. The prepared process document shall be formally signed off by Home Department before completion of final acceptance test.

Note: The Selected SI will ensure Upkeep & Updating of all documentation and manuals during the contractual period. The ownership of all documents, supplied by the Selected SI, will be with Home Department. Documents shall be submitted in two copies each in Printed & in Softcopy Formats.

Success of the project would lie on how professionally and methodically the entire project is managed once the implementation is completed. Successful Integrator thus is required to depute a dedicated team of professionals to manage the project and ensure adherence to the required SLAs.

1.2.12 Helpdesk and Facility Management Services

The details of components at FMS are given in details in **Annexure II**. Selected SI will study the same and scale up as per the requirements of the projects. Licensing and renewals of necessary applications will be responsibility of selected SI.

As part of Facilities Management System (FMS), Command center shall have dedicated, and the mobile vans & police stations shall have shared manpower provided by the SI to troubleshoot & solve issues / problems. The FMS shall be supported by a centralized helpdesk setup within Pune City. This Helpdesk shall be the single point of contact for complaint management & resolution for all the users of the surveillance system. This helpdesk shall be integrated with the Enterprise Management System. The helpdesk shall be designed to meet the SLA response & resolution timelines and provide support to the Police officials in performing their day-to-day functions related to this system.

Functional requirements of the helpdesk management system, fully integrated with the enterprise monitoring and network management system, is provided in **Section 2**. The system will be accessed by the Police officials for raising their incidents and logging calls for support. The detailed service levels and response time which the Selected SI is required to maintain for provisioning of the FMS services are described in the service level agreement of this RFP.

Systems Integrator is also required to depute a dedicated, centralized project management & technical team for the overall project management and interaction with Sr. Police Dept / Home Dept personnel. Indicative resource requirement for this centralized administration of the project during implementation and operation & maintenance phase is as follows:

#	Role	Shifts			Min. Qualification & Experience
		1	2	3	
1	Project Manager	√			BE / BTech/MCA + MBA (preferable) 15+ Years of Experience in IT
2	Asst. Project Manager	√			BE / BTech/MCA + MBA (preferable) 5+ Years of Experience in IT
3	Command center Manager	√			BE / BTech/BCA + MBA (preferable) 5+ Years of Experience in IT & 2+ Years of experience in managing command control center
3	Technical Expert – Network		√		BE / BTech/MCA

#	Role	Shifts			Min. Qualification & Experience
		1	2	3	
					5+ Years of Experience in Network Management
4	Technical Expert – Security		√		BE / BTech/MCA 5+ Years of Experience in IT Security
5	Technical Expert – Server	√			BE / BTech/MCA 5+ Years of Experience in Server Management
6	Technical Expert – Storage	√			BE / BTech/MCA 3+ Years of Experience in Storage Management
7	Technical Expert – EMS		√		BE / BTech/MCA 3+ Years of Experience in EMS
8	Technical Expert – VMS (Video Management System)	√			BE / BTech/MCA 3+ Years of Experience in VMS/ICCC application/ Command control center application

The Systems Integrator is expected to undertake the Facilities Management Service and ensure that all the SLAs are adhered to for 5 years from the date of successful Final Acceptance Testing. In order to ensure that there is no downtime for any of the critical components, the successful Systems Integrator is expected to provision on-site spares and resident engineer(s). Helpdesk Personnel would be located at CP Office Command Center.

Please refer detailed Service Level Agreement (SLA) defined in this RFP to be adhered by the Systems Integrator.

1.2.13 Hand-over of the system at the end of contractual period along with all documentation required to operate and maintain the system.

Systems Integrator shall supply to Project Officer, Pune Police the following before end of 3rd last Quarter (i.e. min. 6 months prior to the expiry of the contract):

- Information relating to the current services rendered and data relating to the performance of the services; Entire Documentation relating to various components of the Project, any other data and confidential information related to the Project;
- All other information (including but not limited to documents, records and agreements) relating to the products & services related to the project to enable Project Officer, Pune Police and its nominated agencies, or its Replacing Vendor to carry out due diligence in order to transition the provision of the Project Services to its nominated agencies, or its Replacing Vendor (as the case may be).
- Source code for all customized applications developed by SI for GoM during this project.

1.2.14 Planning, Suggesting and Submitting the Surveillance System up-grade plan(s) for five years from the date of acceptance along with detailed specifications

As we are aware, constant changes / updates happen in technology, and it is very important that the Surveillance System implemented by the City of Pune keeps its pace with the technology. Home Department would want the Selected SI to submit a report, every 6 months, on the advancements available in technology to make the best use of the existing infrastructure. Any upgradation / augmentation suggested by the SI would be analyzed by the Home Department & it's consultant (if any) and appropriate decision would be taken.

2. Functional Requirements

Functional Requirement of the overall Surveillance System can be categorized into following components:

- Information to be Captured by Edge Devices
- Information to be Managed at the Command Center
- Information to be made available to different Police Personnel
- Operational Requirements
- Storage / Recording Requirements
- Other General Requirements
- Helpdesk Management
- Project Execution Requirements

- Video Analytics

2.1 Information to be Captured by Edge Devices

Cameras being the core of the entire Surveillance system, it is important that their selection is carefully done to ensure suitability & accuracy of the information captured on the field and is rugged, durable & compact. These cameras need to work on 24 X 7 basis and transmit quality video feeds to the centralized data centers and would capture the video feeds at 18 FPS for majority of the time and at 10 FPS for the lean period. However, Police Department shall take the regular review of the requirements for video resolution, FPS and may change these numbers to suit certain specific requirements (for example, there could be a situation when certain cameras are required to be viewed at higher FPS for specific period. It is estimated that not more than 2% of the cameras would be required to be viewed at higher FPS at a given point of time).
Please refer to the benchmark camera specifications given in **Annexure I** and **Annexure II** to understand the detail requirements with respect to camera types & estimated numbers. As clearly identified in the SLA table of this RFP. Cameras need to transmit quality video feed (appropriately focused, clear, unblurred, jitter free, unobstructed, etc.). Cameras should maintain minimum bitrate of 4 Mbps to transmit quality video feed (appropriately focused, clear, un-blurred, jitter free, properly lit etc.). Packet loss should be less than 0.5%.

2.2 Information to be managed at the Command Centers

The proposed Video Management System should provide a complete end-to-end solution for security surveillance application. The control center shall allow an operator to view live / recorded video from any camera on the IP Network. The combination of control center and the IP Network would create a virtual matrix, which would allow switching of video streams around the system.
As informed in the tender, not all the cameras would be simultaneously viewed at Command Control Centers or at Police Stations. Police personnel shall have following access to the video feeds of the cameras of their jurisdiction:

- Viewing rights to all the live Camera Feeds.
- Viewing rights to the stored feeds, stored on Primary / Secondary Storage
- Access to view Alerts / Exceptions / Triggers raised
- Trail Report on specific person / object / vehicle for a specific period / location
- Personalized Dashboard (depending upon grade of police officer)
- Accessibility to advanced analytics on recorded footages
- Provide search of recorded video. Advanced search should be possible based on various filters like alarm / event, area, camera, etc.

Functionality Description:

- The solution should be implemented and compliant to industry open standard commercial-off-the-shelf (COTS) applications that are customizable.
- The solution shall integrate with GIS and map information and be able to dynamically update information on the GIS maps to show status of assets/resources.
- It shall provide complete view of sensors, facilities, video streams and alarms in an easy-to-use and intuitive GIS-enabled graphical interface with a powerful workflow and business logic engine.
- The platform should be able to consume Geospatial data / maps/ services in the form of web-services or direct integration using APIs.
- The platform should be able to disseminate spatial or non-spatial data/information to OGC compliant Geospatial platform.
- The platform should be able to use already available geospatial tools/ widgets/ web services and provide them on CCC interface.
- The solution shall also provide an integrated user interface for all the smart elements & surveillance system implemented
- The solution should provide operators and managers with a management dashboard that provides a real time status and is automatically updated when certain actions, incidents and resources have been assigned, pending, acknowledged, dispatched, implemented, and completed. The above attributes shall be color-coded.
- The solution shall provide the “day to day operation”, “Common Operating Picture” and situational awareness to the center and participating agencies during these modes of operation

- It shall improve scalability for large and geographically distributed environments
- It shall provide a uniform, coherent, user-friendly and standardized interface
- It shall provide possibility to connect to workstations and accessible via web browser
- The dashboard content and layout shall be configurable, and information displayed on these dashboards shall be filtered by the role of the person viewing dashboard
- The solution should allow creation of hierarchy of incidents and be able to present the same in the form of a tree structure for analysis purposes
- The solution shall be available via a VPN as a web-based interface or on mobile app.
- It shall be possible to combine the different views onto a single screen or a multi-monitor workstation
- The solution should maintain a comprehensive and easy to understand audit trail of read and write actions performed on the system
- The solution should provide ability to extract data in desired formats for publishing and interfacing purposes
- The solution should provide ability to attach documents and other artifacts to incidents and other entities
- The solution is required to issue, log, track, manage and report on all activities underway during these modes of operation:
  - anticipation of incident
  - incident or crisis
  - recovery
  - incident simulation

2.3 Information to be made available to different Police stations or viewing centers

Various Police Stations shall be given viewing rights of the feeds of the corresponding cameras from the respective Police Stations. Police personnel shall have following access to the video feeds of the cameras of their jurisdiction:

- Viewing rights to all the live Camera Feeds
- Viewing rights to the stored feeds, stored on Primary / Secondary Storage
- Access to view Alerts / Exceptions / Triggers raised in VMS/CCC application
- Trail Report on specific person / object / vehicle for a specific period / location
- Personalized Dashboard (depending upon grade of police officer). Depending upon the grade of officers, the access rights to different systems should be customizable.

2.4 Role Based Access to the Entire System

Various users should be able to access the system using single sign on and should be role based. Different roles which could be defined (to be finalized at the stage if SRS) could be Administrator, Supervisor, Officer, Operator, etc. Apart from role-based access, the system should also be able to define access based on location. Other minimum features required in the Role Based authentication Systems are as follows:

- The Management Module should be able to capture basic details (including mobile number & email id) of the Police Personnel & other personnel requiring Viewing / Administration rights to the system. There should be met to change these details, after proper authentication.
- Rights to different modules / Sub-Modules / Functionalities should be role based and proper log report should be maintained by the system for such access.
- Biometric authorization coupled with login name & password should be enabled to ensure that only the concerned personnel are able to login into the system.
- Surveillance System should have capability to map the cameras to the Police Personnel from different Police Stations.
- For PTZ cameras, there should be provision to specify hierarchy of operators / officers for control of the cameras from various locations.

2.5 Storage / Recording Requirements

It is proposed that the storage solution is modular enough to ensure compliance to the changes in storage / recording policy, to be evolved upon initial deployment of the system. Following storage requirements are proposed for the project:

a) **The storage solution proposed is that the video feeds would be available for 30 days storage of normal video feeds (fixed, PTZ and mobile surveillance) and 180 days storage of ANPR feeds**. After 30 for normal video feeds (fixed, PTZ & mobile surveillance) & 180 days for ANPR feeds, the video feeds would be overwritten unless it is flagged or marked by the Police for investigation or any other purpose. The video feeds of all relevant cameras capturing the incident in question would be stored until the Police deem it good for deletion.
b) For incidents that are flagged by the Police or any court order, the video of the relevant portion from all relevant cameras should be stored/ archived separately for investigation purposes and a committee at Police Department can decide when this video feed can be deleted.
c) Regardless of the above, the image of the License plate extracted by ANPR software, along with the timestamp and location of the image capture will stored for a period of 6 months
d) Audit trail of the system to be maintained on permanent basis / as per the backup policy defined.
e) Retrieval time for any data stored should be max. 4 hours for critical data & 8 hours for other data.
f) The Recording Servers / System, once configured, shall run independently of the Video Management system and continue to operate in the event that the Management system is off-line.
g) The system shall support the use of separate networks, VLANs or switches for connecting the cameras to the recording servers to provide physical network separation from the clients and facilitate the use of static IP addresses for the devices.
h) The system shall support H.264, H.265 or better, compression formats for all IP cameras connected to the system.
i) The system shall record the native frame rate and resolution supplied by the camera or as configured by the operator from the system administration server.
j) The system should not limit amount of storage to be allocated for each connected device.
k) The on-line archiving capability shall be transparent and allow Clients to browse and archive recordings without the need to restore the archive video to a local hard drive for access.
l) The system shall allow for the frame rate, bit rate and resolution of each camera to be configured independently for recording. The system shall allow the user to configure groups of cameras with the same frame rate, bit rate and resolution for efficient set-up of multiple cameras simultaneously.
m) The system shall support Archiving or the automatic transfer of recordings from a camera's default database to another location on a time-programmable basis without the need for user action or initiation of the archiving process. Archiving shall allow the duration of the camera's recordings to exceed the camera's default database capacity. Archives shall be located on either the recording server or on a connected network drive. If the storage area on a network drive becomes unavailable for recording, the system should have the ability to trigger actions such as the automatic sending of email alerts to necessary personnel.
n) Bandwidth optimization
    - The Recording Server / System shall offer different codec (H.264, H.265 or better) and video resolution options for managing the bandwidth utilization for live viewing on the Client systems.
    - From the Client systems, the user shall have the option of having video images continually streamed or only updated on motion to conserve bandwidth between the Client systems and the Recording Server.
    - The bidder is expected to calculate the bandwidth and storage requirement according to below table:-

Track – 1 (Existing cameras)				
Minimum benchmark parameter		**PTZ Camera**	**Fixed Camera**	**ANPR Camera**
Resolution		1920 X 1080 (Full HD)	1920 X 1080 (Full HD)	1920 X 1080 (Full HD)
Bitrate		4 Mbps	4 Mbps	4 Mbps
	Normal day Time (6 am to 12 am)	18 FPS	18 FPS	25 FPS

Track – 1 (Existing cameras)				
Minimum benchmark parameter		**PTZ Camera**	**Fixed Camera**	**ANPR Camera**
Frame per second (for Viewing & Storage)	Night Time (12 am to 6 am)	10 FPS	10 FPS	25 FPS

Track 2 – New cameras				
Minimum benchmark parameter		**PTZ Camera**	**Fixed Camera**	**ANPR Camera**
Resolution		1920 X 1080 (Full HD)	1920 X 1080 (Full HD)	1920 X 1080 (Full HD)
Bitrate		4 Mbps	2.4 Mbps	4 Mbps
Frame per second (for Viewing & Storage)	Normal day Time (6 am to 12 am)	18 FPS	18 FPS	25 FPS
	Night Time (12 am to 6 am)	10 FPS	10 FPS	25 FPS

o) The Recording Server / System shall support Camera (analogue and IP cameras) devices from various manufacturers.

p) The Recording Server / System shall support the PTZ protocols of the supported devices listed by the camera OEMs.

q) The system shall support full two-way audio between Client systems and remote devices.

r) Failover Support
   - The system shall support automatic failover for Recording Servers. This functionality shall be accomplished by Failover Server as a standby unit that shall take over in the event that one of a group of designated Recording Servers fails. Recordings shall be synchronized back to the original Recording Server once it is back online.
   - The system shall support multiple Failover Servers for a group of Recording Servers.

s) SNMP Support
   - The system shall support Simple Network Management Protocol (SNMP) in order for third-party software systems to monitor and configure the system.
   - The system shall act as an SNMP agent which can generate an SNMP trap as a result of rule activation in addition to other existing rule actions.

2.6 Other General Requirements and Storage

Management / Integration functionality

a) The Surveillance System shall be a fully distributed solution, designed for large multi-site and multiple server installations requiring 24/7 surveillance. The solution shall offer centralized management of all devices, servers and users.
b) The Surveillance System should not have any limit on the number of cameras to be connected for Surveillance, Monitoring, and recording. Any increase in the no. of cameras should be possible by augmentation of Hardware components.
c) The Surveillance System shall support distributed viewing of any camera in the system using Video walls or big screen displays.
d) The Surveillance System shall support alarm management. The alarm management shall allow for the continuous monitoring of the operational status and event-triggered alarms from system servers, cameras and other external devices.
e) It should be possible to integrate the Surveillance System with 3rd-party software, to enable the users to develop customized applications for enhancing the use of video surveillance solution. For e.g. Integrating alarm management to start SMS, E-Mail, VoIP call etc.
f) The Management system shall store the overall network elements configuration in central database, either on the management server computer or on a separate DB Server on the network.

System Administration functionality

a) The System Administration Server shall provide a feature-rich administration client for system configuration and day-to-day administration of the system.
b) The System Administration Server shall support different logs related to the Management Server.
   - The System Log
   - The Audit Log
   - The Alert Log
   - The Event Log
c) Rules
   The system shall support the use of rules to determine when specific actions occur. Rules shall define what actions shall be carried out under specific conditions. The system shall support rule-initiated actions such as:
   - Start and stop recording
   - Set non-default live frame rate
   - Set non-default recording rate
   - Start and stop PTZ patrolling
   - Send notifications via email
   - Pop-up video on designated Client Monitor recipients

Client system

The Client system shall provide remote users with rich functionality and features as described below.

a) Viewing live video from cameras on the surveillance system
b) Browsing recordings from storage systems
c) Creating and switching between multiple of views.
d) Viewing video from selected cameras in greater magnification and/or higher quality in a designated hotspot.
e) Controlling PTZ cameras.
f) Using digital zoom on live as well as recorded video.
g) Using sound notifications for attracting attention to detected motion or events.
h) Getting quick overview of sequences with detected motion.
i) Getting quick overviews of detected alerts or events.
j) Quickly searching selected areas of video recording for motion (also known as Smart Search).

Remote Web Client

a) The web-based remote client shall offer live view of up to 16 cameras, including PTZ control and event / output activation. The Playback function shall give the user concurrent playback of multiple recorded videos with date, alert sequence or time searching.

b) User Authentication – The Remote Client shall support logon using the user name and password credentials.

Matrix Monitor

a) Matrix Monitor – The Matrix Monitor feature shall allow distributed viewing of multiple cameras on the system on any monitor.
b) The Matrix Monitor feature shall access the H.265 stream from the connected camera directly and not sourced through the recording server.

Alarm Management Module

a) The alarm management module shall allow for continuous monitoring of the operational status and event-triggered alarms from various system servers, cameras and other devices. The alarm management module shall provide a real-time overview of alarm status or technical problems while allowing for immediate visual verification and troubleshooting.
b) The alarm management module shall provide interface and navigational tools through the client including.
    i. Graphical overview of the operational status and alarms from servers, network cameras and external devices including motion detectors and access control systems.
    ii. Intuitive navigation using a map-based, hierarchical structure with hyperlinks to other maps, servers and devices or through a tree-view format.
c) The module shall include flexible access rights and allow each user to be assigned several roles where each shall define access rights to cameras.
d) Basic VMS should be capable to accept third party generated events / triggers

Other Miscellaneous Requirements

a) System should have a facility to create CDs or other storage media for submission to Judiciary, which can be treated evidence for legal matters. Such storage media creation should be tampering proof and SI to provide proper technology so that integrity and quality of evidence is kept as per requirements of the judiciary. Bidder is required to specify any additional hardware / software required for this purpose & the same can be listed in miscellaneous section of the commercial bid. SI will also prepare the guideline document to be followed by the Police Personnel for the retrieval of Video / images from the CCTV System so as to maintain integrity of the evidence. Such a guideline document should include methods of retrieval of data, check-list to be followed and flowchart of the entire process to be followed.
b) All the systems proposed, and operationalization of Video Management System should comply with requirements of IT Act.
c) Bidder shall be required to provide a customized Mobile Application to integrate smart phones and tablets for 2-way communication with the Video Management System in a secure manner. Police may provide such tablets / smart phones to the designated Police Personnel. It will be responsibility of SI to configure such tablets / Smartphone with the Surveillance System and ensure that all the necessary access is given to these mobile users so that uploading of video / pictures to the surveillance system is possible.
d) There would be the provision for Third party audit periodically, paid by Police separately.
e) Police shall have collaborative monitoring of CCTV from various system and sub systems of public /private establishments like malls, railway stations, bus stands, temples, local shopkeeper etc. list of establishments will be provided by Police from time to time. SI will be responsible for integrating these cameras to city surveillance system for viewing whenever required.
f) Police shall temporarily install CCTV at various locations during various festivals. SI will be responsible for integrating these cameras into city surveillance system

2.7 Helpdesk Management

It is envisaged that the centralized helpdesk, functioning as proposed below, would be managed by the Systems Integrator and shall serve following objectives:

- Act as the Point of Contact for the users of Surveillance System
- Own an Incident throughout its Lifecycle
- Communicate effectively with Police / Home Dept Officers and IT support teams.
- Maintain high user satisfaction levels

- Maintain the SLA statistics & submit quarterly report to Police / Home Department

A general process flow for the helpdesk management is depicted in the flow-chart given as follows. Systems Integrator shall prepare a detailed Helpdesk Policy in consultation with the Home Department & it's Consultant prior to the Go Live date. System Integrator shall deploy a State-of-Art Enterprise Management System to handle the complexity of Operations & SLA Management defined in the RFP. Benchmark specifications for the Enterprise Management System are given in subsequent Annexure.

2.8 Project Execution Requirements

Actual placement of pole & number of cameras at each location, type of cameras, fixation of height & angle for the cameras to ensure maximum coverage shall be done in consultation with Pune Police. A detailed survey shall be conducted, by the SI along with a team of Pune Police. SI shall prepare the detailed report for field level requirements e.g. Cameras (types & numbers), Camera Mounting requirements, Power Requirements, Connectivity Requirements etc. for perusal of Purchaser. Indicative list of the field level hardware to be provided by SI is as follows:

- Cameras (Fixed Box Cameras, PTZ Cameras, ANPR cameras etc.)
- Local processing unit for ANPR cameras (if required)
- Switches
- Junction box Cabinets
- Pole along with cantilever or gantry for cameras
- UPS Requirement
- Networking and power cables and other related infrastructure

The indicative list of locations for the camera installation is mentioned in the **Annexure III** along with minimum technical requirements of associated hardware (Number of CCTV for that location) to implement a complete Surveillance system.

Based on detailed field survey as mentioned above, SI shall be required to supply, install and commission the surveillance system at the identified locations and thereafter undertake necessary work towards its testing. SI shall use industry-leading practices during the implementation phase w.r.t positioning and mounting the cameras, poles and junction boxes. Some of the check-points that need to be adhered to by the SI while installing / commissioning cameras are as follows:

a) Ensure surveillance objective is met while positioning the camera such that the required field of view is being captured as finalized in field survey
b) Ensure camera is protected from the on-field challenges of weather, physical damage and theft.
c) Make proper adjustments to have the best possible image / video captured.
d) Ensure that the pole is well placed for vibration resistance adhering to the road safety norms.
e) Collusion preventive barriers around the junction box & pole foundation in case it's installed in collision prone place.
f) Appropriate branding or Colour coding of poles and junction boxes, to warn mischief mongers against tampering with the equipment at the junction.
g) SI shall use the industry leading practices while positioning and mounting the cameras and ensure that the pole / mast implementation is vibration resistant. Necessary arrangements for bird scare spikes on top of camera to be also thought of to prevent birds from sitting on top of camera box.
h) The poles shall be installed with base plate, pole door, pole distributor block and cover.
i) Base frames and screws shall be delivered along with poles and installed by the SI.
j) In case the cameras need to be installed beside or above the signal heads, suitable stainless-steel extensions for poles need to be supplied and installed by the SI so that there is clear line of sight.
k) SI shall be responsible to undertake required structural analysis regarding the regulated load conditions and considering the respective wind load while installing the poles / cantilevers for cameras and other components
l) SI shall also be responsible for taking approvals of pole design from Government authorized structural engineer in design phase and should submit structural stability safety certificate of all poles by Government authorized structural engineer after installation of poles.

m) SI shall provide structural calculations and drawings for the approval of Purchaser. The design shall match with common design standards as applicable under the jurisdiction of purchaser/authorized entity.
n) SI shall coordinate with concerned authorities / municipalities for RI / RoW permissions and installation
o) Poles and cabinet shall be so designed that all elements of the field equipment could be easily installed and removed.
p) SI shall ensure that physical look of the installation area returns to neat & tidy conditions after installation of poles, cantilevers etc. The placement shall be designed keeping in mind the normal flow of vehicular traffic and pedestrian movement is not disturbed.
q) UPS shall serve as a backup for commercially available utility power at the intersections and shall ensure no-break functioning of all field components at each intersection in event of failure of utility power supply.
r) SI shall carry out a study and identify locations to provide UPS backup, depending upon power situation across city, to meet the camera uptime requirements.
s) SI shall install UPS at the identified intersections in secure, tamper-proof housing in corrosion resistant cabinets. SI would be needed to have minimum 60-minute backup for cameras.
t) SI shall ensure that the UPS is suitably protected against storms, power surges and lightning.
u) SI shall provide UPS & Other Edge Level equipment for efficient heat dissipation without air conditioning. It shall be able to withstand temperatures prevalent in Pune throughout the year.

2.9 Video Analytics

Surveillance system shall have the capability to deploy intelligent AI based video analytics software on any of the selected cameras. This software shall have the capability to provide various alarms & triggers. The software shall essentially evolve to automate the suspect activity capture and escalation; eliminate the need of human observation of video on a 24x7 basis. The system shall be futuristic and keep improvising its accuracy on the existing behavior of diverse objects and allow it to add more behaviors in the upcoming versions. The proposed backend analytics servers should process necessary GPUs and necessary AI capability to provide the mentioned basic and advance analytics feature. So that probability of raising false alarms is reduced. Video Analytics System/Software should have capability to perform all the Video Analytics (Basis & Advanced) on recorded video feed (feed of Minimum 2 MP Camera). The SI along with the respective OEM has to take entire responsibility of the analytics outcomes as envisaged in this RFP without any issues (like bad image quality, bad video quality, camera focus, environmental disturbance etc.)

Analytics software shall bring significant benefit to review the incidences and look for suspicious activity in both live video feeds and recorded footages. Certain use cases shall be available for online feeds, some would be applicable on offline feeds and some on both feeds.
Bidder must ensure the continuous progress in reduction of false alerts over the period of time. The same must be monitored and will be linked with the SLA. Bidder is expected to provide video analytics-based solution using AI & deep learning and shall ensure that the same is continuously trained (for the contract duration) to increase its accuracy over the time. For analytics-based solution, bidder shall be given a time of two months to train the model and gain the said accuracy of the first year.

The successful bidder is also expected to train their models for other use cases which may come up during the project execution using the raw data generated with the project. These use cases shall not be a part of the SLA.

All the alerts that are raised through Video Analytics or other systems will need to be handled by Command Centre:

i. Rule engine module for event/alarm handling
ii. SOP (Standard Operating Procedure) tool for administrator to configure the SOP responses based on each alert.
iii. Integration with the Incident Management system will be available to the users to log in the incidents and the alerts, view the report from the module about the incidents etc.
iv. Alert processing such as Acknowledging the alert, emergency response, SOP for the alert.
v. Connecting the next steps as per the SOP like informing particular Police personnel \ Fire departments based on the incident etc.

2.9.1 Face Recognition System

Pune police is planning to implement FR system with DB size of min. 1,00,000 images (scalable up to 500000), FR licenses can be dynamically applied to identified cameras as and when required. Face Recognition System (FRS) shall be designed for identifying or verifying a person from various kinds of photo inputs from digital image file to video source. The system shall offer logical algorithms and user-friendly, simple graphical user interface making it easy to perform the facial matching.

The Face Recognition System shall be able to work with any video streaming sensor like IP Camera (Bullet, Fix Box, PTZ,), Body Worn Cameras, Mobile handset cameras etc. as required and shall be able to identify multiple subjects of interest in real-time, through leading face & person recognition technology based on AI & Deep Neural networks and provide an alert when subject of interest in match is found. Primary FR will be configured on fix box camera & PTZ camera. However, the SLA defined for facial recognition (True alerts/Total alerts) will be applicable only for defined Fix box camera, Bullet camera and PTZ camera. FR System/Software should have capability to perform all the functionality on recorded video feed (feed of Minimum 2 MP Camera) of existing project including but not limited to
any other feeds from any other projects.
The software shall essentially evolve to automate the suspect activity capture, other analytics as requested in the RFP and escalation; eliminate the need of human observation. The SI along with the respective OEM has to take entire responsibility of the analytics outcomes as envisaged in this RFP without any issues (like bad image quality, bad video quality, camera focus, environmental disturbance etc.)

The system shall be able to broadly match a suspect/criminal photograph with database created using photograph images available with Scanned photos of criminals / wanted and missing persons, Passport, CCTNS, Prisons, State or National Automated Fingerprint Identification System or any other image database available with police/other entity.

The successful bidder is also expected to train their models for other use cases which may come up during the project execution using the raw data generated with the project. These use cases shall not be a part of the SLA. Certain use cases shall be available for online feeds, some would be applicable on offline feeds and some on both feeds.

The system shall be able to:

1. Face Recognition System shall work on real time and offline mode.
2. The Face Recognition system shall work for matching and identifying criminals:
   a. Videos with images
   b. Images with Videos
   c. Images with Images
3. The system shall have the best-suited technology employed for 1:1 (one to one) and 1: N (one to many) matching application for various purposes for non- voluntary face detection & recognition in the wild/crowd scenarios
4. The Facial recognition system should have capability to integrate to the 3rd party applications such as Big Data frameworks, Open-source platforms, social media platforms such as Twitter, Facebook, Instagram, YouTube, WhatsApp etc. to identify the people.
5. The OEM should have partnership with the offered GPU or graphics acceleration platform to best use the technology. The OEM partnership citation should be submitted along with the technical bid
6. The system shall be able to flag or bookmark an incident for future investigations. Such bookmarked events must override the FIFO cycle for future investigations.
7. The system shall have the provision to take multiple samples of same face belonging to same person and simultaneously detect a minimum of 20 faces in any given frame in real time
8. The facial recognition system (FRS) targeted by this specification must be able to identify multiple persons of interest in real-time or recorded video files, through leading-edge face recognition technology from different video sources like IP cameras, Drones, etc. The solution shall be able to perform in a crowd, on the move, and at a distance, with enrolment on the fly.
9. The FRS system shall be enterprise-grade highly scalable providing the most accurate results for the most demanding real-time and post-event offline mode.
10. The FRS system shall make use of the latest technologies such as AI & Deep learning with a series of latest multiple CPU or GPUs from respective OEMs and not discontinued products.
11. The FRS system shall make use of the stream computing paradigm of the latest GPUs to the limit by performing multiple stages of the face recognition pipeline in parallel. It must efficiently exploit cache

hierarchies of both GPUs and CPUs to greatly improve the computational performance in crowded environments.

12. The FRS system shall have a web-based centralized management module to see all the system logs, the status of all processing servers, search of any face from logs of all connected servers and also managed SYNC of all detections to main DB and vice versa any new face addition in DB to all connected servers.
13. The FRS system shall ensure data privacy and security by proper encryption technologies such as AES-256 standard or better.
14. All External communications between the FRS system and third-party applications support SSL encryption with username and password.
15. While registering a face in the database, the system should also give details of the similar face already present in the database to help in de-duplication of records.
16. The FRS system shall be able to detect multiple faces (from multiple frames) & compare a face against a database with the comparison speed of 10 million within a second.
17. The FRS algorithm of the system shall be able to recognize partial faces with varying angles (up to face rotations of ± 45 ° horizontal and ± 30 ° vertical) & any facial accessories/ hair (mask, glasses, beard, and expressions). The System shall be able to work in the following conditions:
    - Changes in facial expression
    - Changes in facial hair or hairstyle
    - Changes by moderate aging (up to 15 years)
    - Partially hidden faces or occluded faces like fearing dark glasses etc.
    - Changes in lighting conditions
18. The system shall be able for matching suspect faces from pre-recorded video feeds from CCTV deployed at various identified locations.
19. The FRS system shall be able to broadly match a suspect/criminal photograph with a database created using a photograph images database available with different government authorities & local police. The initial database is required to register 1,00,000 subjects.
20. While registering a face in the database, the system should also give details of the similar face already present in the database to help in de-duplication of records.
21. The FRS system shall have an offline inbuilt module which makes it possible to search for a specific subject in one or various recorded videos, existing database records & from previous detections logs of all the cameras at speeds faster than real-time. Analysis of offline videos can be performed faster than a real-time stream. Also, the system shall provide options to give the accuracy threshold parameters for the user and the system should give top five matches for the searched face from the database.
22. The FRS system shall support diverse graphic & video formats as well as live cameras. FRS shall support day/night operation with the ability to detect faces both in colour and in black/white mode by using any H.264. H.265 Fixed IP and PTZ Cameras with IR Illuminators without any special configurations required
23. The FRS system shall able to handle at least a minimum of 20 FRS cameras with atleast 15 frames per server.
24. The FRS system shall have an inbuilt or external tool to synthetically generate rotated views of the face from a single partial image of the suspect face.
25. The FRS system shall how the top 05 nearest matches with a matching percentage score of the captured image.
26. The FRS system shall have the option to enable automatically enroll face images from live / recorded CCTV feeds & also able to check if a new enrolled face is already enrolled in the database before registering the new face in the system.
27. The FRS system shall have an independent module for bulk registration of subjects automatically through folders or files.
28. The FRS system shall be able to utilize any of the file formats like JPEG, PNG, BMP, TIFF etc. format for enrolment for matching.
29. The FRS system shall have the capacity to create different categories of subjects with the option to customize the matching threshold for different categories & cameras.
30. The system should have the functionality to detect the face / person who is not authorised to enter the specific area covered by the FR camera.
31. The system shall be able to work on full HD or better Camera video with maximum performance. The system must perform a full N: N search of the probe image in Max 1 second against a database of up to 25,000 records. (capable to expand up to 1 lacs in future)
32. Capture face images from CCTV feed and generate alerts and pass the alerts to the command control application if a blacklist match is found.
33. Search photographs from the database matching suspect features.

34. Add photographs obtained from newspapers, raids, sent by people, sketches etc. to the criminal's repository tagged for sex, age, etc. for future searches.
35. The system should also allow an attribute search based on criminal profile/criminal persona creation. The feature should consist of the following
    a. Creating criminal profile/criminal persona based on Indian Attributes
    b. criminal profile/criminal persona shall match personality profile of target
    c. criminal profile/criminal persona shall be able to choose from male / female
    d. criminal profile/criminal persona feature shall have ability to choose different kind of attires based on Indian costumes and wearables
    e. criminal profile/criminal persona feature shall have the ability to differentiate between top and bottom wears
36. Investigate to check the identity of individuals upon receiving such requests from Police Stations.
37. Enable Handheld mobile with app to capture a face on the field and get the matching result from the backend server, also provide API's for integrating this app with existing / upcoming mobile application
38. The system shall be able to work on windows or Linux operating system
39. The system shall support database system like MS SQL or Oracle or any other data base system.
40. System shall able to integrate with camera either by RTSP stream directly or shall able to integrate through VMS provided by SI, preference will be given to tight integration through VMS.
41. OEM of facial recognition system shall certify the number of cameras that can run at a given point of time on a proposed physical server/virtual machine proposed by the bidder, on peak load and with maximum performance.
42. The facial recognition system shall be enabled at cameras identified by the Police. Police / Intelligence Agencies shall dynamically identify these cameras, as and when required.
43. The Facial recognition software must send an alert to VMS which shall highlight the camera location on the city map (uploaded in VMS) for each detection (Whenever required).
44. The Facial recognition engine shall process the matching facial profiles at the backend Facial recognition system and only alerts/results should be display on mobile devices. The Facial recognition system shall be highly optimised to be able to support and run on feeds received from mobile handset/body worn cameras, and, if required, send the alerts to field teams on match.
45. The FRS system shall support a user management module that enables different user level groups & permission levels to them.
46. The user interface of the system shall be user-friendly & have a report management tool without the installation of any additional client software. It shall be able to generate real-time different reports in different file formats.
47. The FRS system shall be able to connect multiple local systems permitting centralized management, synchronization of multiple databases & Load balancing.
48. The FRS system should have a flexible deployment architecture such as centralised, distributed and cloud deployment or a mix of all the architectures as per the user requirements
49. The FRS system shall allow the regular backup of its database files, so the customer can rebuild the system quickly in any system failure situations.
50. The FRS system shall have ready to serve a mobile app for Alarm monitoring and cross-check the identity of any suspect from the field. The mobile app should be provided for at least 20 android-based devices.
51. The FRS system shall support to push identification alarms on email. The email format shall include subject ID, captured & matched face, frame image, date and time of the match, a score of the match, watchlist name & camera details etc.
52. The Face Recognition Algorithm vendor should appear in top 15 Vendors of latest FRVT 1:N report.
53. The FRS system shall provide an SDK/API for integration with any third-party software like C4I Command & Control Centre. API must be available with a full set of documentation of each method with accompanying sample code. All-important FRS functions shall be fully accessible via API.
54. The system should be highly secure and should have been tested for vulnerabilities and should have been penetration tested by the CERT-IN empanelled auditors.

2.9.2 Video Summarization

GoM/ Pune Police is planning to procure about 30 floating licenses of Video Summarization System, these licenses can be dynamically applied to identified cameras as and when required and 2 standalone hardware licenses for at least 8 cameras simultaneously. The Video Synopsis analytics system shall be based on computer vision and AI (Artificial Intelligence) technology enabling rapid video review, search, quantitative video insights and smart

alerting, thereby shortening time-to-target to detect and mitigate security threats and enhancing safety and operational optimization significantly. System shall enable generation of Summarization of video sourced from both recorded offline-video files and online VMS platforms with full case management, multi-camera search, appearance similarity and face recognition (optional).

The system shall be able to:

- The Video Summarization System (VSS) shall be a truly open platform and can be interfaced with any existing or new ONVIF complied VMS system selected by the customer.
- The Video Summarization System (VSS) to make available live & recorded video surveillance feeds searchable, quantifiable, and actionable. The VSS shall be based on computer vision and AI (Artificial Intelligence) technology enabling rapid video review, search, quantitative video insights, and smart alerting, thereby shortening time-to-target to detect and mitigate security threats and enhancing safety and operational optimization significantly. The VSS system shall enable the generation of usable metadata from recorded offline-video files and online VMS platforms with full case management, multi-camera search, appearance similarity.
- The VSS system shall facilitate leveraging of quantitative video analysis-derived intelligence for informed, data-driven decision-making, including advanced trend and dimensional (area, path, duration, and other) KPI analysis as well as full dashboarding and scheduling capabilities.
- The VSS system shall support different video file formats & codecs: AVI, MKV, MPEG4, MOV, WMV, DVR, ASF, RT4, DIVX, 264, GE5, TS, 3GP, DAV, XBA, MP4, FLV, H.264, H.265/HEVC, MPEG-4, and H.263/ codecs most commonly used and as per the project requirements
- The VSS system shall be able to generate live alerts thereby delivering a proactive response to critical user-defined events ensuring enhanced safety and security.
- The VSS system shall show the timestamp at which a specific object appeared in the video. This timestamp shall be visible on the image of that specific subject / object.
- The VSS system shall automatically extract all the moving objects (People/Vehicle/Animal) from the original video and without alerting the original image/video/data, simultaneously displaying events that have occurred at different times.
- The VSS system shall rapidly pinpoint people and vehicles of interest, using a range of appearance and movement filters, across video feeds from multiple cameras.
- The VSS system shall have the facility to create inclusion & exclusion zones on the recorded video. Only objects in selected zones shall be shown or removed.
- The VSS system shall be able to follow movement of people across multiple cameras selected by the operator in the archived video systematically and quickly, in a time synchronized fashion.
- The VSS system shall dynamically visualize Key performance (KPI) Indicators (Traffic Patterns, Visitor/Pedestrian count, general trends & Insights, behavior trending, object movement, demographic segmentation, and people interaction with objects, plus an out-of-the-box extensible library of vertical-specific analytics tools) to enable visual correlation with video metadata and their classification on customizable dashboards.
- **Indicative Object Search Criteria**
    - **Time Range** – Shall be able to limit the search criteria to specific time ranges
    - **Source** – Shall be able to limit the objects to specific CCTV Camera feed or offline pre-recorded multiple video files
    - **Classes** – Post-analysis, reviewed video shall be shown based on People, Two-Wheeled Vehicles, Other Vehicles with the following categories:
        - People Class: Man, Woman, child etc.
        - 2-Wheeled Vehicle Class: Bicycle, Motorcycle etc.
        - Other Vehicles Class: Car, Pickup, Van, Truck, Bus, auto-rikshaw, three-wheeler pickup trucks, ambulance, etc.
        - Capturing license plates of Vehicles appeared in the video feeds.
    - **People Attributes** – Shall be able to select the attributes within the people class to refine the search Like:
        - Count : Total count of people in a pre-defined range of view or area.
        - Bags: Backpacks, Handheld Bags
        - Hats: Hats, No Hats, Niqab, Pagree, caps
        - Upper Wear: Short/No Sleeves, Long Sleeves
        - Lower Wear: Long, Short

        - Colour - Identify objects according to any combination of various colors like Yellow, Brown, Red, Orange, Green, Lime, Grey, Black, Cyan, Purple, Pink and White.
        - Faces of appeared people in the input video feeds.
    - **Size** - Select objects based on their actual (real-life) size specified in meters.
    - **Speed** - Select objects based on their actual speed specified in KMPH
    - **Dwell** - Select objects dwelling for longer than a certain period in a scene
    - **Area** - Identify objects included or excluded within one or more user-defined areas.
    - **Path** - Identify objects traveling along one or more user-defined paths"
- The VSS system shall be customizable to meet the specific needs and operational processes of a given user environment and to support unique requirements specific to forensic search and alert response.
- The VSS system shall allow the user to bookmark any event clip for ready reference at any later point of time also it shall allow the user to tag critical Event clips so that they do not get removed from the storage based on retention period settings.
- The VSS system shall be able to link together in a chain of separate video clips into a single video. All the video data shall be stored in a database for easy and quick review by operators.
- The VSS system shall be also capable of generating Near-Real-Time alerts based on any of the above-mentioned similar searches and filter criteria & shall able to trace the subject/object along with its path on the map.
- The VSS system shall be able to identify the CROWDING HOTSPOTS AND TRENDS by utilizing dashboards to visually represent object movement, behavior trending, hotspots, and object interactions.
- The VSS system shall also able to pinpoint people of interest using digital images extracted from video or external sources, either by a specific person image selected or per watchlist.
- The VSS system shall be able to results dwell time, changes to scene background; and path lines for object travel, apply visual layers and heat maps to identify the movement of objects/subjects depending upon the global filters subject to the situation.
- The VSS system can be deployed on physical or virtual system architectures.
- The VSS system shall have a centralized administration module to view and activate cameras, configure hosts and any other related services.
- The VSS system shall have mobile application, which runs on both Android and iOS devices which derives exponential value from surveillance camera investments by making video searchable on-the-go. It shall have on the move Access to Processing servers to achieve unmatched accuracy and seamless extensibility .
- The VSS system client module shall be accessible and able to work in a web browser such as Google Chrome and Firefox or through a client software interface etc.
- The VSS system shall be capable of enabling end-users to log in just once and access all needed applications without having to log in separately to each one.
- The VSS system shall also be interfaced for alerts into third-party alerting infrastructures such as APP or QRT systems.
- The VSS system shall provide an OPEN API to interface any third party systems without any additional cost to the customer.
- The system should be highly secure and should have been tested for vulnerabilities and penetration tested by the CERT-IN empanelled auditors.

The solution is recommended to enable simultaneous digital video recording from network, intelligent video analysis and remote access to live and recorded images from any networked computer. It shall be able to automatically track and classify objects such as cars and people and push content to the respective security personnel as required for real time analysis. The system shall also have display of timeline, customizable site map, live video, video playback, integrated site map, remote live view, multi-site capability, encryption, watermarking and event-based recording.
All cameras shall support motion detection, camera tampering and audio analytics. Solution shall also have Automated PTZ camera control for zooming in on interesting events like motion detection etc. as picked up by camera without the need of human intervention. It shall be completely scalable, with a many-to-many client-server model allowing multiple physical systems to be used in an array of servers.

2.9.3 Advanced Analytics

- The Behavioral understanding systems shall provide an automatic real-time understanding of video content captured by widespread surveillance cameras located throughout traffic control centers, roads, public areas, buildings, commercial and industrial zones while providing insights and alerts for a variety of actions, events, and scenes of interest.

- The system should be able to analyse the videos in real time as well as offline and on-demand videos for forensic analysis purpose.
- The system shall be an AI/deep learning based Behavioral understanding systems for Real-Time video intelligence in the most dynamic and challenging urban environment on a 24x7 basis.
- The system shall apply holistic and temporal analysis for scene participants through a top-down approach (scene -> event(s) -> action(s) -> object(s) -> attribute(s) by using the time dimension to connect between the video's image sequence and the different modalities.
- The system shall detect objects, actions, events, and scenes where the size of the participating objects is at least 10% of the frame. The objects and behaviors should not be obscured.
- The system shall be capable to get video streams, analyzes them and produces alerts for events of interest to the user.
- The system shall allow applying multiple or all behaviors on a camera or a set of behaviors on the camera group using a simple user interface.
- The system shall be a True open platform and should be easy to integrate using OpenAPI standards and capable to get video streams, analyzes them and produces alerts for events of interest to the user. The system should not be OS dependent in order to ensure the system is not compromised when the operating system is compromised
- The system shall provide charts that will be useful in measuring the system's throughput and health.
- The system shall detect any pre-defined behaviors within 03-05 seconds in front of the camera.
- The system shall be automated and does not require typical rule or filter drawings to detect any interest in alarm.
- The system shall offer insights and alerts for a variety of actions, events, and scenes of interest, such as detection of any Violent activities, suspicious activities, unusual grouping activities, environment activities, etc.
- The system shall offer at least the following type of alerts recognition under the above-mentioned respective activity group:
    - **Violence & Violent Behavior Activity:**
        - The human fighting.
        - The human holding a weapon.
        - human instream tracking.
    - **Suspicious Behavior Activity:**
        - The human lies or falls on the ground.
        - The human abandons an object.
    - **Grouping Behavior Activity:**
        - The Human or Humans group running.
        - Humans Gathering in a group.
        - Riot/Brawl
    - **Vehicle Behavior Activity:**
        - Vehicle Collison (such as Car, Bus & Truck).
        - Vehicle park on sidewalks or at no-parking areas.
        - A car in a non-parking or stopping areas.
    - **Vehicle Collison (such as Car, Bus & Truck).**
        - Vehicle park on sidewalks or at no-parking areas.
        - A person gets out of a car on non-parking or stopping areas.
    - **Environment Activities**
        - Any kind of fire or smoke detection on the streets.
    - **Other Features**
        - People not wearing masks.
        - People not keeping social distance
        - Encroachment of footpaths
        - **No helmet detection - System should be able to identify two-wheeler vehicles along with their vehicle registration number plate with riders and pillion not wearing a helmet.
        - **Wrong way driving direction - System should be able to identify vehicles (2W, 3W and 4W) which are going against the prescribed flow of traffic for particular roads with their vehicle registration plate
        - **Triple seat riding on 2 wheelers - System should be able to identify two wheelers with more than two persons riding on the vehicle with their vehicle registration plate
        - **Improper/ Illegal parking - System should be able to detect vehicles (2W, 3W and 4W) parked at prescribed 'No Parking' zones.

- **Vehicle overspeeding detection with spot speed and configurable average speed for the identified patch of the road.
- **The system should capture vehicles with driver not wearing seatbelts.

- The system shall be seamlessly integrated with the proposed VMS and showcase the generated metadata on the live & playback video stream.
- The system must have a consolidated alert dashboard where each alarm type can be accompanied by a video snippet in a form of different color codes.
- The system shall be able to handle up to 30 streams on a single GPU enabled server machine.
- The system shall also retain the alarms and the detections for a period of time set in the system setting, subject to storage capacity. Minimum period of 30 days
- The system shall be based on Deep Learning technology and should have the provision to train the datasets to improve its accuracy on the existing behavior of diverse objects and allow it to add more behaviors as per the user requirements
- The system should have integration APIs for sending the alerts to the ICCC / C4i platforms.
- The system should be highly secure and should have been tested for vulnerabilities and should have been penetration tested by the CERT-IN empaneled auditors.

The solution is recommended to enable simultaneous digital video recording from network, intelligent video analysis and remote access to live and recorded images from any networked computer. It shall be able to automatically track and classify objects such as cars and people and push content to the respective security personnel as required for real time analysis. The system shall also have display of timeline, customizable site map, live video, video playback, integrated site map, remote live view, multi-site capability, encryption, watermarking and event-based recording.
All cameras shall support motion detection, camera tampering and audio analytics. Solution shall also have Automated PTZ camera control for zooming in on interesting events like motion detection etc. as picked up by camera without the need of human intervention. It shall be completely scalable, with a many-to-many client-server model allowing multiple physical systems to be used in an array of servers.

**All the alerts generated for above use cases should be flagged along with multiple images of violation. These images should be saved in .jpg or .png format (or format compatible with eChallan system) which can be passed onto the eChallan system for generating echallans.

2.9.4 Automatic Number Plate Recognition

- SI shall provide Automatic Number Plate Recognition (ANPR) solution at the identified locations. SI shall describe in detail, the design, operational and physical requirements of the proposed ANPR system, to demonstrate compliance with all the specified requirements in this RFP.
- The ANPR server shall process the image using OCR software for getting the registration number of the vehicle with highest possible accuracy. The system shall be able to detect, normalize and enhance the image of the number plate for detection of alpha numerical characters. System shall be able to identify stolen/ suspected vehicles by cross checking the numbers with vehicle database. ANPR software shall be integrated with video management system.
- The Accuracy of ANPR system should be more than 80% in case of standard English Alphanumeric Font and High Security Registration plates, the system should be able to capture the Registration Plate of a vehicle moving at a speed of up to 80 Km/hour with the same level of accuracy.
- On successful recognition of the number plate, ANPR system should be able to generate automatic alarm to alert the control room for vehicles which have been marked as "Wanted", "Suspicious", "Stolen", "Expired". (System should have provision/expansion option to add more categories for future need).
- The ANPR system should be able to plot the camera locations on to the city map in case of any stolen/suspected/wanted/Hot listed vehicle has been identified at any or multi camera locations, such that the pictorial view of the data on the map gives a meaningful insight to the alert generated and the monitoring of the same can be done in a better way. Vehicle of interest should be triggered.
- The ANPR system shall provide a user interface with live view of vehicle entry point 24x7, event notification, image captured, number detection and recognition, event reports customized report generation etc.
- The ANPR system must generate post analysis report with Full or partial plate number, Capturer Type, Zone, Camera, Capturer ID, Make, Model, Color, Sticker, any identification mark ,Date from - Date to, Time

from - Time to, Status, Direction, Minimum Latitude, Maximum Latitude, Minimum Longitude, Maximum Longitude

- The post-event forensic search shall be able to search vehicle by number (full or partial), by date and time, by location or zone, by alarm, by the camera, by the speed etc.
- The ANPR system on its city dashboard able to show the following daily/weekly statistics: Number of vehicles on the road at a given time, day wise alarms of blacklisted vehicles etc.
- The system should provide partial number plate search and vehicle search based on various parameters such as vehicle speed, vehicle classification (at least bus, heavy vehicle, car, bike and auto-rikshaw) and vehicle color (to be captured during the day time).
- The analysis of the image captured shall be done in real time. The database so created from the images captured & analysis shall store the following:
    1. Details of vehicle
    2. Number and time of entries and exits
    3. License plate numbers
    4. Validation/Analysis results etc.

2.9.5 IP based Public Address Speakers will be needed for Crowd Management and Incident Management

Coverage

Public Addressing Speakers will need to be deployed across the city at different type of locations as required by the Pune City Police.

List of locations which should be installed with Public Address speakers is mentioned in the Annexure III. There are total 1462 (455+1007 location which includes existing & proposed locations. The 500 IP based PA speakers would be installed in these locations. The final location and count of PA speaker per location would be finalized during AS-IS / Requirement Documentation Phase in discussion with Pune police.

Functional Requirements

- The IP based Public Address Speakers (PA) should be capable of addressing citizens at specific locations from the Command & control center.
- The proposed system shall contain an IP-based announcing control connected to the Command and control Center.
- Public Address speakers shall be used at intersections, public places, marketplaces or those critical locations as identified by Authority to make important announcements for the public. It shall be able to broadcast messages across all PA speakers or specific announcement could be made to a location supporting single zone / multi zone operations. The system shall also deliver pre-recorded messages to the loudspeakers attached to them from CD/DVD Players & Pen drives for public announcements.
- The system can contain an IP-based amplifier and uses PoE power or IP-based horn directly powered by PoE that could drive the speakers. The system shall also contain the control software that could be used to control/monitor all the components of the system that includes Controller, Calling Station & keypad, Amplifier (Mixing & Booster).
- PA system's master controller should have function keys for selecting the single location, group of locations or all locations, simple operation on broadcasting to any terminal or separated zones.
- PA system's master controller should facilitate multiple MIC inputs and audio inputs.
- Public addressing system outdoor speakers shall be mounted on the same Pole where cameras are installed at the height of 15 Feet with proper mounting kit.
- Installed PA speakers shall be discoverable in the tree of VMS or command control center application.
- Police/Authorized personnel shall be able to use the system for public announcement on the ground with the help of ECB installed on the same pole (Optional).

3. Annexures I - Technical Specification

3.1 Fixed Cameras

#	Parameter	Minimum Specifications or better
1.	Video Compression	H.265 or better
2.	Video Resolution	FULL HD 1920 X 1080
3.	Frame rate support[1]	25/30 FPS @ 1080P
4.	Image Sensor	1/3" or ½.8" Progressive Scan CMOS
5.	Lens Type	Manual Vari-Focal (Autofocus), IR Correction
6.	Lens *	Auto IRIS, 5 – 50 mm (+- 1mm), F1.2/F1.4
7.	IR	IR with minimum 50 mtr Range
8.	Minimum Illumination	Colour: 0.1 lux, B/W: 0.01 lux (at 50 IRE), IR: 0 lux
9.	IR Cut Filter	Automatically Removable IR-cut filter
10.	Day/Night Mode	Colour, Mono, Auto
11.	Multiple Streams	Triple streaming with individually configurable streams
12.	Auto adjustment + Remote Control of Image settings	Colour, brightness, sharpness, contrast, white balance, exposure control, backlight compensation, Gain Control, Remote Focus, Remote Zoom
13.	Wide Dynamic Range	True WDR >= 120 dB
14.	Audio	Audio Capture Capability (Optional)
15.	Local storage	Micro SD card up to 64 GB. In the event of failure of connectivity to the central server the camera shall record video locally on the SD card automatically. After the connectivity is restored these recordings shall be automatically merged with the recording server such that no manual intervention is required to transfer the SD card-based recordings to recording server (1 Nos. of 64 GB SD card to be provided with each camera)
16.	Network Interface	RJ45 10 Base-T/100 Base-TX, POE
17.	Alarm I/O	Minimum 1 Input & 1 Output contact for 3rd party interface
18.	Protocol	IPV4, IPV6, HTTP, HTTPS, FTP, RTSP, RTP, TCP, UDP, RTCP, DHCP, UPnP, QoS, ONVIF profile S & G
19.	Onboard Analytics	Motion Detection & Tampering alert
20.	Security	Password Protection, IP Address filtering, User Access Log, HTTPS encryption, TSL/SSL. Camera should not be having GB 28181, GBT 28181, GB/T 28181 standard
21.	Power	POE /POE+ to follow IEEE 802.3 af /at
22.	Operating conditions	0 to 50°C (temperature), 50 to 90% (humidity)
23.	Casing	NEMA 4X/IP-67 rated, IK 10
24.	Certification	UL/EN/CE, FCC, BIS

[1] ANPR Cameras shall record videos up to 30 FPS

3.2 Fixed Cameras for FR

#	Parameter	Minimum Specifications or better
1.	Video Compression	H.265 or better
2.	Video Resolution	4MP or higher
3.	Frame rate support[2]	25/30 FPS @ 1080P
4.	Image Sensor	1/3" or ½.8" Progressive Scan CMOS
5.	Lens Type	Manual Vari-Focal (Autofocus), IR Correction
6.	Lens *	Auto IRIS, 5 – 50 mm (+- 1mm), F1.2/F1.4
7.	IR	IR with minimum 50 mtr Range
8.	Minimum Illumination	Colour: 0.1 lux, B/W: 0.01 lux (at 50 IRE), IR: 0 lux
9.	IR Cut Filter	Automatically Removable IR-cut filter
10.	Day/Night Mode	Colour, Mono, Auto
11.	Multiple Streams	Triple streaming individually configurable
12.	Auto adjustment + Remote Control of Image settings	Colour, brightness, sharpness, contrast, white balance, exposure control, backlight compensation, Gain Control, Remote Focus, Remote Zoom
13.	Wide Dynamic Range	True WDR >= 120 dB
14.	Audio	Audio Capture Capability (Optional)
15.	Local storage	Micro SD card up to 64 GB. In the event of failure of connectivity to the central server the camera shall record video locally on the SD card automatically. After the connectivity is restored these recordings shall be automatically merged with the recording server such that no manual intervention is required to transfer the SD card-based recordings to recording server (1 Nos. of 64 GB SD card to be provided with each camera)
16.	Network Interface	RJ45 10 Base-T/100 Base-TX, POE
17.	Alarm I/O	Minimum 1 Input & 1 Output contact for 3rd party interface
18.	Protocol	IPV4, IPV6, HTTP, HTTPS, FTP, RTSP, RTP, TCP, UDP, RTCP, DHCP, UPnP, QoS, ONVIF profile S & G
19.	Onboard Analytics	Motion Detection & Tampering alert
20.	Security	Password Protection, IP Address filtering, User Access Log, HTTPS encryption, TSL/SSL. Camera should not be having GB 28181, GBT 28181, GB/T 28181 standard
21.	Power	POE /POE+ to follow IEEE 802.3 af /at
22.	Operating conditions	0 to 50°C (temperature), 50 to 90% (humidity)
23.	Casing	NEMA 4X/IP-67 rated, IK 10
24.	Certification	UL/EN/CE, FCC, BIS

[2] ANPR Cameras shall record videos up to 30 FPS

3.3 PTZ Camera

#	Parameters	Minimum Specifications or better
1.	Video Compression	H.265 or better
2.	Video Resolution	FULL HD 1920 X 1080
3.	Frame rate	Up to 25/30 fps @ 1080p
4.	Image Sensor	1/3" or ½.8" Progressive Scan CMOS
5.	Lens	Auto-focus, 4.3 – 129 mm (+/- 2 mm) (corresponding to 30 X or better)
6.	Multiple Streams	Support Multiple streams simultaneously & each stream should be individually configurable.
7.	Minimum Illumination	Colour: 0.2 lux, B/W: 0.01 lux (at 30 IRE, F 1.6) or better
8.	IR Illuminator	In built up to 200 meters
9.	Day/Night Mode	Colour, Mono, Auto
10.	Wide Dynamic Range	WDR >= 120 dB
11.	PTZ	Pan: 360° endless/continuous, up to 100°/s (auto), up to 100°/s (Manual) Tilt: 90°, up to 100°/s (Auto), up to 40°/s (Manual) 30 optical zoom and 10x digital zoom Pre-set tour 100 preset positions, Guard tour
12.	Auto adjustment + Remote Control of Image settings	Colour, brightness, sharpness, contrast, white balance, exposure control, backlight compensation, Gain Control, Electronic Image Stabilization
13.	Protocol	HTTP, HTTPS, RTSP, RTP, TCP, UDP, RTCP, DHCP, ONVIF Profile S & G
14.	Security	Password Protection, IP Address filtering, User Access Log, HTTPS encryption, TSL/SSL, Camera should not be having GB 28181, GBT 28181, GB/T 28181 standard
15.	Local Storage	Micro SD card up to 64 GB. In the event of failure of connectivity to the central server the camera shall record video locally on the SD card automatically. After the connectivity is restored these recordings shall be automatically merged with the recording server such that no manual intervention must transfer the SD card-based recordings to recording server. (1 Nos. of 64 GB SD card to be provided with each camera)
16.	Onboard Analytics	Motion Detection & Tampering alert, Out of focus
17.	Audio	Audio Capture Capability (Optional)
18.	Network Interface	RJ45 10 Base-T/100 Base-TX, POE
19.	Alarm I/O	Minimum 1 Input & 1 Output contact for 3rd party interface
20.	Operating conditions	0° to 50°C
21.	Casing	NEMA 4X/IP-67 rated & IK10
22.	Power	802.3at PoE+ (Class 4) or 24VDC/24AC
23.	Certification	UL/EN/CE, FCC, BIS

3.4 External IR Illuminator (If required)

#	Parameters	Minimum Specifications or better
1.	Range	Min. 100 mars.
2.	Minimum Illumination	High sensitivity at Zero Lux
3.	Angle of illumination	Adjustable
4.	Power	Automatic on/off operation
5.	Casing	NEMA 4X / IP-66 rated
6.	Operating conditions	0° to 50°C
7.	Certification	UL / CE / FCC / EN

3.5 ANPR Cameras

The basic specifications of the cameras would remain similar to the Fix Box Camera specifications given in this RFP. SI would need to ensure the placement of these cameras at appropriate height to ensure the optimum results. Fixed boxed camera used for ANPR shall record videos up to 30 FPS if required.

3.6 Drone with Public Address (PA) System

Specifications:

1. **Vehicle Specifications:**
   - Endurance: >60 minutes at MSL (Mean Sea Level)
   - Range (LoS): >5 km
   - Weight: <6 kg
   - Wind Resistance: 36 km/h (~20 knots)
   - Ingress Protection: IP53 Rated for Dust & Water Resistance
   - Operating Altitude: 500 m AGL (Above Ground Level)
   - Max Launch Altitude: 3000 m AMSL (Above Mean Sea Level)
   - Vehicle Size: <1.5 m x 1.5 m
   - Live HD Video: Remote Monitoring with Live HD video streaming over

2. **Compatible Payloads/Camera Options**
   - HD Daylight Payload: 1280 x 720 pixels, 10x Optical Zoom
   - Thermal Payload (Black/White hot): 320 x 240, 640 x 480 pixels
   - Thermal Payload (Radiometric): 640 x 480 pixels
   - Photogrammetry Payload

3. **Fail Safe Features**
   - Multiple GPS
   - Return home on low battery
   - Return home on high winds
   - Return home on communication failure

4. **Ground Control Station (GCS)**
   - Laptop/Tablet
   - Handheld controller for pan, tilt, zoom
   - High range Communication Box
   - Encrypted uplink & downlink

5. **GCS Application Software**
   - Pre-flight self-checks
   - Simultaneous real-time map tracking &
   - Video streaming integration with Verint SMC at Command & Control Center
   - In-flight UAV health monitoring
   - Live flight parameters

6. **Megaphone Capability**

- Live Announcement, Pre-recorded Messages, Inbuilt Siren &
- Aux Input, Audible up to 75m from drone

3.7 Mobile Vans & Related Equipment

#	Parameters	Minimum Specifications or better
1.	Mobile Vans	• Cushioned seats for 5 personnel plus driver & co-driver • Siren unit and revolving light on body roof • 500W floodlight • Observation hatch • Mobility: High • Gradeability: 20° • Maximum Speed: at least 100 kmph • Ground Clearance: 175mm or better. • The vehicle should be pre-fabricated with fitments for all IT equipment and racks required.
2.	PTZ	• 1 No. as per specifications mentioned above
3.	Fixed	• 4 No. as per specifications mentioned above
4.	Mobile NVR	• H.265 and H.264 both support • Min 4-port POE Ethernet Port • Incoming throughput 100 Mbps • Recording throughput 100 Mbps • 3840x2160, 1920 x 1200, 1920 x 1080, 1680 x 1050, 1600 x 1200 • Manual, motion detection, timing and alarm • Supports 4 channels 1080P synchronous playback • 1ch, HDMI0 resolution 1920×1080P • 1× RJ45, 10/100/1000M self-adaptive Ethernet port • Supports TCP/IP, UDP, DHCP, DNS, NTP, DDNS, FTP, UPNP, multicast, etc. • MIL-STD-810G • Integration with proposed VMS solution
5.	Body worn Camera	• Full HD, Fixed non-rotatable lens. • Min. 5 hours of continuous recording. • 30 second pre-record buffering. • Min. IP65 rated. • Build-in GPS. • Min.64GB internal storage. • 4G/5G connectivity.
6.	Laptop -Rugged	• 14" screen or more and Full HD with sunlight-viewable • Processor: 11th Generation Intel Core i5 / AMD Ryzen 5 with dedicated GPU. • Memory: 16GB RAM expandable up to 32 GB • 1TB HDD. • Long battery life of up to 18 hours. • Windows 11 Pro • IP66 water and dust resistance, MIL-STD 810H certified.
7.	5G/4G Router	• 4 x 10/100 Ethernet ports: 1 x WAN,3 x LAN ports for local connectivity at vehicle • NAT, Static/Dynamic routing, Firewall, OpenVPN, IPsec, H.323 and SIP-alg protocol NAT helpers, allowing proper routing of video packets • Two level LTE cellular networks redundancy via Dual 4G/5G SIM based between SIM1 to SIM2 auto switchover.

		• Should have inbuilt GPS receiver for the RTC, the RTC clock should be made available over four Ethernet ports for external hardware use • IPv6 & IPv4 Management, UDP, TCP, IP, ICMP, SNMPv2c, HTTPs, STP,
8.	Invertor	• 1 KVA With 150 AH battery for 45 minutes backup

Note : Selected bidder to ensure that Mobile VANs must have provision of mounting V-SAT. If needed ,department may decide to mount VSAT at their level.

3.8 Video Management System

1. VMS shall be used for centralized management of all field camera devices, video servers and client users.
2. VMS shall be deployed in a clustered server environment or support inbuilt mechanism for high availability and failover. (No loss of data/recording during failure of directory/recording servers.)
3. VMS shall support a flexible rule-based system driven by schedules and events.
4. VMS shall be supported for fully distributed solution for monitoring and control function, designed for limitless multi-site and multiple server installations requiring 24/7 surveillance with support for devices from different vendors.
5. VMS shall support ONVIF compliant internet protocol (IP) cameras from different vendors.
6. The bidder shall clearly list in their proposal the make and models that can be integrated with the VMS, additionally all the offered VMS and cameras must have Open Network Video Interface Forum (ONVIF) compliance.
7. VMS shall be enabled for any standard storage technologies and video wall system integration.
8. VMS shall be capable of being deployed in a virtualized server environment without loss of any functionality.
9. All CCTV cameras locations shall be overlaid in graphical map in the VMS Graphical User Interface (GUI). The cameras selection for viewing shall be possible via clicking on the camera location on the graphical map. The graphical map shall be of high-resolution enabling operator to zoom-in for specific location while selecting a camera for viewing.
10. VMS shall have an administrator interface to set system parameters, manage codecs, manage permissions and manage storage.
11. Day to day control of cameras and monitoring on client workstations shall be controlled through the VMS administrator interface.
12. Whilst live control and monitoring is the primary activity of the monitoring workstations, video replay shall also be accommodated on the GUI for general review and also for pre- and post-alarm recording display.
13. The solution design for the VMS shall provide flexible video signal compression, display, storage and retrieval.
14. VMS shall support H.264 , H.265 or better video compression.
15. All CCTV camera video signal inputs to the system shall be provided to various command control center(s), viewing center etc., and the transmission medium used shall best suit the relative camera deployments and access to the CCTV Network.
16. VMS client shall have the capability to work with touch enabled multi-monitor workstations. It shall be capable of displaying videos in up to three (3) monitors simultaneously.
17. VMS shall be capable of transferring recorded images to recordable media (such as CD/DVD and/or tapes) in tamper evident and auditable form. All standard formats shall be supported including, but not limited to:
    - AVI files
    - JPEG Images
    - WAV audio
    - MP4 or SEF (Secured Export Format) (or latest)
18. All streams to the above locations shall be available in real-time and at full resolution. Resolution and other related parameters shall be configurable by the administrator in order to provide for network constraints.
19. The VMS shall support field sensor settings. Each channel configured in the VMS shall have an individual setup for the following settings, the specific settings shall be determined according to the encoding device:
    - Resolution
    - Frame Rate
    - Bit Rate
20. The VMS shall support bookmarking the videos. Thus, allowing the users to mark incidents on live and/or playback video streams.
21. The VMS shall support the following operations:
    - Adding an IP device

- Updating an IP device
- Updating basic device parameters
- Adding/removing channels
- Adding/removing output signals
- Updating an IP channel
- Removing an IP device
- Enabling/disabling an IP channel
- Refreshing an IP device (in case of firmware upgrade)
- Multicast at multiple aggregation points
- PTZ functions

22. The VMS shall support retrieving data from edge storage. Thus, when a lost or broken connection is restored, it shall be possible to retrieve the video from SD card and store it on central storage. System should support to view the recordings available over cameras local storage device (such as an SD card) and copy them to the server.
23. VMS shall support automatic failover for recording.
24. VMS should also support dual recording or mirroring if required.
25. VMS shall support manual failover for maintenance purpose.
26. VMS shall support access and view of cameras and views on a smartphone or a tablet (a mobile device).
27. VMS shall support integration with the ANPR application.
28. VMS shall support integration with other online and offline video analytic applications.
29. VMS shall be able to accept alerts from video analytics built into the cameras, other third-party systems, sensors etc.
30. VMS shall support alarm management module that shall allow for continuous monitoring of the operational status and event-triggered alarms from various system servers, cameras and other devices. The alarm management module shall provide a real-time overview of alarm status or technical problems while allowing for immediate visual verification and troubleshooting. Alarm Filtering option should be available, for example, it should be possible to "silence" alarms for a desired period.
31. VMS should not have any limit on the number of cameras to be connected for Surveillance, Monitoring and Recording. Any increase in the no. of cameras should be possible by augmentation of Hardware components & camera license.
32. VMS shall support distributed viewing of any camera in the system using Video walls or big screen displays.
33. It should be possible to integrate the VMS with 3rd-party software, to enable the users to develop customized applications for enhancing the use of video surveillance solution. For e.g., integrating alarm management to initiate SMS, E-Mail etc.
34. The system shall support the use of rules to determine when specific actions occur. Rules shall define what actions shall be carried out under specific conditions. The system shall support rule-initiated actions such as:
    - Start and stop recording
    - Set non-default live frame rate
    - Send notifications via email
    - Pop-up video on designated Client Monitor recipients
35. PSCDCL has setup PA-ECB system over 135+ locations across the city. VMS shall integrate feeds of PA-ECB system to display closest/ on location cameras whenever call from ECB or activity from PA system gets triggered.

Client System Features:

1. Viewing live video from cameras on the surveillance system.
2. Browsing recordings from storage systems.
3. Creating and switching between multiple of views.
4. Viewing video from selected cameras in greater magnification and/or higher quality in a designated hotspot.
5. Using digital zoom on live as well as recorded video.
6. Using sound notifications for attracting attention to detected motion or events.
7. Getting quick overview of sequences with detected motion.
8. Getting quick overviews of detected alerts or events.
9. Quickly searching selected areas of video recording for motion (also known as Smart Search).

Remote Web Client

1. The web-based remote client shall offer live view of up to 9 / 16 (configurable) cameras, including PTZ control (if applicable) and event / output activation. The Playback function shall give the user concurrent playback of multiple recorded videos with date, alert sequence or time searching.
2. User Authentication – The Remote Client shall support logon using the username and password credentials

Mobile Client

1. The bidder shall be required to provide a standardised Mobile Application to integrate smart phones and tablets for 2-way communication with the Video Management System in a secure manner. It will be responsibility of SI to configure such tablets / Smartphone with the Surveillance System and ensure that all the necessary access is given to these mobile users.
2. Communication with mobile client and server shall be encrypted with Digital Certificate.

3.9 Poles for Mounting Cameras

#	Parameter	Minimum Specifications
1.	Pole type	Hot Dip Galvanized after Fabrication with Silver coating of 86 micron as per IS:2629; Fabrication in accordance with IS-2713 (1980)
2.	Height	5-7 Meters (or higher), as-per-requirements, for different types of cameras & Site conditions
3.	Pole Diameter	Min. 10 cm diameter pole (bidder to choose larger diameter for higher height)
4.	Cantilevers	Based on the location requirement suitable size cantilevers to be considered with the pole
5.	Bottom base plate	Minimum base plate of size 30x30x1.5 cm
6.	Mounting facilities	To mount CCTV cameras, ANPR Cameras, Traffic Signals, Switch, etc.
7.	Pipes, Tubes	All wiring must be hidden, through tubes/pipes. No wires shall be visible from outside.
8.	Foundation	Casting of Civil Foundation with foundation bolts, to ensure vibration free erection (basic aim is to ensure that video feed quality is not impacted due to winds in different climatic conditions). Expected foundation depth of min. 100cms. Please refer to earthing standards mentioned elsewhere in the document.
9.	Protection	Lightning arrester shall be provided, to protect all field equipment mounted on pole.
10.	Sign Board & Number plates	A sign board describing words such as “This area under surveillance” and with serial number of the pole.
11.	Compliance	• Bidder shall submit the pole design during design phase, this design should be certified by Gov. authorised structural engineer. • After installation bidder shall submit structural stability and safety certificate for all poles from Gov. authorised structural engineer.

3.10 Edge Level Network switch

#	Parameter	Minimum Specifications
1.	Type	Managed Outdoor switch
2.	Total Ports	• Minimum 8 or 4 10/100 ⁄TX PoE/PoE+, 2x SFP Ports (can have 4xSFP Ports in certain locations) • May require higher port density at some locations, depending upon site conditions • May require Fibre ports (for devices or for uplinks) at some locations, depending upon site conditions/distances.
3.	PoE Standard	IEEE 802.3af/ IEEE 802.3at or better
4.	Protocols	• IPV4, IPV6 • Support 802.1Q VLAN • DHCP support • IGMP • SNMP Management • Should support Loop protection and Loop detection • Should support Ring protection / STP & RSTP • End point Authentication

		• Should support NTP
5.	Access Control	• Support port security • Support 802.1x (Port based network access control). • Support for MAC filtering • Support security group access control list
6.	PoE Power per port	Enough to operate the CCTV cameras/edge devices connected (If the device is POE)
7.	Enclosure Rating	IP 30 or equivalent Industrial Grade Rating (to be housed in Junction box)
8.	Operating Temperature	0°C ~ 50°C
9.	Multicast support	IGMP Snooping V1, V2, V3
10.	Management	Switch needs to have RS-232/USB/RJ45 console port for management via a console terminal or PC, Web GUI NTP, Syslog for log capturing SNMP V1, V2, V3
11.	Compliance	UL/EN/IEC or equivalent

3.11 Video wall Screen

#	Parameter	Minimum Specifications
2	Back Light Type/Panel Technology	Full HD IPS LED Display/Direct LED backlight
3	Screen Size	65”
4	Basel Size	0.44 mm Even Bezel and 0.88 mm Bezel TO Bezel
5	Resolution	Full high definition (1920 x 1080) 16:9 Widescreen
6	Cube Depth	Less than 100 mm
7	Contrast ratio	1100:1
8	Uniformity	Screen should have uniformity of the colors
9	Brightness	700 nit or Cd/m2
10	Viewing angle	178 degree/178 degree (H/V)
11	Input	HDMI and other inputs as per proposed Video Wall solution offered
12	Control	- On Screen Display (OSD)
		- IR remote control, LAN daisy chain control
13	Operations	24 x 7
14	LED Lifespan	60000 hours
15	General	Installation of video wall should seamless
16	Certification	FCC, UL/ETL/CB

3.12 Video wall controller

#	Parameter	Minimum Benchmark Specifications
1	Display controller	Controller to control 4*2 displays
2	Chassis	Industrial grade Rack mount
3	Operating System Platform	Windows 10 - 64 bit
4	Processor options	Xeon/ i7
5	RAM	16 GB DDR3
6	HDD	Support up to minimum 2 HDD
7		500 GB
8	Networking	Inbuilt Dual-port Gigabit Ethernet Controller
9		Serial ATA

#	Parameter	Minimum Benchmark Specifications
10	Input / Output supported	LAN
11		2x RJ45 LAN ports
12		USB 2.0 port or better
13	RAID	RAID 1 support
14	Power Supply	(1+1) Redundant hot swappable
15	Accessories	DVD +RW, Keyboard and mouse
16	Voltage	100-240V @ 50/60 Hz
17	Redundancy support	Power Supply
18		HDD
19		Cooling FAN
20		LAN ports
21		Bidder to provision controller in Redundancy
22	Scalability	Display multiple source windows in any size, anywhere on the wall
23	Control functions	Brightness / contrast / saturation/ Hue/ Filtering/ Crop / rotate
23	Outputs	should have one output card for every cube
24	Resolution support	DVI: 1920x1200 RGB: 2048x1536
25	Inputs	Capable of connecting all operator work stations in CCC. Fiber connectivity is suggested between workstation and controller for seamless performance.

3.13 Video wall management software

#	Parameter	Minimum Benchmark Specifications
1	Scaling and display	Software to enable the user to display multiple sources in any size and anywhere on the display wall
2	Auto Source Detection	Software should support for auto source detection
3	Layout Management	Should support for Video, RGB, DVI, Internet Explorer, Desktop Application and Remote Desktop Monitoring Layouts
4	Scenarios	Software should able to Save and Load desktop layouts from Local or remote machines
5	Layout Scheduler	All the Layouts can be scheduled as per user convenience
6		Software should support auto launch of Layouts according to specified time or event by user
7	Layout Preview	Software should support layout preview option
8	Launch Application	Software should be able to support
9	Integration with 3rd party devices	System should offer interface to enable control from 3rd party devices
10	Live Preview	Software should able to provide live preview of videowall
11	Work space allocation	System should provide functionality to the administrator to define and allocate workspace for a particular operator or a group of operators when working on a Video wall
12	Authentication	Software should offer levels of authentication
13	Offline Layouts	It should be possible to create offline layouts
17	SNTP	System should support SNTP function

3.14 Junction Box

#	Parameter	Minimum Specifications
1.	Size	Suitable size as per site requirements to house the field equipment
2.	Cabinet Material	GI
3.	Material Thickness	Min 1.2mm
5.	Number of Locks	Two
6.	Protection	IP 55, Junction Box design should ensure to keep the temperature within suitable operating range for equipment's and should also avoid intentional water splash and dust intake
7	Mounting	On Camera Pole / Ground mounted on concrete base

8.	Form Factor	Rack Mount/DIN Rail
9.	Other Features	Rain Canopy, Cable entry with glands, proper earthing and Fans/any other accessories as required for operation of equipment's within junction box.

3.15 Application / Database / Analytics / Recording / Other Sever

#	Parameter	Minimum Specifications
1.	Processor	Application/Database/Recording Servers: Latest series/ generation of 64-bit x86 processors with 16 (or higher) Cores each and native clock speed minimum 2.6 Ghz or higher. Minimum 2 Processors for each physical server. Analytics Servers: Latest series/ generation of 64-bit x86 processors with 18 (or higher) Cores each and native clock speed minimum 2.9 Ghz or higher. Minimum 2 Processors for each physical server. These are minimum requirements. SI can offer higher number of Cores and clock speed as per their proposed solution. Note: Native clock speed means that the clock speed shall be without Turbo Boost or equivalent technology. However, bidder is allowed to use Turbo boost or equivalent technology for boosting the clock speed above the min. requirement.
2.	RAM	Application /Recording Servers: Minimum 4 Gb DDR4 DIMM per core scalable to 8 Gb DDR4 DIMM per core for each processor. Analytics /Database Servers: Minimum 8 Gb DDR4 DIMM per core scalable to 16 Gb DDR4 DIMM per core. Bidder to submit undertaking from VMS / VA (FR & Video Summarization) OEM on recommendation of the most suitable Compute (Processor and RAM, etc.).
3.	Internal Storage	2 TB SAS (SSD) Server grade/enterprise disk, (OS drive: 512 Gb SSD for analytics servers)
4.	GPU	Analytics Server: Minimum 4 nos. of GPU cards per server scalable to 8 nos. of GPU cards hosted on PCIe cards. The required number of GPUs to be populated in the server only. Graphics card with min 8 GB video memory (non-shared). Bidder to submit undertaking from VMS / VA (FR & Video Summarization) OEM on recommendation of the most suitable graphic card
5.	Network interface	Dual Integrated Gigabit Ethernet ports (or 10 G ports as required for any server type) (Minimum 2 Integrated Gigabit Ethernet ports) Optional: Fiber channel adapter (if required) FCOE ports are acceptable
6.	Power supply	Dual, Redundant Power Supply
7.	RAID support	As per requirement/solution
8.	Operating System	Licensed version (Enterprise Edition) of 64 bit latest version of Linux/ Unix/Microsoft® Windows based Operating system, matching with the processor(s)
9.	Form Factor	Rack Server (For Video Analytics Servers); Blade for other application server
10.	Virtualization	Shall support Industry standard virtualization hypervisor like Hyper-V, VMware and Citrix.

11.	Compliance	The Server configuration recommended by the all solutions must be submitted (signed and Stamped by OEM) and bidder must ensure to comply with same. The server must be sized with 10% spare capacity and same must be submitted

3.16 Core Switch (L 3 Switch)

#	Parameter	Minimum Specifications
1	Ports	• 24 or 48 (as per requirements) 10/100/1000 Base-TX/FX ports and extra 2 or 4 Nos of 10G Base SX/LX/LR ports as per network solution offered. • TX/FX Split as per field/site requirement • All ports can auto-negotiate between 10Mbps/ 100Mbps/ 1000Mbps, half-duplex or full duplex and flow control for half-duplex ports.
2	Switch type	Layer 3
3	MAC	32K or more
4	Backplane	Properly sized Switching fabric capacity (as per network configuration to meet performance requirements of wire speed switching for the connected devices)
5	Port Features	Must support Port Mirroring, Port Trunking and 802.3ad LACP Link Aggregation port trunks
6	Flow Control	Support IEEE 802.3x flow control for full-duplex mode ports.
7	Protocols	• IPV4, IPv6 • Support 802.1D, 802.1S, 802.1w, Rate limiting • Support 802.1Q VLAN encapsulation, IGMP v1, v2 and v3 snooping • 802.1p Priority Queues, port mirroring, DiffServ • DHCP support • Support upto 1024 VLANs • Support IGMP Snooping and IGMP Querying • MLD Snooping • Support Multicasting • Should support Loop protection and Loop detection, • Should support Ring protection • OSPF, BGP • PIM SM, PIM DM
8	Access Control	• Support port security • Support 802.1x (Port based network access control). • Support for MAC filtering. • Should support TACACS+ and RADIUS authentication
9	VLAN	• Support 802.1Q Tagged VLAN and port based VLANs and Private VLAN • The switch must support dynamic VLAN Registration or equivalent • Dynamic Trunking protocol or equivalent
10	Protocol and Traffic	• Network Time Protocol or equivalent Simple Network Time Protocol support • Switch should support traffic segmentation • Traffic classification should be based on user-definable application types: TOS, DSCP, Port based, TCP/UDP port number
11	Management	• Switch needs to have console port for management via PC • Must have support SNMP v1, v2 and v3 • Should support 4 groups of RMON • Should have accessibility using SSH, Console access, easier software upgrade through network using TFTP etc. Configuration management through CLI, GUI based software utility and using web interface

3.17 Layer 2 Switch

#	Parameter	Minimum Specifications
1.	Ports	• 24 or 48 (as per requirements) 10/100/1000 Base-TX ports and extra 2 or 4 Nos. of 10G Base SR/LR ports as per network solution offered. • All ports can auto-negotiate between 10Mbps/ 100Mbps/ 1000Mbps, half-duplex or full duplex and flow control for half-duplex ports.
2.	Switch type	Layer 2
3.	MAC	8 K or more
4.	Backplane	56 Gbps capacity (as per network configuration to meet performance requirements of wire speed switching for the connected devices)
5.	Forwarding Rate	Packet Forwarding Rate should be 70.0 Mpps or better
6.	Port Features	Must support Port Mirroring, Port Trunking and 802.3ad LACP Link Aggregation port trunks
7.	Flow Control	Support IEEE 802.3x flow control for full-duplex mode ports.
8.	Protocols	• IPV4, IPv6 • Support 802.1D, 802.1S, 802.1w, Rate limiting • Support 802.1Q VLAN encapsulation, IGMP v1, v2 and v3 snooping • 802.1p Priority Queues, port mirroring, DiffServ • DHCP support • Support upto 1024 VLANs
9.	Access Control	• Support port security • Support 802.1x (Port based network access control). • Support for MAC filtering. • Should support TACACS+ and RADIUS authentication
10.	VLAN	• Support 802.1Q Tagged VLAN and port based VLANs and Private VLAN • The switch must support dynamic VLAN Registration or equivalent • Dynamic Trunking protocol or equivalent • The switch should support Rapid Per-VLAN Spanning tree
11.	Protocol and Traffic	• Network Time Protocol or equivalent Simple Network Time Protocol support • Switch should support traffic segmentation • Traffic classification should be based on user-definable application types: TOS, DSCP, Port based, TCP/UDP port number
12.	Management	• Switch needs to have console port for management via PC • Must have support SNMP v1, v2 and v3 • Should support 4 groups of RMON • Should have accessibility using SSH, Console access, easier software upgrade through network using TFTP etc. Configuration management through CLI, GUI based software utility and using web interface

3.18 Edge Level Switch (For Cameras & other edge device)

#	Parameter	Minimum Specifications
13.	Type	Managed Outdoor switch
14.	Total Ports	• 4/8 10/100/1000 ƒTX PoE/PoE+, 2x SFP Ports (can have 4xSFP Ports in certain locations) • May require higher port density at some locations, depending upon site conditions • May require Fibre ports (for devices or for uplinks) at some locations, depending upon site conditions/distances.
15.	PoE Standard	IEEE 802.3af/ IEEE 802.3at or better
16.	Protocols	• IPV4, IPV6 • Support 802.1Q VLAN

		• DHCP support • IGMP • MLD • SNMP Management • Should support Loop protection and Loop detection • Should support Ring protection / STP & RSTP • End point Authentication • Should support NTP • OSPF, BGP • PIM SM, PIM DM
17.	Access Control	• Support port security • Support 802.1x (Port based network access control). • Support for MAC filtering • Support security group access control list
18.	PoE Power per port	Sufficient to operate the CCTV cameras/edge devices connected (If the device is POE)
19.	Enclosure Rating	IP 30 or equivalent Industrial Grade Rating (to be housed in Junction box)
20.	Operating Temperature	0°C ~ 75°C
21.	Multicast support	IGMP Snooping V1, V2, V3, MLD V1 & V2
22.	Management	Switch needs to have RS-232/USB/RJ45 console port for management via a console terminal or PC, Web GUI NTP, Syslog for log capturing SNMP V1, V2,V3
23.	Compliance	UL/EN/IEC or equivalent
24.	Compliance/Immunity Standards	Electrostatic Discharge: IEC/EN 61000-4-2 Radiation: 1EC/EN 61000-4-3 Electrical Fast Transient/ Burst Immunity: IEC/EN 61000-4-4 Surge Immunity: 1EC/EN 61000-4-5 Power Frequency Magnetic Field: IEC/EN 61000-4-8 Shock and Vibration: IEC/EN 60068-2-27

3.19 Storage (Primary & Secondary)

#	Parameter	Minimum Specifications
1.	Solution/ Type	• Bidder shall provide the appropriate Storage Solution to meet the Project Requirements. • Existing storage of 2.2 PB is present onsite (Isilon) • Bidder is expected to carry out the video storage requirement estimation and supply as per the solution proposed. Solution proposed should yield low cost per TB, while meeting the required performance parameters
2.	Hardware Platform	• Rack mounted form-factor • Modular design to support controllers and disk drives expansion
3.	Controllers	• At least 2 Controllers in active/active mode and upgrade upto 8 controller pairs either in same chassis or with federation of arrays to deliver higher scalability and performance. • The controllers / Storage nodes should be upgradable seamlessly, without any disruptions / downtime to production workflow for performance, capacity enhancement and software / firmware upgrades. Storage to be Unified Storage / combination of NAS and SAN storage.
4.	RAID support	• Should support various RAID Levels to meet the SLA requirements
5.	Cache	• Minimum 256 GB of useable System memory /cache across all controllers. If cache is provided in any additional hardware for the storage solution, then cache must be over and above 256GB.
6.	Redundancy and High Availability	• The Storage System should be able to protect the data against single point of failure with respect to hard disks, connectivity interfaces, fans and power supplies

7.	Management software	• All the necessary software (GUI Based) to configure and manage the storage space, RAID configuration, logical drives allocation, snapshots etc. are to be provided for the entire system proposed. • Licenses for the storage management software should include disc capacity/count of the complete solution and any additional disks to be plugged in in the future, up to max capacity of the existing controller/units. • A single command console for entire storage system. • Should also include storage performance monitoring and management software • Should provide the functionality of proactive monitoring of Disk drive and Storage system for all possible disk failures • Should be able to take "snapshots" of the stored data to another logical drive for backup purposes
8.	Data Protection	The storage array must have complete 256 GB cache protection mechanism either by de-staging data to disk or providing complete cache data protection with battery backup for up to 4 hours
9.	Licensing	There should not be any additional license requirements arising in future during capacity augmentation of the Storage solution. The proposed Storage should be scalable by merely adding the desired number of HDDs and Disk Enclosures in future along with necessary connecting back-end/front-end cables till the maximum scalability is reached within the proposed Appliance.
10.	Data Availability	Proposed storage solution should provide the data availability of 99.99%
11.	Security	Storage must support file protection against accidental, premature malicious deletion and modification of data using file locking mechanism.
12.	Application data & Video data	Application data (block data) should be on SSD/SAS and Video data (File data) should be an SAS/NL-SAS

3.20 Firewall

#	Parameter	Minimum Specifications
1.	Physical attributes	• Should be mountable on 19” Rack • Modular Design • Internal redundant power supply
2.	Interfaces	• 4 x GE, upgradable to 8 GE • Console Port 1 number • 4 x 10 G Ports as per network design (if required)
3.	Performance and Availability	• Encrypted throughput: minimum 1 Gbps • Concurrent connections: up to 100,000 • Simultaneous VPN tunnels: 2000
4.	Routing Protocols	• Static Routes • RIPv1, RIPv2 • OSPF
5.	Protocols	• TCP/IP • RTP • IPSec, DES/3DES/AES • FTP, HTTP, HTTPS, SNMP, SMTP • DHCP, DNS, Support for IP v4 & IPv6 • IPSEC
6.	Other support	• 802.1Q, NAT, PAT, IP Multicast support, Remote Access VPN, URL Filtering, support VLAN, Radius/ TACACS, Support multilayer firewall protection
7.	QoS	• QoS features like traffic prioritisation, differentiated services, committed access rate. Should support for QoS features for defining the QoS policies.
8.	Management	• Console, SSHv2, Browser based configuration • SNMPv1, SNMPv2, SNMPv3

9.	Certifications	• ICSA / NDPP Labs or equivalent

3.21 Intrusion Prevention System

This can be offered as separate unit or as a module in firewall

#	Parameters	Required Specifications
1.	Performance	Should have an aggregate throughput of no less than 500Mbps Total Simultaneous Sessions – 100,000
2.	Features	• IPS should have Dual Power Supply • IPS system should be transparent to network, not default gateway to Network • IPS system should have Separate interface for secure management • IPS system should be able to protect Multi Segment in the network, should be able to protect 4 segments.
3.	Real Time Protection	• Web Protection • Mail Server Protection • Cross Site Scripting • SNMP Vulnerability • Worms and Viruses • Brute Force Protection • SQL Injection • Backdoor and Trojans • Denial of service (DoS)
4.	Stateful Operation	• TCP Reassembly • IP Defragmentation • Bi-directional Inspection • Forensic Data Collection • Access Lists
5.	Signature Detection	Should have provision for Real Time Updates of Signatures, IPS Should support Automatic signature synchronization from database server on web Device should have capability to define User Defined Signatures
6.	Block attacks in real time	• Drop Attack Packets • Reset Connections • Packet Logging • Action per Attack
7.	Alerts	• Alerting SNMP • Log File • Syslog • E-mail
8.	Management	• SNMP v1, v2, v3 • HTTP, HTTPS • SSHv2, Console
9.	Security Maintenance	• IPS Should support 24/7 Security Update Service • IPS Should support Real Time signature update • IPS Should support Provision to add static own attack signatures • IPS should have provision for external bypass Switch

3.22 KVM Module

#	Parameters	Minimum Specifications
1.	KVM Requirement	Keyboard, Video Display Unit and Mouse Unit (KVM) for the IT Infrastructure Management at Data Centre
2.	Form Factor	19” rack mountable
3.	Ports	minimum 8 ports
4.	Server Connections	USB or KVM over IP.
5.	Auto-Scan	It should be capable to auto scan servers
6.	Rack Access	It should support local user port for rack access
7.	SNMP	The KVM switch should be SNMP enabled. It should be operable from remote locations
8.	OS Support	It should support multiple operating system
9.	Power Supply	It should have dual power with failover and built-in surge protection
10.	Multi-User support	It should support multi-user access and collaboration

3.23 Network Racks

42 U Racks for DC

#	Parameter	Minimum Specifications
1.	Type	• 19”, 42U racks mounted on the floor • Floor Standing Server Rack - 42U with Heavy Duty Extruded Aluminium Frame for rigidity. • Top & Bottom cover with cable entry gland plates. Heavy Duty Top and Bottom frame of MS. • Two pairs of 19" mounting angles with 'U' marking. Depth support channels - 3 pairs with an overall weight carrying Capacity of 500Kgs. • All racks should have mounting hardware Packs, Blanking Panels as per requirements. • Stationery Shelf (2 sets per Rack) • All racks must be lockable on all sides with unique key for each rack • Racks should have Rear Cable Management channels, Roof and base cable access
2.	Wire managers	Two vertical and four horizontals minimum
3.	Power Distribution Units	• 2 per rack • Power Distribution Unit - Vertically Mounted, 32AMPs with 25 Power Outputs. (20 Power outs of IEC 320 C13 Sockets & 5 Power outs of 5/15 Amp Sockets), Electronically controlled circuits for Surge & Spike protection, LED readout for the total current being drawn from the channel, 32AMPS MCB, 5 KV AC isolated input to Ground & Output to Ground
4.	Doors	• The racks must have steel (solid / grill / mesh) front / rear doors and side panels. Racks should NOT have glass doors / panels. • Front and Back doors should be perforated with at least 60% or higher perforations.

		• Both the front and rear doors should be designed with quick release hinges allowing for quick and easy detachment without the use of tools.
5.	Fans and Fan Tray	• Fan 90CFM 230V AC, 4" diameter (4 Nos. per Rack) • Fan Housing Unit 4 Fan Position (Top Mounted) (1 no. per Rack) - Monitored - Thermostat based - The Fans should switch on based on the Temperature within the rack. The temperature setting should be factory settable. This unit should also include - humidity & temperature sensor
6.	Metal	Aluminium extruded profile
7.	Side Panel	Detachable side panels (set of 2 per Rack)

42 U Smart Rack for Secondary Storage at DCP office

Smart Cabinet Server Rack consists of following inbuilt features: -

- Integrated Racks & containment
- Size 42 U
- Precision Cooling system
- UPS & batteries (Optional)
- High Sensitive Smoke Detection System
- Environment monitoring system
- Biometric Access Control

6 U Network Rack

#	Parameters	Minimum Specifications
1.	Maximum Height	330.0 mm
2.	Maximum Width	560.00 mm
3.	Maximum Depth	400.00 mm
4.	Maximum Mounting Depth	286.00 mm
5.	Rack Height	6U
6.	Rack Width	19”
7.	Others	1. The front door shall be insulated metallic door fitted with rubber gasket and a central glass for clear visibility of all components installed in the rack
8.		2. The 6U rack shall have one cable manager for network cables.
9.		3. The 6U Rack shall have provision for two separate top entries one for power and one for network cables.

Networking Standards

- ANSI/TIA-942, Telecommunications Infrastructure Standard for Data Centres
- ANSI/TIA/EIA/568-C.1, Commercial Building Telecommunications Cabling Standard – 2009
- ANSI/TIA/EIA 568-C.2, Copper Cabling Components Standard
- ANSI/TIA/EIA 568-C.3, Optical Fibre Cabling Components Standard
- ANSI/TIA/EIA-569-B, Commercial Building Standard for Telecommunications Pathways and Spaces

- ANSI/TIA/EIA-606-A, Administration Standard for the Telecommunications Infrastructure of Commercial Buildings
- ANSI/J-STD-607-A, Commercial Building Grounding (Earthing) and Bonding Requirements for Telecommunications
- Building Industries Consulting Services International (BICSI) Telecommunications Distribution Methods Manual (TDMM) – Preferred

3.24 Access Control System

The Access Control System shall be deployed with the objective of allowing entry and exit to and from the premises to authorized personnel only. The system deployed shall be based on Biometric Technology. An access control system consisting of a central PC, intelligent controllers, power supplies and all associated accessories is required to make a fully operational online access control system. Access control shall be provided for entry / exit doors of CCC & DC. These doors shall be provided with electric locks and shall operate on fail-safe principle. The lock shall remain unlocked in the event of a fire alarm or in the event of a power failure. The fire alarm supplier shall make potential free contacts available for releasing the locks in a fire condition especially for staircase and main doors. Entry to the restricted area shall be by showing a proximity card near & Biometric details to the reader and exit shall be using a push button installed in the secure area. The system shall monitor the status of the doors through magnetic lock contacts. The system should be designed and implemented to provide following functionality:

- Controlled Entries to defined access points
- Controlled exits from defined access points
- Controlled entries and exits for visitors
- Configurable system for user defined access policy for each access point
- Record, report and archive each and every activity (permission granted and / or rejected) for each access point.
- User defined reporting and log formats
- Fail safe operation in case of no-power condition and abnormal condition such as fire, theft, intrusion, loss of access control, etc.
- Day, Date, Time and duration-based access rights should be user configurable for each access point and for each user.
- One user can have different policy / access rights for different access points.

3.25 WorkStation

#	Parameter	Minimum Specifications
1.	Processor	Latest generation 64bit X86 Quad core processor (3.33 Ghz) or better
2.	Chipset	Latest series 64bit Chipset
3.	Motherboard	OEM Motherboard
4.	RAM	Minimum 16 GB, upgradable up to 32GB
5.	Graphics card	WHQL certified Graphics card with 4 GB video memory (non- shared). It should support 3 monitors simultaneously with no degradation in video quality considering HD video quality and 25 fps Bidder to submit undertaking from VMS / VA OEM on recommendation of the most suitable graphic card
6.	HDD	2 TB SATA-3 Hard drive @7200 rpm
7.	Media Drive	NO CD / DVD Drive
8.	Network interface	1000BaseT, Gigabit Ethernet (10/100/1G auto sensing)

#	Parameter	Minimum Specifications
9.	Audio	Line/Mic IN, Line-out/Spr Out (3.5 mm)
10.	Ports	Minimum 6 USB ports (out of that 2 in front). These would be disabled for data transfer.
11.	Keyboard	104 keys minimum OEM keyboard
12.	Mouse	2 button optical scroll mouse (USB)
13.	Monitor[3]	Three monitors of 22" TFT LED monitor, Minimum 1920 x1080 resolution, 5 ms or better response time, TCO 05 (or higher) certified
14.	Certification	Energy star 5.0/BEE star certified
15.	Operating System	Windows (Latest version) Licensed Enterprise edition ,64 bit pre-loaded OS with recovery disc
16.	Security	BIOS controlled electro-mechanical internal chassis lock for the system.
17.	Antivirus feature	Advanced antivirus, antispyware, desktop firewall, intrusion prevention (comprising of a single, deployable agent) which can be managed by a central server. (Support, updates, patches and errata for the entire contract/ project period)
18.	Power supply	SMPS; Minimum 400-watt Continuous Power Supply with Full ranging input and APFC. Power supply should be 90% efficient with EPEAT Gold certification for the system.
19.	DRM Software	There would be DRM Software/application installed on these workstations that would prevent unauthorized copying of video feeds and other sensitive data.

[3] MSI Shall provide 1 Nos of Monitor at proposed viewing centers of same configuration.

3.26 IP based Public Address Speakers

1	PA Speakers	• Should have the capability to control individual PAS i.e. to make an announcement at select location (1:1) and all locations (1: many) simultaneously. • The PAS should also support both, Live and Recorded inputs • It should allow localised usage (for the utility of the on-field constable / any officer to make localised announcements) • All announcements made shall be stored for 90 days
2	Speaker	Minimum 2 speakers, to be used for Public Address System, to deliver min. 75 dB at the listening level *(Final count would be decided during AS-IS / Requirement Documentation Phase in discussion with Pune Police)*
3	Connectivity	IP Based
4	Access Control	Access control mechanism would be also required to establish so that the usage is regulated.
5	Integration	With VMS and Command and Control Centre
6	Construction	Cast Iron Foundation and M.S. Pole, Sturdy Body for equipment
7	Battery	Internal Battery with different charging options (Solar/Mains)
8	Power	Automatic on/off operation
9	Casing	IP-65 rated for housing
10	Operating conditions	0° to 50°C

4. Annexure II - Existing Project Components

Note: With reference to the Manufacturer Authorization Form (MAF) submitted by the OEMs at the time of bid submission for previous RFP, below mentioned components have reached End of Life/Support. Bidder may have to confirm from respective OEMs, for additional support on below mentioned items. Bidder shall consider reusing existing hardware, wherever possible by extending the OEM support (MAF from respective OEMs shall be provided by the bidder in their bid proposal) for the contract period.

Annexure A: Existing Surveillance Equipment Infrastructure

Sr. No.	Description	Make	Model	Unit	Qty.
1	Outdoor Box camera	Axis	P1354	Nos.	2
2	Outdoor Box camera	Axis	P1375	Nos.	22
3	Outdoor Box camera	Dvtel	CF-3211-00	Nos.	723+230
4	Outdoor Box camera	Infinova	V6202-TA220	Nos.	16
5	Outdoor Box camera	Infinova	VT210-A2-B0	Nos.	27+3
6	Outdoor Box camera	CP Plus	CP-UNC-BE21-VMD	Nos.	10
7	PTZ Camera	Dvtel	CP-3211-181	Nos.	149+3
8	PTZ Camera	Infinova	VT231-A222-A361	Nos.	22+1
9	PTZ Camera	Infinova	VT231-A230-A061	Nos.	26
10	PTZ Camera	CP Plus	CP-UNP-F3021L15-DA	Nos.	24
11	PTZ Camera	CP Plus	CP-UNP-F3021L20-DAP	Nos.	5
12	ANPR Camera	Axis	P1354	Nos.	49+22
13	ANPR Camera	Axis	P1375	Nos.	7
8	Poles for Cameras and Equipment	Type: Hot Dip Galvanized after Fabrication with Silver coating of 86 micron as per IS:2629	Customized	Nos.	593 (470+123)
9	Provisioning of Electrical Power	MSEDCL	-	Nos.	358+84
10	Routers	CISCO	CGR1120/K9	Nos.	358+84
11	Junction Box	Customized	Customized	Nos.	360+85
12	6 port LIU	Molex	WFR-00028-02	Nos.	940+246

Sr. No.	Description	Make	Model	Unit	Qty.
13	OFC (6C MM)	Reputed	AFOUN006OM2	Mtrs.	45752
14	6 Mtrs. STP Patch Cord CAT6	D link/Molex	PCD-07026-XX	Nos.	1763
15	SC-SC 50/125, Multimode, 3 Mtrs patch chord OM2	Molex	91.99.522.00300	Nos.	940
16	Pipes (Included in excavation)	HDPE 50mm DWC pipe	-	Mtrs.	36165
17	Multi-Port Media Converter	MOXA	EDS-P206A-4PoE	Nos.	481+129
18	Media Converters	MOXA	IMC-21A-M-SC	Nos.	470+123
19	UPS per Junction (1 KVA with 1 Hr. Backup)	EATON	UPS 1000 IN XL 1 KVA	Nos.	359+84
20	Supply of IR Illuminators	Raytec	VAR-I8-1	Nos.	100
21	16 Channel Encoder	Verint	S1816e	Nos.	1

Annexure B: Existing Network Connectivity Infrastructure

Below are the components of existing infrastructure through which the network connectivity is established. Current network is of BSNL using MPLS networking. A network of 2.4 MBPS is provided currently amongst the Edge Devices and Data Center.

S. No.	Description	Units Installed	OEM	Count	Specifications
1	Bandwidth for Outdoor Box Cameras (Camera - Data Centre)	790+241	BSNL	NA	BSNL MPLS Connectivity
2	Bandwidth for Outdoor PTZ Cameras (Camera - Data Centre)	197+33			
3	ANPR Cameras	56+22			
4	CP Office Command Center	1			
5	PMC	1			
6	PCMC	1			
7	Collectorate	1			
8	Addl. CP Offices 1 (Addl. CP East)	1			
9	Addl. CP Offices 2 (Traffic Control Room, Yerwada)	1			
10	Zonal DCP Offices 1	1			

S. No.	Description	Units Installed	OEM	Count	Specifications
11	Zonal DCP Offices 2	1			
12	Zonal DCP Offices _PCPC Zone	1			
13	Zonal DCP Offices 4	1			
14	Police Station - Data Center	40			
15	Bandwidth for connecting existing cameras setup by PMC or Pune Smart City	-			

Annexure C: Existing Data Center Infrastructure

Below are the components of Data Center

Sr. No.	Description	Make	Model	Unit	Qty.
1	Server (Inclusive of Operating System)				
1.a	Application Server	IBM	x240	Nos.	3
1.b	Recording Server	IBM	x240	Nos.	5
1.c	Analytics Server	IBM	x240	Nos.	6
1.d	Database Server (included)	-	-	Nos.	-
1.e	Management Server	IBM	x240	Nos.	3
1.f	Enterprise Backup Server	IBM	x3550M4	Nos.	1
1.g	Domain Controller (LDAP Server) (included)			Nos.	0
1.h	Any Other Server required Cater to the scope of work mentioned in Vol-II			Nos.	1
1.h.1	IP Telephony server	CISCO	UCUCS-EZ-C220M3S	Nos.	9
2	Application and System Software			Nos.	0
2.a	Video Management System (Included in 2b)				0
2.b	Software for Recording, Viewing of Videos (Pl give the break-up if different s/w required)			Lumpsum	1

Sr. No.	Description	Make	Model	Unit	Qty.
2.b.1	Master Server Software	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	2
2.b.2	Master Recorder Server Software	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	4
2.b.3	Camera Licenses	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1771
2.b.4	Failover Camera Licenses	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1771
2.b.5	Camera Licenses for Existing Cameras	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	210
2.b.6	Failover Camera Licenses for Existing Cameras	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	210
2.b.7	Review / Client Licenses	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	60
2.b.8	Multisite Directory Server Software	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1
2.b.9	Virtual Matrix Software License	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1
2.b.10	SDK Connection License	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1
2.b.11	Nextiva Client SDK Development Kit	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1
2.b.12	Investigation Management Site License	Verint	Nextiva SMC - Version 7.6.747.0	Nos.	1
2.c	Updated Base Map for Pune & Pimpri Chinchwad (min. 0.6 mtr resolution, min. 1:500) (GIS Software)	Arminda	-	Sq. Kms.	659.46
2.d	Viewing Software for GIS	MapMyIndia	-	Nos.	1
2.e	RDBMS (if Required)			Nos.	0
2.f	Backup Solution	IBM	Tivoli Storage Manager	Nos.	1
2.g	Enterprise Management System (Give breakup of SLA Mngt, Helpdesk Mngt, Network Mngt, BMS etc. if applicable)	BMC	1. BMC Remedy : Version 8.1.02 2. BMC Entuity: Version 15.0 3. BMC ProactiveNet Performance Management: Version 9.6 4. SAP BusinessObjects BI: Version: 14.1.1.1036 5. BMC TM ART Central: Version 4.2 6. BMC Dashboards: Version 7.7.02	Nos.	1
2.h	Anti-virus Software (included)	Symantec	SYMC PROTECTION SUITE ENTERPRISE EDITION	Nos.	0

Sr. No.	Description	Make	Model	Unit	Qty.
2.i	LDAP Software (Not Applicable)	Microsoft	Windows 2012 Datacenter edition		0
2.j	Customized Software • Dashboard for Police Department (for various levels) • Integration with CCTNS, VAHAN			Lumpsum	1
2.j.1	PSIM Server License	Verint	Verint NEXP-7.6 Build 747.1	Nos.	1
2.j.2	PSIM Client Licenses	Verint	Verint NEXP-7.6 Build 747.1-PSIMS	Nos.	9
2.j.3	Licenses to Manage 9 Sq.km city area	Verint	Verint Nextiva PSIM 4100 City Zone	Nos.	50
2.j.4	PSIM Monitored Devise Software License	Verint	Verint NEXP-7.6 Build 747.1 DEV-S	Nos.	1500
2.j.5	Interface to License Plate Recognition	Verint	Verint VIS-SV-PS	Lot	1
2.j.6	Interface to GPS enabled police cars	Verint	Verint VIS-SV-PS	Lot	1
2.j.7	Special Professional Services - Customized Professional services Interface to Vahan	Verint	Verint VIS-SV-PS	Lot	1
2.j.8	Special Professional Services - Customized Professional services SMS gateway integration	Verint	Verint VIS-SV-PS	Lot	1
2.j.9	Interface to video analytics	Verint	Verint VIS-SV-PS	Lot	1
2.j.10	Interface to 2D GIS mapping of Pune City	Verint	Verint 2D modelling	Lot	1
2.j.11	Installation services	Verint	Verint VIS-INST-PS	Nos.	50
2.j.12	PS - integration development - Vahan System	Verint	-	Nos.	30
2.j.13	PS - integration development - SMS Gateway	Verint	-	Nos.	30
2.j.14	PS - integration development - PIU interface	Verint	-	Nos.	30
2.j.15	PS - integration development - CCTNS system	Verint	-	Nos.	30
2.j.16	Training per student, cost is per student	Verint	VIS-ONST-TRAIN	Nos.	5
2.j.17	Nextiva Project Management – Verint professional services for project management	Verint	NEX-5-PM-PS	Nos.	50
2.j.18	Gold Maintenance Charges (PSIM)	Verint	One-year Maintenance & Support	Lot	1
2.j.19	Added 4-year Gold Maintenance Charges	Verint	Four-year Maintenance & Support	Lot	1

Sr. No.	Description	Make	Model	Unit	Qty.
2.k	Any other Server-side Software needed to cater to the scope of work mentioned in Vol II				
2.k.1	IP Telephony Application	Cisco		Nos.	1
2.k.2	Video Analytics Licenses	Agent VI	Agent VI - Version 5.3.0.222	Nos.	100
2.k.2.1	Analytics Redundant server Licenses	Agent VI	Agent VI - Version 5.3.0.222	Nos.	2
2.l	ANPR Software and License	Vehant	-	Nos.	0
2.l.1	Automatic number plate recognition	Vehant 7.0	7.0	Nos.	78
2.l.2	ANPR Central Server Software License	Vehant 7.0	7.0	Nos.	1
3	Primary Storage	EMC	NL-400	Nos.	1
4	Secondary Storage	EMC	NL-400	Nos.	1
5	Storage Management System (Included IN 3 and 4)			Nos.	0
6	Tape Library	IBM	TS3500	Lumpsum	1
7	Core Router			Nos.	2
7.1	ASR 1004 - Core Router	CISCO	ASR1004-10G/K9	Nos.	2
8	Switches			Nos.	0
8.1	L2 Switches			Nos.	1
8.1.1	Cisco 2960 - 24 TCL	CISCO	WS-C2960-24TC-L	Nos.	1
8.1.2	Cisco 2960S - 24 DMZ	CISCO	WS-C2960S-24TS-L	Nos.	2
8.2	L3 Switches			Nos.	2
8.2.1	Nexus 7004 - Core Switch	CISCO	N7K-C7004-S2	Nos.	2
9	KVM Switch (Not Applicable)			Nos.	0
10	Firewall	CISCO	ASA5545-IPS-K9	Nos.	2
11	Intrusion Prevention System (Included in Firewall)	CISCO	INCLUDED	Nos.	0
12	Racks (Caged) (Refer Sch-R)	Reputed		Nos.	0
13	Indoor Fixed Dome Cameras			Nos.	6

Sr. No.	Description	Make	Model	Unit	Qty.
13	Indoor Fixed Dome Cameras	Dvtel	CM-3211-00	Nos.	6
13.1	5" x 5" PVC Junction Box for Dome Camera	Reputed		Nos.	6
14	Fire-Proof Enclosure for Media Storage	Godrej		Nos.	1
15	Networking Cost (Passive Components) (Pl. specify the details)	-	-	Nos.	0
15.1	Patch Cord	-	-	Lot	1
15.1	LC-LC MM 3-meter OM3	Reputed	-	Nos.	26
15.2	LC-LC MM 10-meter OM3	Reputed	-	Nos.	15
15.3	Fiber Patch cord laying	Allied Digital	-	Nos.	41
15.4	3 Mtr. UTP Patch Cord	Reputed	-	Nos.	17
15.5	UTP patch cord laying	Allied Digital	-	Nos.	17
15.6	Miscellaneous Components like ferrules, cable ties etc.	Allied Digital	-	LS	0
16	Any Other Component(s) to Cater to the requirements of Scope of Work mentioned in Vol II			Nos.	0
16.1	Network Analyze Module (Included)			Nos.	1
16.2	Identity Service Engine	CISCO	ISE-3355-K9	Nos.	3

Annexure D: Existing Application Portfolio

Below are the applications used for the project

Integration Type	S. No	Application Description	Make and Model	License Count	License Validity	Status
Internal	1	Video Management System	Nextiva Review – Version 7.6.747.0	1	Perpetual Licenses	Running
	2	**Software for Recording, Viewing of Videos (give break-up if different s/w required) & integration with other applications**				Running
	2.1	Software for Recording, Viewing of Videos (Pl give the break-up if different s/w required)		1		Running
	2.2	Master Server Software	Nextiva SMC - Version 7.6.747.0	2		Running
	2.3	Master Recorder Server Software	Nextiva SMC - Version 7.6.747.0	4		Running
	2.4	Camera Licenses	Nextiva SMC - Version 7.6.747.0	1771		Running
	2.5	Failover Camera Licenses	Nextiva SMC - Version 7.6.747.0	1771		Running
	2.6	Camera Licenses for Existing Cameras	Nextiva SMC - Version 7.6.747.0	210		Running
	2.7	Failover Camera Licenses for Existing Cameras	Nextiva SMC - Version 7.6.747.0	210		Running
	2.8	Review / Client Licenses	Nextiva SMC - Version 7.6.747.0	60		Running
	2.9	Multisite Directory Server Software	Nextiva SMC - Version 7.6.747.0	1		Running
	2.10	Virtual Matrix Software License	Nextiva SMC - Version 7.6.747.0	1		Running
	2.11	SDK Connection License	Nextiva SMC - Version 7.6.747.0	1		Running
	2.12	Nextiva Client SDK Development Kit	Nextiva SMC - Version 7.6.747.0	1		Running
	2.13	Investigation Management Site License	Nextiva SMC - Version 7.6.747.0	1		Running
	3	Automatic Number Plate Recognition (ANPR)	Vehant 7.0	1		Running
	4	Video Analytics Software	Agent VI - Version 5.3.0.222	100		Running
	5	Enterprise Management System (System Performance Management System Performance Management, Network Alerts, For Helpdesk & Ticketing, SLA Mngt Network Mngt) BMC Remedy - Ticketing BMC Proactive Net Performance Management: Alerts BMC Entuity - Network Monitoring	1. BMC Remedy: Version 8.1.02 2. BMC Entuity: Version 15.0 3. BMC Proactive Net Performance Management: Version 9.6 4. SAP BusinessObjects BI: Version: 14.1.1.1036 5. BMC TM ART Central: Version 4.2 6. BMC Dashboards: Version 7.7.02	1		Running

Integration Type	S. No	Application Description	Make and Model	License Count	License Validity	Status
	6	Anti-virus Software	Symantec - SYMC PROTECTION SUITE ENTERPRISE EDITION	Lumpsum		Running
	7	Backup Solution	IBM Tivoli Storage Manager	1		Running
	8	**Customized Software**				Running
	8.1	PSIM Server License	Verint NEXP-7.6 Build 747.1	1		Running
	8.2	PSIM Client Licenses	Verint NEXP-7.6 Build 747.1-PSIMS	9		Running
	8.3	Licenses to Manage 9 Sq.km city area	Verint Nextiva PSIM 4100 City Zone	50		Running
	8.4	PSIM Monitored Devise Software License	Verint NEXP-7.6 Build 747.1 DEV-S	1500		Running
	8.5	Interface to License Plate Recognition	Verint VIS-SV-PS	1		Running
	8.6	Interface to GPS enabled police cars	Verint VIS-SV-PS	1		Running
	8.7	Special Professional Services - Customized Professional services Interface to Vahan	Verint VIS-SV-PS	1		Running
	8.8	Special Professional Services - Customized Professional services SMS gateway integration	Verint VIS-SV-PS	1		Running
	8.9	Interface to video analytics	Verint VIS-SV-PS	1		Running
	8.10	Installation services	Verint VIS-INST-PS	50		Running
	9	Nextiva Project Management – Verint professional services for project management.	NEX-5-PM-PS	50		Running
	10	Gold Maintenance Charges (PSIM)	-	1		Running
	11	Additional 4-year Gold Maintenance Charges	-	1		Running
	12	IP Telephony Application	Cisco	1		Running
External	1	Updated Base Map for Pune & Pimpri Chinchwad (min. 0.6 mtr resolution, min. 1:500) (GIS Software)	MapMyIndia	659.46 sq.km.		Running
	2	Viewing Software for GIS	MapMyIndia	1		Running
	3	Integration with CCTNS for Stolen Vehicles and Missing Persons data	CCTNS	1		Running
	4	Integration with Vahan	Vahan	1		Running
	5	Integration of 1 Identified Public Private Collaborative Monitoring	Collaborative Monitoring	1		Running

Annexure E: Existing Command Center and Viewing Centers Infrastructure

Command Center

Sr. No.	Description	Make	Model	Unit	Qty.
1	**Video Wall (along with hardware & software) (8 (4 x 2) Cubes)**	-	-	-	1
1.1	4 X 2 configuration of 70" diagonal rear projection Video Wall having FULL HD resolution.	Delta	DVS-7088R8 (LED LAMP)	Nos.	8
1.2	Controller offered with 8 outputs can take mirror images through Dual LAN, preloaded with wall management s/w. It has 20 Universal Inputs.	Delta	Icon Pro	Nos.	1
1.3	Mounting stand	Delta	-	Nos.	4
1.4	DVI Cable 10m to connect controller to Cubes	Delta	Custom	Nos.	8
1.4	Installation Charges for Video Wall	Allied Digital	-	Lumpsum	1
2	Monitoring Workstations (3 Monitors)	HP	Z220 CMT	Nos.	10
3	Network Color Laser Printers	HP	CP5525n	Nos.	2
4	IP Phones	Cisco	CP-7942G=	Nos.	5
5	Indoor Fixed Dome Cameras			Nos.	6
5	Indoor Fixed Dome Cameras	Dvtel	CM-3211-00	Nos.	6
5.1	PVC Junction Box for Dome Cameras (5" x 5")	Reputed	-	Nos.	6
6	**Switches**			Nos.	2
6.1	Layer 2 24 Port Managed Switch POE Switch	CISCO	WS-C2960S-24PS-L	LOT	2
7	**Router**				
7.1	ASR 1001 - (350 Mbps - 2.5 Gbps) Router	CISCO	ASR1001	LOT	2
8	Networking/IT Racks	WQ	-	LOT	1
9	Networking Cost (Passive Components) (Pl. specify the details)	D-Link	-	LOT	1
9.1	Lan cabling per Node	-	-	Nos.	25
10	Civil Work (Give separate breakup for Masonry work, Flooring, Ceiling etc.)	-	-	LOT	1
11	Electrical Cabling & Necessary Illumination Devices	-	-	LOT	1
12	Fire Safety System with alarms	Ravel	-	LOT	1
13	Public Address System	Bosch	-	LOT	1
14	Access Control System (RFID/Proximity based, for all staff)	Solus	-	LOT	1
15	Partitioning System	-	-	LOT	1

Sr. No.	Description	Make	Model	Unit	Qty.
16	Office Workstations (Furniture and Fixtures)	-	-	LOT	1
17	Comfort AC	Samsung	AR24HV2USNBNNA	LOT	1
18	UPS (30 Minutes Backup)	Eaton	Eaton	Nos.	1
19	DG set	KOEL Green	KGL-40WS/40	Nos.	1

PMC & PCMC Viewing Center

Sr. No.	Description	Make	Model	Unit	Qty.
1	LED TV	LG	42WS10	Nos.	2
2	Monitoring Workstations (2 Monitors)	HP	Z220 CMT	Nos.	2
3	Network Colour Laser Printers	HP	CP5525n	Nos.	1
4	IP Phones	Cisco	CP-7942G=	Nos.	1
5	Switches	-	-	-	-
5.1	Layer 3 Managed Switch POE Switch	CISCO	WS-C3560CG-8PC-S	Nos.	1
6	Router			Nos.	1
6.1	ISR G2 Router 3925 for Speeds (for above 75 Mbps)	CISCO	CISCO3925-SEC/K9	Nos.	1
7	Networking Cost (Passive Components)	-	-	-	-
7.1	Lan cabling per Node	Reputed	-	Nos.	4
7.2	Electrical cabling	Reputed	-	Lot	1
8	Chairs and Computer Tables	Reputed	-	Nos.	2
9	Comfort AC	Samsung	AR24JC3TFUR	Nos.	1
10	UPS (30 Minutes Backup)	Eaton	9145 Series 1000 IN	Nos.	1

2 Viewing Centers Additional CP Offices

Sr. No.	Description	Make	Model	Unit	Qty.
1	LED TV	LG	42WS10	Nos.	2
2	Monitoring Workstations (1 Monitors)	HP	Z220 CMT	Nos.	2
3	Electrical Cabling	-	-	LOT	1
4	IP Phones	CISCO	CP-7942G	LOT	1
5	Computer Table and Chairs	Customized	Customized	LOT	1
6	**Networking cost (Active & Passive Components)**	-	-	-	-
6.1	Lan cabling per Node	Reputed	-	Nos.	3
6.2	Switch	-	-	Nos	1
6.2.1	Layer 3 8 Port Managed Switch POE Switch	CISCO	WS-C3560CG-8PC-S	Nos	1
6.3	Router	-	-	Nos	1
6.3.1	ISR G2 Router 3925 for Speeds (for above 75 Mbps)	CISCO	CISCO3925-SEC/K9	Nos	1
7	UPS as per requirement with 30 Min Backup	Eaton	9145 Series 2000 IN XL	Nos	1

Viewing Centre at Collectorate, 4 Zonal DCP Offices and 40 Police Stations

Sr. No.	Description	Make	Model	Unit	Qty.
1	LED TV	LG	42WS10	nos.	1
2	Monitoring Workstations (1 Monitor)	HP	Z220 CMT	nos.	1
3	Electrical Cabling	Reputed		nos.	1
4	IP Phones	CISCO	CP-7942	nos.	1
5	Computer Tables and Chairs	Customized	Customized	nos.	1
6	**Networking Cost (Active and Passive Components)**				
6.1	Lan cabling per Node	Customized	-	Nos.	2
6.2	**Switch**			nos.	1
6.2.1	Layer 3 Managed Switch POE Switch	CISCO	WS-C3560CG-8PC-S		1
6.3	**Router**			nos.	1
6.3.1	ISR G2 Router 3925 for Speeds (for above 75 Mbps)	CISCO	CISCO3925-SEC/K9		1
6.3.2	2951 router for less than 75 maps	CISCO	CISCO2951-SEC/K9		1
7	UPS as per requirement (for 30 minutes back up)	Eaton	9145 Series 2000 IN XL	nos.	1

5. Annexure III - Locations List

Annexure A: List of Existing Locations

Existing CCTV Count as a part of PCS Project	
PoliceStation	**Count of Cameras**
Zone 1	**346**
Deccan	49
Faraskhana	57
Khadak	58
Samarth	37
Shivaji Nagar	47
Vishrambaug	98
Zone 2	**199**
Bharti Vidhyapeeth	28
Bundgarden	49
Koregaonpark	27
Lashkar	44
Sahakarnagar	17
Swargate	34
Zone 3	**108**
Alankar	6
Dattawadi	32
Kothrud	37
Uttam Nagar	4
Warje Malwadi	29
Zone 4	**233**
Chandan Nagar	13
Chaturshringi	87
Khadki	19

Existing CCTV Count as a part of PCS Project	
PoliceStation	**Count of Cameras**
Vimantal	40
Vishrantwadi	28
Yerawada	46
Zone 5	**191**
Bibwewadi	15
Hadapsar	44
Kondhawa	49
Marketyard	17
Mundhwa	29
Wanavadi	37
Pimpri Chinchwad Area	**264**
Bhosari	31
Chinchwad	20
Hinjewadi	22
Midc Bhosari	33
Nigdi	55
Pimpri	44
Sangvi	51
Wakad	08
Grand Total	**1341**

Police station wise list of locations is provided on the next page.

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
	Zone 1		**0**	**272**	**74**	**346**	
	Deccan		**0**	**35**	**14**	**49**	
1	Apte road shreyas hotel	18.5197750,73.843421	0	2	0	2	17/09/2014
2	Athwale chowk law college road pune	18.510583,73.829567	0	1	1	2	01/01/2015
3	Bhide bridge(drawing merged with Deccan PMT)	18.515886,73.845798	0	0	1	1	08/09/2014
4	Deccan PMT near auto stand J.M. road	18.5160920,73.842735	0	2	1	3	10/09/2014
5	Ferguson college gate f.c.road pune	18.522800,73.841033	0	1	0	1	25/12/2014
6	Ghodke chowk Prabhat road	18.5144510,73.833854	0	2	1	3	30/01/2015
7	Good luck Chowk f.c.road pune	18.5175780,73.841751	0	3	1	4	01/01/2015
8	Kamla Nehru Park Near A.T.M. center Bhandarkar road pune	18.516917,73.834033	0	1	1	2	05/01/2015
9	KFC hotel j.m.road pune	18.516250,73.8425	0	1		1	17/09/2014
10	Khanduji baba chowk	18.514650,73.842233	0	4	1	5	16/01/2015
11	Lane no.15,bhandarkar road	18.5191970,73.830146	0	1	1	2	05/01/2015
12	Nalstop chowk karve road pune	18.508617,73.83115	0	2	1	3	25/12/2014
13	Near Vinayak Mitra Mandal Bhandarkar	18.5180810,73.83773	0	2	0	2	05/01/2015
14	Near fashion zone shop opp. mcdonld j.m.road pune Z-Bridge	18.5185680,73.844467	0	0	1	1	10/09/2014
15	Near Marathwada college B.M.C.C road pune	18.519867,73.838167	0	2	0	2	24/01/2015
16	Opp.Bata shop j.m.road pune	18.5170550,73.84317	0	2	0	2	08/09/2014
17	Petrico Hotel near dagadi Building j.m.road pune	18.5193000,73.845008	0	1	0	1	03/11/2014
18	Prabhat Junction Law college road pune	18.515467,73.828817	0	1	1	2	15/12/2014
19	Ras shala chowk karve road pune	18.511800,73.837317	0	2	0	2	16/01/2015
20	Roopali hotel f.c.road pune	18.519483,73.8412	0	0	1	1	05/01/2015
21	Sambhaji brige f.c.road	18.515883,73.842	0	1	1	2	15/12/2014
22	Savarkar stachu opp. Garvare shool karve	18.5129790,73.840316	0	1	0	1	16/01/2015
23	Shubham Hotel J.M. road Pune	18.520563,73.846252	0	1	0	1	26/11/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
24	V.s.khandekar chowk s.b.road pune	18.5201870,73.830413	0	2	0	2	05/01/2015
25	Vaishali hotel f.c.road pune	18.5207060,73.841178	0	0	1	1	01/12/2014
	Faraskhana		**0**	**44**	**13**	**57**	
26	Phadake haud chowk	18.518921,73.859777		2	1	3	10/09/2014
27	Belbaug Chowk Laxmi Road	18.5156400,73.856254		2	1	3	15/08/2014
28	Budhwar chowk, Shivaji road	18.5149590,73.85733		3	1	4	16/08/2014
29	Devji Baba Chowk Laxmi Road	18.5190040,73.861526		2		2	14/08/2014
30	Dulya Maruti Chowk Laxmi Road	18.516150,73.862717		2		2	14/08/2014
31	Gadgil Chowk Shivaji Road	18.521900,73.855731		2	1	3	08/09/2014
32	Gotiram Bhaiya Chowk Shivaji Road	18.5127390,73.857574		3		3	16/08/2014
33	Hamzekhan Chowk Laxmi Road	18.5159810,73.861831		2		2	14/08/2014
34	Juna Bazar	18.527850,73.86325		2	1	3	08/09/2014
35	Kagdipura Chowk	18.5240260,73.859924		2		2	08/09/2014
36	Kasat Chowk (Manish Market)	18.513733,73.860167		2		2	10/09/2014
37	Kranti Hotel Chowk.	18.514983,73.857467		2	2	4	14/08/2014
38	Kumbharves Chowk '(Gadgil Chowk to R.T.O.)	18.524083,73.857917		2		2	10/09/2014
39	Lal Mahal Chowk Shivaji Road	18.5184720,73.856147		2	1	3	16/08/2014
40	Moti Chowk (Raviwar Peth Area)	18.5169350,73.859382		2		2	08/09/2014
41	Pawale Chowk (Kasba Peth Area)	18.5219840,73.858268		2	1	3	26/11/2014
42	Rameshwar Chowk Shivaji Road	18.5134980,73.857147		2	1	3	16/08/2014
43	Sonya Maruti Chowk Laxmi Road	18.5160230,73.85942		3	1	4	15/08/2014
44	Tambaoli Masjid Chowk Laxmi Road	18.5160670,73.85997		3	1	4	15/08/2014
45	Vijay Maruti Chowk Laxmi Road	18.516037,73.858236		2	1	3	14/08/2014
	Khadak		**0**	**47**	**11**	**58**	
46	Babu mamdi chowk lohiyanagar	18.504917,73.86615		2		2	21/11/2014
47	Bismilla Hotel, Ghorpade Peth	18.5057460,73.862617		5	2	7	26/11/2014
48	Chamanshah, chowk, bhawa I peth	18.505633,73.870667		3	1	4	03/11/2014
49	Chudaman chowk, bhawani peth	18.508117,73.875267		2	1	3	26/11/2014
50	Dalal chowk	18.5047230,73.862278		3	1	4	22/03/2018
51	Dhammapal Chowk Harka Nagar	18.5073180,73.872413		1	1	2	21/11/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
52	Ekbote colony, Ghorpade Peth	18.501600,73.866183		1	1	2	08/09/2014
53	Juna Motor Stand Bhawani Peth	18.5094940,73.872238		2		2	10/09/2014
54	Maharana pratap chowk, bajirao road	18.5079680,73.853775		2		2	03/02/2014
55	Mahatama Phule Wada Ganj Peth	18.5060440,73.864166		2		2	08/09/2014
56	Mamlear kacheri, shivaji road, pune	18.506583,73.858067		1	1	2	03/02/2014
57	Mase Ali, Ganj Peth	18.5089070,73.86325		5		5	15/09/2014
58	Nanaware chowk, ghorpade peth	18.5017100,73.863548		2	1	3	12/08/2014
59	Natubag junction, bajirav road, pune	18.510567,73.8538		2		2	12/08/2014
60	Old ramoshi gate chowki, neharu road	18.5104770,73.868392		2		2	12/08/2014
61	Phadget Chowky Shivaji Road	18.5084380,73.857925		3		3	10/09/2014
62	RashtraBhushanChowk,shivajiroad,pune	18.503933,73.858983		2	1	3	12-Aug-14
63	Seven love chowk,shankarshet road, pune	18.501650,73.869133		3	1	4	12/11/2014
64	Shivaji maratha high school ,shukrawar peth	18.506717,73.856533		1		1	12/08/2014
65	Sinha Garage Chowk, Ghorpade Peth	18.5038100,73.861373		2		2	15/08/2014
66	Sonawane hospital	18.5061910,73.868194		1		1	15/09/2014
	Samarth		**0**	**30**	**7**	**37**	
67	A d camp chowk	18.513167,73.868717		3	1	4	26/04/2015
68	Apolo chowk	18.5195270,73.864881		2	1	3	01/06/2015
69	Inamdar chowk to ghodepir chowk	18.5155820,73.864459		4	1	5	06/06/2015
70	M.s.e.b chowk	18.5198920,73.868553		4	1	5	15/03/2015
71	Mahatmaphule chowki chowk	18.516267,73.871		3	1	4	20/05/2015
72	Nana chawadi to sant kabir chowk	18.515533,73.868683		4	1	5	16/03/2015
73	Narpatgiri chowk	18.5229800,73.867237		3	1	4	22/04/2015
74	Palkhi vithoba mandir chowk	18.512500,73.866417		2		2	18/03/2015
75	Shantai hotel chowk	18.520050,73.871033		2		2	16/03/2015
76	Somwarpeth daruwala pul	18.5195350,73.862272		3		3	06/03/2015
	Shivaji Nagar		**0**	**38**	**9**	**47**	
77	Ayangaar Institute	18.533717,73.8386		2		2	05/01/2015
78	B. D. D. S.	18.526067,73.847067		1		1	26/11/2014
79	Balgandharav Chowk	18.5226440,73.84822		3	1	4	17/09/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
80	Central Mall Chowk	18.5347250,73.83898		2		2	10/09/2014
81	Dengale Bridge	18.525367,73.856917		2		2	12/11/2014
82	dyaneshwar paduka chowk	18.5269550,73.842275		2		2	17/09/2014
83	Engineering College Chowk	18.516301,73.858045		3	1	4	21/08/2020
84	Hare Krishna Mandir Road	18.530033,73.837183		2		2	03/11/2014
85	Kude Chowk	18.522867,73.853117		2	1	3	08/09/2014
86	Mangala Talies Chowk	18.523400,73.854717		2	1	3	15/09/2014
87	Omkareshwar Ghat Rasta	18.5221500,73.850608		2		2	12/11/2014
88	P. M. C	18.5246960,73.85321		1	1	2	14/12/2014
89	Police H. Q. (Gate)	18.5308660,73.84552		2		2	21/11/2014
90	S. G. Barave Chowk	18.5269960,73.850966		2	1	3	08/09/2014
91	Sakhar Sankul	18.5345420,73.848014		1		1	26/11/2014
92	Shivajingar Court	18.527167,73.85565		4	1	5	01/12/2014
93	Simla office Chowk	18.5301510,73.849948		3	1	4	17/09/2014
94	Vir chafekar chowk	18.531617,73.84475		2	1	3	01/01/2015
	Vishrambaug		**0**	**78**	**20**	**98**	
95	A.B.Chowk, Kelkar road	18.516567,73.853783		4	1	5	10/09/2014
96	Extension of Bhide pull	18.5147600,73.845619		3	1	4	31/08/2017
97	Fadtare Chowk, kumthekar road	18.5135780,73.85234		6		6	15/09/2014
98	Ganjave chowk, Shastri road	18.5095150,73.843566		2	1	3	10/09/2014
99	Ganpati chowk	18.5150380,73.855057		2	1	3	04/09/2017
100	Holkar chowk	18.5133050,73.845382		4	1	5	30/08/2017
101	Kunjir talim chowk	18.5134860,73.850341		2	1	3	02/09/2017
102	Mandai	18.511883,73.856		3	1	4	08/09/2014
103	Narayan peth police chowky	18.5161600,73.847999		2		2	31/08/2017
104	New english schoool	18.5113250,73.845158		3	1	4	01/09/2017
105	Omkareshwar Chowk, Shaniwar peth	18.519917,73.849783		2		2	15/09/2014
106	Poona Hospital	18.511250,73.842033		2		2	15/09/2014
107	R.S.S. office Motibag shaniwar peth	18.518983,73.851683		2	1	3	15/09/2014
108	Sahitya parishad chowk ,tilak road	18.5105010,73.845954		3	1	4	15/09/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
109	Sakal press	18.5193330,73.854278		2		2	03/09/2017
110	Senadatta chowk,shashtri road	18.505100,73.84475		3	1	4	08/09/2014
111	Sevasadan chowk	18.514633,73.853733		4	1	5	10/09/2014
112	Shanipar Chowk, bajirao road	18.513350,73.854		4	1	5	08/09/2014
113	Shaniwar Wada	18.5203640,73.854454		1	1	2	10/09/2014
114	SP college	18.507831,73.850461		4	1	5	30/08/2017
115	Takli haveli chowk	18.5163130,73.849906		4	1	5	31/08/2017
116	Tambadee Jogeshwari Lane	18.5167160,73.854881		1		1	15/09/2014
117	Tilak chowk/ Alka Takiz chowk	18.5129280,73.843719		4	1	5	08/09/2014
118	Tilak smarak	18.5087010,73.849151		2		2	01/09/2017
119	Tilak Statue Chowk, Mandai	18.513433,73.856217		4	1	5	08/09/2014
120	Umbrya ganpati	18.5142100,73.850082		3	1	4	02/09/2017
121	Vijay talkies	18.5139200,73.847243		2	1	3	01/09/2017
	Zone 2		**6**	**160**	**33**	**199**	
	Bharti Vidhyapeeth		**6**	**17**	**5**	**28**	
122	B.vidhypith main gate	18.4580660,73.858062		1	1	2	15/07/2015
123	Gokul nagar	18.4496310,73.874298		2	1	3	02/07/2015
124	Leketown society	18.4589630,73.863408		1	1	2	30/06/2015
125	PMPL Depot Katraj	18.447483,73.858633	2	4	1	7	17/06/2015
126	Rajaram gas agency chowk	18.454633,73.853583		1	1	2	21/06/2015
127	Rajiv gandhi zoo	18.454383,73.858767		3		3	09/07/2015
128	Santosh nagar bhaji market (KATRAJ CHOWKY, PUNE–SATARA ROAD)	18.446699,73.859183		1		1	24/06/2015
129	Santosh nagar bhaji market (ANPR -KATRAJ CHOWKY, PUNE–SATARA ROAD)		4			4	
130	Shelar mala(katraj talao)	18.445800,73.864017		2		2	28/06/2015
131	Trimurthi chowk	18.457108,73.850442		2		2	11/07/2015
	Bundgarden		**0**	**42**	**7**	**49**	
132	Alankar cinema chowk	18.528250,73.876333		3		3	15/03/2015
133	Blue nile chowk	18.521483,73.877333		3		3	24/02/2015
134	Bolai chowk	18.523900,73.869983		4	1	5	24/02/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
135	Collector kacheri chowk	18.525380,73.872058		3	1	4	01/05/2015
136	Dhole patil ganpati chowk	18.5350170,73.8763		2		2	20/05/2015
137	Iskcon tempal chowk	18.520347,73.880282		2		2	12/04/2015
138	Juna Bazar	18.527850,73.86325		3		3	08/09/2014
139	Kirad chowk	18.522100,73.8761		3	1	4	06/03/2015
140	Maldhakka chowk	18.525417,73.86735		4	1	5	09/03/2015
141	Neharu memorial hall chowk	18.519939,73.87539		3		3	13/05/2015
142	Pune railway station	18.5283750,73.874239		2	1	3	09/06/2015
143	Pune s.t.stand	18.527367,73.872217		2		2	09/03/2015
144	R.t.o chowk	18.5298540,73.863184		3	1	4	06/03/2015
145	Rajguru chowk	18.5346450,73.873707		2		2	15/03/2015
146	Vidhan bhavan chowk	18.524483,73.87855		3	1	4	09/03/2015
	Koregaonpark		**0**	**24**	**3**	**27**	
147	Hemant karkare chowk	18.5400190,73.883044		1		1	17/06/2015
148	I.b.bangala chowk	18.5284670,73.879303		3		3	24/06/2015
149	Jahangir chowk	18.530850,73.875617		3	1	4	13/05/2015
150	K.p.chowk n.m. road	18.539467,73.8855		3	1	4	28/06/2015
151	Mahatma gandhi garden chowk	18.5415790,73.885185		3		3	02/07/2015
152	Mobaz chowk	18.5348010,73.879078		3	1	4	21/06/2015
153	Morawada bridge	18.523119,73.887781		3		3	30/06/2015
154	New circuit house	18.527201,73.886185		2		2	05/07/2015
155	Old circuit house	18.528483,73.885367		3		3	09/07/2015
	Lashkar		**0**	**35**	**9**	**44**	
156	Babajan chowk	18.511450,73.87645		3	1	4	24/02/2015
157	Bhopale chowk	18.511367,73.877817		3		3	09/03/2015
158	Fashon street	18.509333,73.879233		3	1	4	22/02/2015
159	Kohinoor chowk	18.511517,73.878767		3		3	24/01/2015
160	Kureshi masjid	18.508550,73.8778		3	1	4	09/06/2015
161	Lashkar water supply	18.5087280,73.8832		2	1	3	24/02/2015
162	M.Gandhi bus stand	18.5065020,73.881836		3	1	4	22/02/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
163	Mahavir chowk	18.547017,73.878817		3	1	4	30/01/2015
164	Naz chowk	18.517550,73.878917		3		3	24/01/2015
165	Shivaji market	18.513700,73.87725		3	1	4	15/03/2015
166	Try luck chowk	18.507183,73.87855		3	1	4	24/07/2015
167	West end chowk m g road	18.518417,73.87925		3	1	4	06/03/2015
	Sahakarnagar		**0**	**13**	**4**	**17**	
168	Ahilya devi chowk	18.4684180,73.858073		2	1	3	24/06/2015
169	Ellora Palace Satara road	18.4656180,73.858205		1		1	21/06/2015
170	Gulab nagar chowk	18.475288,73.858339		1	1	2	05/07/2015
171	Last bus stop dhankawadi	18.4654490,73.850333		1	1	2	18/07/2015
172	Padmavati pumping station	18.476300,73.856233		2		2	09/07/2015
173	Shahu bank chowk	18.464703,73.858077		2		2	28/06/2015
174	Tin hatthi chowk	18.468633,73.852933		2	1	3	02/07/2015
175	Tresure park	18.481017,73.854183		2		2	30/06/2015
	Swargate		**0**	**29**	**5**	**34**	
176	Ganeshkala krida rangmanch	18.5008490,73.855456		2		2	08/06/2015
177	Giridhar bhuvan chowk	18.493033,73.869283		4	1	5	22/07/2015
178	Jedhe Chowk	18.5006800,73.85885		5	2	7	02/09/2017
179	Mitramandal chowk	18.4972570,73.854309		2		2	21/06/2015
180	P.m.t vehicle depot swargate	18.4965200,73.868317		3		3	10/04/2015
181	Puram chowk	18.5054460,73.853421		2		2	24/06/2015
182	Sarasbaugh chowpati	18.502367,73.851967		3	1	4	19/07/2015
183	Sarasbaugh ganesh mandir	18.5023830,73.853402		3		3	15/07/2015
184	Shani mandir chowk	18.4981210,73.849532		2		2	17/06/2015
185	Swargate S.T.stand	18.4998510,73.856514		3	1	4	11/07/2015
	Zone 3		**8**	**81**	**19**	**108**	
	Alankar		**0**	**5**	**1**	**6**	
186	Dahanukar circle	18.495433,73.813167		1		1	01/12/2014
187	Kokan express hotel	18.498383,73.813767		1		1	28/11/2014
188	Mhatare bridge	18.5052470,73.835242		2	1	3	28/11/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
189	Mritunjay mandir	18.5029140,73.818367		1		1	01/12/2014
	Dattawadi		**4**	**23**	**5**	**32**	
190	Dandekar pul chowk	18.5009260,73.845715		1	1	2	30/06/2015
191	L.r.highschool chowk	18.485183,73.8498		2	1	3	25/03/2017
192	Lakshmi narayan chowk	18.496650,73.858117		2		2	15/05/2015
193	Mangir baba chowk	18.502733,73.84535		3	1	4	02/07/2015
194	Muktagan collage chowk	18.4898420,73.854219		3		3	19/07/2015
195	Navratna hotel chowk	18.4857170,73.84997		2	1	3	22/07/2015
196	Panchami hotel chowk	18.492467,73.857883		1	1	2	09/07/2015
197	Panchshil chowk	18.4933160,73.854226		2		2	05/07/2015
198	Parvati jal kendra panmala	18.4973210,73.839731		2		2	28/06/2015
199	Surabhi mangal karyalay	18.487683,73.85395		2		2	11/07/2015
200	Ustav chowk	18.489350,73.8574		3		3	20/03/2015
201	Rajaram bridge	18.489017, 73.826750 18.489450, 73.826983	4			4	
	Kothrud		**0**	**27**	**10**	**37**	
202	Anand nagar	18.5100230,73.815582		2	1	3	15/12/2014
203	ARAI (shifted to kelewadi chowk)	18.507550,73.8232		2		2	01/12/2014
204	City pride	18.499067,73.820717		2		2	28/11/2014
205	Doordarshan	18.5169420,73.817543		2		2	15/12/2014
206	Fly over bridge	18.505800,73.82525		2	1	3	28/11/2014
207	Karishma soc.	18.504350,73.8211		1	1	2	28/11/2014
208	Karve putala	18.502100,73.8153		2	1	3	15/12/2014
209	Kelewadi Pumping	18.510600,73.823517		2		2	28/11/2014
210	Kinara chowk	18.5086670,73.807743		1	1	2	15/08/2014
211	Kothrud bus depot	18.508100,73.7948		2	1	3	28/11/2014
212	Mahadeo Mandir	18.505850,73.832833		1	1	2	15/12/2014
213	Mangeashkar Hospital	18.503167,73.831767		1		1	15/12/2014
214	Mayur colony , HDFC bank	18.504833,73.817867		2		2	01/12/2014
215	Nal stop	18.508500,73.830983		2	1	3	16/08/2014

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
216	Right Bhusari Colony	18.508161,73.788816		1	1	2	27/08/2020
217	Shivaji putala	18.502267,73.813583		1	1	2	15/12/2014
218	Sutar hospital	18.5039500,73.807807		1		1	28/11/2014
	Uttam Nagar		**0**	**4**	**0**	**4**	
219	Kondawa Gate N.D.A. road	18.454550,73.766983		2		2	23/03/2015
220	Uttam Nagar Market, N.D.A. road	18.464750,73.7735		2		2	18/11/2014
	Warje Malwadi		**4**	**22**	**3**	**29**	
221	Bapugi Buwa Chowk Ram Nagar	18.475633,73.80335		2		2	28/11/2014
222	dukkar khind	18.495867,73.787717		2		2	15/04/2015
223	Ganapati Matha	18.475750,73.795667		2		2	28/11/2014
224	Karve Nagar Chowk	18.489000,73.814517		2		2	28/11/2014
225	Old Octroi Naka (Ambedkar chowk)	18.483883,73.80985		3	1	4	15/12/2014
226	Opposite Mai Mangeshkar Hospital, Mumbai – Bangalore Highway (Entry point for service road)	18.484517,73.799933		2		2	20/10/2014
227	Rajaram Bridge	18.489017,73.82676		3	1	4	15/12/2014
228	Spencer Chowk	18.4923800,73.821825		2		2	01/12/2014
229	Under Warje Bridge	18.4801310,73.804656	4	4	1	9	26/04/2015
	Zone 4		**30**	**166**	**37**	**233**	
	Chandan Nagar		**0**	**10**	**3**	**13**	
230	Radisson Hotel, Chowk Kharadi	18.552500,73.9375		3	1	4	23/07/2015
231	Sangharsh Chowk, chandan nagar	18.559400,73.931167		2	1	3	01/06/2015
232	Shivaji Chowk, chandan nagar	18.5619720,73.93109		3	1	4	08/06/2015
233	Vegetable market Vadgoansheri	18.539500,73.9205		2		2	28/06/2015
	Chaturshringi		**8**	**68**	**11**	**87**	
234	Ambedkar Chowk, Aundh	18.560200,73.8031		3		3	08/03/2015
235	Arun kadam Chowk Janwadi	18.529400,73.824833		3	1	4	24/06/2015
236	Balewadi Octroi Naka	18.574817,73.76395		3		3	07/02/2015
237	Balewadi Phata	18.563467,73.782833		3		3	07/02/2015
238	Baner Pashan Link Road	18.555067,73.7977		2		2	15/02/2015
239	Baner Phata	18.549983,73.8095		3	1	4	07/02/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
240	Breman Chowk Aundh Pune	18.5602600,73.814953		4	1	5	05/07/2015
241	Chatushrungi Mata Mandir	18.539250,73.829833		2		2	22/07/2015
242	Deep Bungalow Chowk Pandavnagar Pune	18.5299220,73.835433		4	1	5	30/06/2015
243	Hotel sadanand	18.568550,73.767	4			4	17/06/2015
244	Hotel Sadanand javel baner Road	18.5693520,73.767366		2		2	16/03/2015
245	International Student Hosstel Pune Vidhyapeeth	18.557117,73.822417		2		2	15/02/2015
246	Opp Medipoint Hospital	18.565583,73.802567		2		2	07/02/2015
247	Ozan Moll samor ITI Road Pune	18.5557060,73.809264		3	1	4	11/07/2015
248	Parihar Chowk Aundh Pune	18.560500,73.80905		4	1	5	09/07/2015
249	Piramid Chowk Senapati Bapat Road	18.5316500,73.830436		4	1	5	02/07/2015
250	Pune Vidhyapeeth Chowk	18.5427530,73.828774		5	2	7	15/07/2015
251	Rajiv Gandhi Bridge, Aundh	18.566783,73.81175		2		2	22/02/2015
252	Shetaki Mahamandal Chowk	18.525267,73.829733		4		4	19/07/2015
253	Shivaji Chowk Pashan	18.537833,73.796517		4	1	5	15/02/2015
254	Suskhind	18.546350,73.77455	4	3		7	15/02/2015
255	Sutarwadi Junction	18.543517,73.785133		2		2	16/03/2015
256	Vetal Baba Chowk	18.5290260,73.829986		4	1	5	28/06/2015
	Khadki		**0**	**17**	**2**	**19**	
257	Adjacent to Holkar Bridge	18.556833,73.859067		1		1	24/06/2015
258	Ambedkar Chowk, Aundh Road	18.562550,73.830233		2	1	3	17/06/2015
259	Bopodi Chowk	18.5718350,73.837124		2	1	3	10/05/2015
260	Church Chowk, Opp. Police Station	18.5574020,73.84643		4		4	06/06/2015
261	Egg Hatching Chowk	18.5491540,73.850934		3		3	24/07/2015
262	Holkar Bridge	18.554617,73.8599		2		2	21/06/2015
263	MSEB Chowk	18.5651400,73.850287		2		2	15/07/2015
264	Road towards Ammunition Factory	18.562733,73.852333		1		1	28/06/2015
	Vimantal		**10**	**22**	**8**	**40**	
265	509 Chowk, Viman Nagar	18.575367,73.8994		2	1	3	13/07/2017
266	Ahmednagar toll naka	18.564350,73.9432	4			4	24/07/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
267	Airport ingate / parking	18.5780020,73.908034		3	1	4	25/10/2017
268	Dutta mandir Chowk	18.5653170,73.913869		2	1	3	09/06/2015
269	Ganpati mandir Chowk	18.567433,73.918583		2		2	24/06/2015
270	Kharadi bypass junction	18.562200,73.938783		3	1	4	24/07/2015
271	Nexa showroom	18.562308,73.906951	2	2	1	5	22/10/2020
272	Nico garden Kashmir Chowk	18.5618490,73.913959		2	1	3	21/06/2015
273	Petrol pump Chowk	18.532590,73.933014	4	2	1	7	25/10/2017
274	Shrikrishna Chowk	18.5651470,73.918108		2		2	28/06/2015
275	Suzuki showroom Chowk	18.560683,73.918083		2	1	3	17/06/2015
	Vishrantwadi		**4**	**19**	**5**	**28**	
276	Bharat Dhaba, Dhanori	18.593483,73.8925		2		2	10/12/2016
277	Chandrama Chowk	18.5486260,73.871117		3	1	4	30/06/2015
278	Dighi chowk	18.633783,73.8748	2			2	19/07/2015
279	BANACHA ODHA, ALANDI–BHOSARI ROAD	18.633783,73.8748	2			2	19/07/2015
280	Mental Corner Chowk	18.5652880,73.878104		3	1	4	02/07/2015
281	Navrang plaza, Ektanagar	18.573233,73.88395		2	1	3	06/04/2017
282	R and DE	18.578400,73.876317		2		2	19/07/2015
283	Sambhaji Chowk	18.575900,73.876383		3	1	4	09/07/2015
284	Vishrantwadi Chowk	18.5725050,73.878352		4	1	5	05/07/2015
	Yerawada		**8**	**30**	**8**	**46**	
285	Chitra Chowk Vegetable market	18.548133,73.885783		3	1	4	11/07/2015
286	Gold adlabs Chowk	18.545850,73.906133		4	1	5	24/06/2015
287	Golf Club Chowk	18.5524240,73.889631	4	3	1	8	05/07/2015
288	Gunjan Chowk	18.544850,73.8847		3	1	4	02/07/2015
289	Kranti Chowk Laxmi nagar	18.549417,73.916783		2	1	3	15/07/2015
290	Opp yerwada jail	18.5603010,73.891922		3	1	4	19/07/2015
291	Parnkuti Chowk	18.545033,73.884717		4	1	5	30/06/2015
292	Road coming from New sangamwadi Bridge	18.544817,73.877167		3		3	09/07/2015
293	Shastrinagar Chowk	18.552433,73.896717	4	3		7	17/06/2015
294	Vadgoansheri jn on nagar road	18.558633,73.911533		2	1	3	21/06/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
	Zone 5		**20**	**132**	**39**	**191**	
	Bibwewadi		**0**	**12**	**3**	**15**	
295	Anna bhau sathe chowk	18.478800,73.861367		2		2	09/06/2015
296	Bhagali hospital chowk	18.474833,73.8578		2	1	3	17/06/2015
297	Mahesh society	18.468883,73.864133		2	1	3	19/07/2015
298	Pasalkar chowk	18.461283,73.87765		3		3	21/06/2015
299	PMPL depot Indiranagar Bibvewadi	18.461117,73.8712		2		2	18/03/2015
300	Upper Indira Nagar last bus stop	18.460967,73.867483		1	1	2	22/07/2015
	Hadapsar		**6**	**31**	**7**	**44**	
301	No.15 Manjri phata /15 number phata (ANPR)	18.575141,73.808638 18.499450,73.951483	4	2		6	05/07/2015
302	Adj to Vidya petrol pump	18.5040180,73.921325		2	1	3	21/06/2015
303	Amnora park junction	18.5186530,73.932559		4	1	5	09/07/2015
304	A-Z Hotel junction	18.5239320,73.932375		3		3	19/07/2015
305	Gandhi Chowk	18.5020900,73.933598		3	1	4	24/06/2015
306	Kasai Hotel, D P Road	18.5111080,73.941514		2	1	3	11/07/2015
307	Magarpatta Chowk	18.502433,73.927517		3	1	4	17/06/2015
308	Mantarwadi phata	18.4660940,73.962078	2			2	15/09/2015
309	PMT Depot	18.500817,73.938567		4	1	5	28/06/2015
310	Ravi Darshan Bldg Chowk	18.5005860,73.943838		2		2	02/07/2015
311	Sadhana high school	18.5070510,73.937753		2		2	22/07/2015
312	Saraf katta hadapsar goan	18.5022500,73.93387		2	1	3	15/07/2015
313	Shani mandir chowk	18.4981210,73.849532		2		2	17/06/2015
	Kondhawa		**6**	**31**	**12**	**49**	
314	Ashirwad Chowk	18.4706510,73.888996		1	1	2	21/06/2015
315	Bhagawa Chowk	18.4747270,73.884373		2	1	3	02/07/2015
316	Bhagwan Hotel junction	18.4797270,73.902056		2		2	15/06/2015
317	Dorabji Chowk	18.4817880,73.896451		2	1	3	11/07/2015
318	Ganesh Chowk	18.4724730,73.886258		1	1	2	28/06/2015
319	Jyoti Chowk	18.4779600,73.890293		2	1	3	05/07/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
320	Kausar Bag	18.4732150,73.893392		2		2	22/07/2015
321	Khadi Mission Chowk	18.452567,73.891017	4			4	03/11/2017
322	Khan- E - Khas Chowk	18.4711230,73.886242		2	1	3	17/06/2015
323	N I B M Chowk	18.4831100,73.897775		2	1	3	09/07/2015
324	Navazish Chowk	18.4716600,73.883152		3	1	4	15/05/2015
325	Nyati Chowk	18.4672610,73.919689		2	1	3	15/07/2015
326	Pitashri Ashram	18.4729480,73.885331		2		2	30/06/2015
327	PMT bus Stop Bhagyoday nagar	18.4731290,73.88906		2	1	3	24/06/2015
328	Sec. No 42 Madarsa Baitul junction	18.4687070,73.887766		1	1	2	13/05/2015
329	Sheetal Chowk	18.4702090,73.889063		2	1	3	01/06/2015
330	Undri Chowk	18.4506760,73.916778	2			2	15/09/2015
331	ZamZam Hotel Chowk	18.4736800,73.89123		3		3	19/07/2015
	Marketyard		**0**	**12**	**5**	**17**	
332	Dr.ambedkar nagar zopadpatti	18.4842790,73.867542		2		2	09/04/2015
333	Gangadham chowk	18.480900,73.872233		3	1	4	19/07/2015
334	Magar statue	18.4895290,73.864315		3	1	4	11/07/2015
335	Marketyard chowk ustav hotel	18.489217,73.857533		1	1	2	15/07/2015
336	P.m.t depot marketyard	18.4839080,73.871527		2	1	3	22/07/2015
337	Wakhar mahamandal chowk	18.4893880,73.870124		1	1	2	04/02/2015
	Mundhwa		**4**	**21**	**4**	**29**	
338	Adj to B T kawade Rd Bridge	18.518550,73.906367		2		2	21/06/2015
339	Anant Theater/ Rly Crossing	18.5295170,73.902566		2		2	24/06/2015
340	Hadpsar Rly Station Road	18.5345730,73.929329		2		2	09/07/2015
341	Kalyani nagar jn, ABC Road	18.5390370,73.903651		3	1	4	28/06/2015
342	Phule Chowk Keshav nagar	18.5335070,73.933648		4	1	5	02/07/2015
343	Phule Chowk Keshav nagar(ANPR)	18.532631,73.932833	4			4	12/07/2020
344	Rly Gate Ghorpadi	18.5273690,73.897272		2		2	11/07/2015
345	Shivaji Chowk Keshav nagar	18.5321250,73.937062		2		2	05/07/2015
346	Tadigutta Chowk	18.5321740,73.917007		2	1	3	30/06/2015
347	Wad Chowk Ghorpadi Gaon	18.5282640,73.899882		2	1	3	17/06/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
	Wanavadi		**4**	**25**	**8**	**37**	
348	B T Kawade Rd Junction	18.506450,73.907067		2		2	05/07/2015
349	Bhairoba Nala Chowk	18.506683,73.8986		3		3	02/07/2015
350	Fatima Nagar chowk (ANPR)	18.506584,73.900857	4			4	26/07/2020
351	Golibar maidan Chowk	18.5030500,73.878697		4	1	5	30/06/2015
352	Jagtap Chowk	18.492217,73.9		2	1	3	03/08/2015
353	Lulla nagar kondhawa	18.485950,73.8889		3	1	4	20/05/2015
354	Ramtekdi junction	18.5063140,73.912777		2	1	3	09/07/2015
355	Ramtekdi water Plant	18.5007530,73.913885		2	1	3	26/07/2015
356	Saiyadnagar Rly Crossing Chowk	18.4888510,73.93205		2	1	3	19/07/2015
357	Shivarkar Road 10 sq building	18.502533,73.90075		2	1	3	02/08/2015
358	Sub Area Headquarters junction	18.521567,73.890883		3	1	4	28/02/2017
	Total		**64**	**811**	**202**	**1077**	

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
Pimpri Chinchwad area			**14**	**221**	**29**	**264**	
	Bhosari		**2**	**27**	**2**	**31**	
359	Alandi Road Chowk Bhosari	18.623100,73.851517		4		4	30/06/2015
360	Ambedkar puttala Dapodi	18.579333,73.830417		4		4	24/06/2015
361	Banacha Odha	18.6280700,73.866367	2			2	02/07/2015
362	Mayuri Mangal Karyalaya	18.625500,73.861717		2		2	19/07/2015
363	Nashik Phata / Kasarwadi Rly Stn	18.6071900,73.822459		6	1	7	09/07/2015
364	Nehru Chowk Dapodi	18.579967,73.834583		3		3	05/07/2015
365	PCMT Chowk	18.578377,73.83457		2		2	11/07/2015
366	PMT Chowk	18.622400,73.851067		2	1	3	15/07/2015
367	Shivaji puttala Dapodi	18.5788780,73.829429		2		2	28/06/2015
368	Solanki Jwellers Dhigi Road	18.6214500,73.851188		2		2	02/07/2015
	Chinchwad		**0**	**20**	**0**	**20**	

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
369	220 K.V.Power House Bijalinagar Chinchwad	18.640050,73.774467		2		2	31/01/2015
370	Axis Bank Ahinsa Chowk Chinchwad	18.632933,73.785217		4		4	07/02/2015
371	Empire Estate Pune Mumbai Road	18.6369810,73.796798		1		1	11/08/2017
372	Gurudwar Chowk Chinchwad	18.645567,76.766733		4		4	07/02/2015
373	MIDC Water Tank	18.641950,73.777617		1		1	04/02/2015
374	Morya Gosavi Mndir Chinchwad	18.6264800,73.779304		2		2	30/01/2015
375	Opp Talera Hospital	18.625350,73.784967		2		2	04/02/2015
376	Shradha garden	18.635350,73.790083		4		4	04/02/2015
	Hinjewadi		**4**	**17**	**1**	**22**	
377	Bhoomkar Chowk	18.6050830,73.751853		4		4	27/05/2015
378	Kalewadi Phata	18.6036310,73.777702		4	1	5	28/06/2015
379	Mamta nagar Jakat naka Opp. Indira Collage Express Highway	18.608167,73.75145		2		2	21/06/2015
380	Mankar Chowk Wakad Pune	18.589867,73.7733		3		3	17/06/2015
381	Wakad Chowky Javel Express Highway	18.5929430,73.756805	4	4		8	24/06/2015
	Midc Bhosari		**6**	**22**	**5**	**33**	
382	Balajinagar Zopadpatti	18.632550,73.83005		1	1	2	24/01/2015
383	Borhade wasti Juna Jakatnaka	18.6616170,73.848449		3		3	30/01/2015
384	C Ward Office	18.631850,73.826567		2		2	04/02/2015
385	Moshi Tol Naka	18.448400,73.863833	6	2	1	9	07/02/2015
386	Rakesh Sweet Mart Indrayaninagar	18.6353030,73.844022		3		3	24/01/2015
387	Shivaji puttala Landewadi	18.6199590,73.841732		3	1	4	04/02/2015
388	Visheweshwar Chowk MIDC	18.6428740,73.832171		4	1	5	07/02/2015
389	Wakhar mahamandal chowk	18.4893880,73.870124		4	1	5	04/02/2015
	Nigdi		**2**	**47**	**6**	**55**	
390	Akurdi Gaothan, Behind Ganga Vision	18.6510210,73.779389		3		3	06/06/2015
391	Akurdi Railway Station chowk	18.648783,73.764967		3	1	4	06/03/2015
392	Ambedkar Chowk, Sector 22	18.6701870,73.777451		1		1	10/05/2015
393	Anand Hospital Akurdi chowk	18.653817,73.764767		4	1	5	16/03/2015
394	Bhel chowk	18.655717,73.7713		4	1	5	22/02/2015
395	Chinchwad 'Railway Station	18.640567,73.791833		2		2	15/03/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
396	Ganga vision	18.6475290,73.785202		2		2	24/06/2015
397	Kandoba mal chowk	18.650317,73.785333		4	1	5	24/02/2015
398	Kasturi Market	18.6642220,73.795194		2		2	16/03/2015
399	Lokmanya 'Hospital chowk	18.655817,73.773433		3		3	09/04/2015
400	Mahalsakant Collage chowk	18.6492500,73.776114		4		4	15/03/2015
401	Thermax chowk	18.660483,73.79345		4	1	5	09/03/2015
402	Tilak chowk	18.658900,73.777333		4	1	5	08/03/2015
403	Triveninagar chowk	18.668133,73.785583	2	3		5	16/03/2015
404	Ward Office	18.656017,73.771017		2		2	25/03/2017
405	Water Purification Plant, Sector 23	18.664150,73.7649		2		2	21/06/2015
	Pimpri		**0**	**39**	**5**	**44**	
406	Ajmera Masulkar colony Mukhya Rasta	18.6330600,73.815994		2		2	02/07/2015
407	Baudhanagar Bhattnagar Corner	18.625433,73.79785		3		3	19/07/2015
408	Dr.Babasaheb Ambedkar Putala chowk Pimpri Pune	18.626383,73.805717		4	1	5	28/06/2015
409	Finolex Chowk	18.6299300,73.802932		4	1	5	10/04/2015
410	Kalewadi pullajavel Chowk	18.6187780,73.798591		2		2	22/07/2015
411	Karrachi Chowk	18.6213690,73.801628		3	1	4	11/07/2015
412	Kasarwadi	18.603000,73.825333		2		2	20/04/2015
413	Mahavir Chowk Pimpri	18.638717,73.794767		4	1	5	08/06/2015
414	Pimple Courts Road	18.637317,73.805167		2		2	17/06/2015
415	Pimpri Chinchwad mahanagar palika karyalay Pimpri	18.627717,73.80505		2		2	30/06/2015
416	Pimpri Railway Samoril Bhaji Market	18.624150,73.8009		3		3	15/07/2015
417	Shagun Chowk	18.622267,73.80195		4	1	5	09/07/2015
418	Vallabh Nagar	18.6137210,73.816791		2		2	21/06/2015
419	Yashwantrao Chavan Hosptial Praveshdwar & Parking	18.622500,73.820533		2		2	05/07/2015
	Sangvi		**0**	**41**	**10**	**51**	
420	Dore Chowk	18.574250,73.820917		3	1	4	09/07/2015
421	Ganpati Chowk Juni Sangavi	18.5726960,73.817192		4	1	5	09/07/2015
422	Kokanenagar Masjid javal	18.613267,73.7927		2	1	3	24/06/2015

SN	Location Name	Geo Location	ANPR	Fixed	PTZ	Total Cameras	Location Go Live
423	Kranti Chowk Navi Sangavi	18.582483,73.812617		3		3	11/07/2015
424	Krushana Chowk, Navi Sangavi	18.581767,73.815483		3		3	19/07/2015
425	Lucky Bakery Pimpri Pull Javel Kalewadi	18.6193520,73.796371		3		3	22/07/2015
426	Mhadev Mnadir Chowk Pimpri Pull Pimpale saudagar	18.601750,73.799333		3	1	4	10/04/2015
427	Pimpale Gurav Bus Stop	18.592967,73.819233		2	1	3	05/07/2015
428	Rakshak Chowk	18.582067,73.79745		4	1	5	11/07/2015
429	Sai Chowk	18.5790150,73.815971		2	1	3	30/06/2015
430	Sangvi Phata (ES Hosp)	18.573633,73.806117		4	1	5	02/07/2015
431	Shrusthi Chowk	18.596833,73.817317		3	1	4	09/06/2015
432	Vasantdada Putala Chowk Dapodi pull Juni Sangavi	18.5751850,73.825675		3		3	15/07/2015
433	Vishalnagar Chowk Pimpale Nilakh	18.584317,73.782333		2	1	3	21/06/2015
	Wakad		**0**	**8**	**0**	**8**	
434	16 Number Bus Stop Thergaon Pune	18.6065080,73.7732		2		2	15/05/2015
435	Birla Hosp Chowk	18.624567,73.775467		2		2	30/06/2015
436	Tapkir Chowk kalewadi	18.612333,73.795083		2		2	17/06/2015
437	Wakad bridge	18.592367,73.757067		2		2	30/08/2015
	Grand Total		**78**	**1032**	**231**	**1341**	

Annexure B: List of Proposed Locations

Proposed locations for augmentation of Pune City Surveillance Project is listed below:

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
Zone 1									
1	Vishrambaug	Shinde Paar chowk	18.51803806	73.850081	New Location	1	3	0	0
2	Vishrambaug	Dakshin mukhi maruti mandir	18.51836202	73.853787	New Location	1	2	0	0
3	Vishrambaug	Mandar Lodge kelkar road	18.51641636	73.852268	New Location	1	4	0	0
4	Vishrambaug	Modi ganpati	18.51528107	73.848316	New Location	1	2	0	0
5	Vishrambaug	Awshad gali	18.51219736	73.848647	New Location	1	3	0	0
6	Vishrambaug	Karpe Chowk	18.51103644	73.850727	New Location	1	4	0	0
7	Vishrambaug	Nimbalkar talim (Sujata mastani)	18.51188722	73.852142	New Location	1	3	0	0
8	Vishrambaug	Ramesh dine chowk	18.51264992	73.85255	New Location	1	2	0	0
9	Vishrambaug	Baurangari	18.51753177	73.855518	New Location	1	2	0	0
10	Vishrambaug	Tamble Jogeshweri	18.51681394	73.854904	Addl Cameras only	1	2	0	0
11	Vishrambaug	Shanivar wada entry exit	18.52036583	73.854456	Addl Poles and Cameras	2	4	0	0
12	Vishrambaug	Omkareshwer mandir	18.51977451	73.849614	Addl Poles and Cameras	2	2	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
13	Vishrambaug	existing bide bridge	18.51472693	73.845539	Addl Cameras only	0	1	0	0
14	Vishrambaug	Z bridge exit	18.51388543	73.844671	New Location	1	1	0	0
15	Vishrambaug	Alka chowk (tilak sambhaji chowk)	18.51281525	73.843811	Addl Poles and Cameras	4	0	0	7
16	Vishrambaug	Alka theater	18.51228268	73.843632	New Location	1	3	1	0
17	Vishrambaug	Dhramveer Sambhaji chowk	18.50837265	73.84398	New Location	1	3	0	0
18	Vishrambaug	Patrakar bavan	18.50923777	73.841925	New Location	1	3	0	0
19	Vishrambaug	Vaikunth Shamshan bhumi	18.50792229	73.84116	New Location	1	2	1	0
20	Vishrambaug	Tendulkar Garden	18.50454213	73.839907	New Location	1	3	0	0
21	Vishrambaug	Rajendra Nagar	18.50319373	73.842223	New Location	1	1	1	0
22	Vishrambaug	SPM english	18.50472955	73.847262	New Location	1	2	0	0
23	Vishrambaug	Lokmanya jogging park	18.50566111	73.84572	New Location	1	3	1	0
24	Vishrambaug	Vithal Mandir (navi peth)	18.50988207	73.846037	New Location	1	2	0	0
25	Vishrambaug	Maharashtra elecronics	18.51013131	73.846718	New Location	1	1	0	0
26	Vishrambaug	tilak smarak	18.5087254	73.849071	Addl Cameras only	1	1	1	0
27	Vishrambaug	bhave high school	18.51136613	73.847545	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
28	Vishrambaug	Dnyan prabudin khau gali	18.51229448	73.845657	New Location	1	3	0	0
29	Vishrambaug	City pride chowk	18.51325812	73.848541	New Location	1	1	1	0
30	Vishrambaug	tathastu chowk	18.5131284	73.847354	New Location	1	1	1	0
31	Vishrambaug	Zondhale chowk	18.51297103	73.845503	New Location	1	1	0	0
32	Vishrambaug	Kala shetram chowk	18.51401907	73.848472	New Location	1	2	0	0
33	Vishrambaug	Common wealth laxmi road	18.51442049	73.851703	New Location	1	2	0	0
34	Vishrambaug	Huzurbaga lokhande talim chowk	18.51564763	73.853778	New Location	1	2	0	2
35	Vishrambaug	Tulshi baug (Bank of Mahrashtra)	18.51404849	73.853799	New Location	1	1	0	0
36	Vishrambaug	Tulsi baug ganpati	18.51418868	73.855299	New Location	2	3	0	0
37	Vishrambaug	tulshibaug kaka halwai	18.51441709	73.856148	New Location	1	2	0	0
38	Faraaskhana	Gadgil chowk	18.52152653	73.855258	New Location	1	2	0	0
39	Faraaskhana	Tambat chowk	18.5205019	73.858954	New Location	1	1	1	0
40	Faraaskhana	Pawale chowk	18.52193377	73.858268	Addl Poles and Cameras	2	2	0	0
41	Faraaskhana	Chota shaikh salla darga	18.52301381	73.858258	New Location	1	2	0	0
42	Faraaskhana	kumbarwada	18.52392483	73.857932	Addl Poles and Cameras	3	4	1	4
43	Faraaskhana	Shivaji Akhada	18.52628935	73.861272	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
44	Faraaskhana	Gogadev chowk	18.52517132	73.863721	New Location	1	2	1	0
45	Faraaskhana	Shreekrishna chowk mangalwar peth	18.52445699	73.863185	New Location	1	1	1	0
46	Faraaskhana	bhimnagar smrat chowk	18.52328501	73.861571	New Location	1	2	1	0
47	Faraaskhana	Kamla Nehru	18.52248646	73.861633	New Location	2	3	1	0
48	Faraaskhana	Devji baba	18.51925854	73.861502	Addl Cameras only	0	1	0	0
49	Faraaskhana	RCM college	18.51917111	73.860888	New Location	1	1	0	0
50	Faraaskhana	Phadke haud chowk	18.51865673	73.859682	Addl Poles and Cameras	1	2	0	0
51	Faraaskhana	Adinath chowk	18.51427897	73.859498	New Location	1	2	0	0
52	Faraaskhana	Mirza galib chowk	18.51307792	73.860303	New Location	1	2	1	0
53	Faraaskhana	Burudi pul	18.51379096	73.864231	New Location	1	1	0	0
54	Faraaskhana	Panguli chowk	18.51371021	73.862819	New Location	1	3	0	0
55	Faraaskhana	Dulya maruti chowk	18.51617856	73.862792	New Location	1	2	0	0
56	Faraaskhana	Dagdi nagaboda	18.51799737	73.861686	New Location	1	2	1	0
57	Faraaskhana	Dangekar office	18.5168376	73.860158	New Location	2	4	1	0
58	Faraaskhana	Ubyan maruti mandal	18.51781298	73.857482	New Location	1	3	0	0
59	Faraaskhana	Old tapkir gali	18.5172439	73.858045	New Location	2	2	0	0
60	Faraaskhana	Pasodeya vithal mandir	18.51690246	73.858011	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
61	Faraaskhana	Shrikrishna chowk	18.5157873	73.85692	New Location	1	2	1	0
62	Faraaskhana	New building kranti hotel	18.51514404	73.858078	New Location	1	0	1	0
63	Faraaskhana	Mohan Icecream ravivar peth	18.5155211	73.859402	New Location	1	1	0	0
64	Faraaskhana	tarachand market budhwar peth	18.51556751	73.858064	New Location	1	2	1	0
65	Faraaskhana	Nehru Chowk	18.51427929	73.858819	New Location	1	2	0	0
66	Faraaskhana	Shukwar peth police chowki	18.5142103	73.858212	New Location	1	3	0	0
67	Faraaskhana	Shreenath talkies	18.51416215	73.857075	New Location	1	1	1	0
68	Faraaskhana	Datta mandir chowk	18.51473933	73.856649	New Location	2	3	1	0
69	Faraaskhana	belbaug chowk	18.51566384	73.856268	New Location	1	2	0	4
70	Faraaskhana	Budwar chowk	18.51687862	73.856235	New Location	1	2	0	0
71	Khadak	Daware chowk	18.5123924	73.86435	New Location	1	2	0	0
72	Khadak	Govind Halwai chowk	18.51310615	73.861845	New Location	1	3	1	0
73	Khadak	Shubhasha darga	18.5129421	73.859631	New Location	1	2	1	0
74	Khadak	ganesh mandir ekbote chowk	18.50167124	73.866107	Addl Cameras only	0	3	0	0
75	Khadak	Kumarpacific 7 luv bridge	18.50173385	73.871701	New Location	1	2	0	0
76	Khadak	rohidas garden	18.5030578	73.873682	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
77	Khadak	Maharashtra mitra mandal	18.50541853	73.873099	New Location	1	2	0	0
78	Khadak	damapal chowk	18.50735477	73.872307	Addl Poles and Cameras	1	2	0	0
79	Khadak	Junamotar stand	18.50937747	73.872243	Addl Cameras only	0	2	1	0
80	Khadak	Old ramoshi gate	18.51068328	73.868403	Addl Poles and Cameras	1	1	1	0
81	Khadak	timber market (bahubali chowk)	18.50721361	73.868358	New Location	3	6	0	0
82	Khadak	Sonawane hospital	18.5062954	73.868573	Addl Poles and Cameras	1	2	0	0
83	Khadak	Lohiya nagar police chowki	18.50606394	73.865965	New Location	1	3	1	0
84	Khadak	Krantveer lauji salve	18.50970526	73.865517	New Location	1	3	0	0
85	Khadak	Angarshah baba chowk	18.51069689	73.863259	New Location	1	2	1	0
86	Khadak	Kasturi chowk	18.51054015	73.862056	New Location	2	4	0	0
87	Khadak	Jain temple	18.51053824	73.859648	New Location	1	3	1	0
88	Khadak	Shitladevi chowk	18.50858009	73.859714	New Location	1	2	0	0
89	Khadak	Vega center chowk shanker seth road	18.49992379	73.863313	New Location	4	3	1	4
90	Khadak	Jagdewadi entry	18.50018464	73.861971	New Location	1	2	0	0
91	Khadak	Sushil lodge	18.50174055	73.85857	New Location	1	2	0	0
92	Khadak	HBC tower chowk	18.50403834	73.857009	New Location	1	4	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
93	Khadak	Pabhalkar chowk	18.50586014	73.856126	New Location	1	3	1	0
94	Khadak	Misaal Chowk	18.51061391	73.857881	New Location	1	1	0	0
95	Khadak	Ankur Corner (gadi khana)	18.51154799	73.857931	New Location	1	1	1	0
96	Khadak	Anna Saheb chowk	18.51132162	73.857276	New Location	1	3	0	0
97	Khadak	Badami Houd	18.51067282	73.855152	New Location	1	0	1	0
98	Khadak	Kalahud chowk	18.50992145	73.855316	New Location	1	2	0	0
99	Khadak	Swamis hotel	18.50856642	73.854168	New Location	1	1	0	0
100	Khadak	Prestige Point	18.50835988	73.855278	New Location	2	4	1	0
101	Khadak	Sanas Plaza	18.50588844	73.853628	New Location	1	2	1	0
102	Khadak	Puram chowk	18.50546399	73.853577	Addl Cameras only	0	0	1	0
103	Samarth	Sheral seth chowk	18.5182091	73.867115	New Location	1	4	0	0
104	Samarth	Jadhav Bakery	18.52164871	73.865087	New Location	1	3	0	0
105	Samarth	Sadanad nagar jopadpati	18.52349312	73.864688	New Location	1	2	1	0
106	Samarth	Dadasaheb gaikwad dawakhana	18.52484011	73.864877	New Location	1	2	0	0
107	Samarth	Parage Chowk	18.52358467	73.86385	New Location	1	1	0	0
108	Samarth	Shetty Corner	18.52253239	73.863863	New Location	1	4	1	0
109	Samarth	khadi ch maidan	18.52021196	73.864112	New Location	1	4	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
110	Samarth	Kadarbai chowk	18.5175732	73.868354	New Location	2	4	1	0
111	Samarth	Azaad Ali chowk	18.51672463	73.867123	New Location	1	3	0	0
112	Samarth	Dhavle chowk	18.51675302	73.865973	New Location	1	3	0	0
113	Samarth	Machi market nana peth	18.51563205	73.864494	Addl Poles and Cameras	1	1	0	0
114	Samarth	Sakhilipeer	18.51467256	73.867128	New Location	1	3	0	0
115	Samarth	Samadhan chowk	18.51470207	73.865956	New Location	1	4	0	0
116	Samarth	Dokeralin	18.5146484	73.865133	New Location	1	0	1	0
117	Samarth	Ashok chowk	18.51400653	73.865172	New Location	1	3	1	0
118	Samarth	Shivpratap chowk	18.51310204	73.864782	New Location	1	3	0	0
119	Samarth	Aina Masjid	18.51306323	73.867859	New Location	1	2	0	0
120	Samarth	Rajewadi chowk exit	18.51363994	73.871295	New Location	1	1	0	0
121	Samarth	Rajewade chowk entry	18.51369324	73.868753	New Location	1	1	0	0
122	Samarth	Patrechawl	18.51270957	73.869384	New Location	1	1	1	0
123	Samarth	Barat Talkies	18.51128219	73.870333	New Location	1	2	1	0
124	Samarth	Manjula bai chawl	18.51105289	73.874752	New Location	1	1	1	0
125	Samarth	Padumji chowk	18.5106883	73.87211	New Location	2	4	0	0
126	Samarth	Christ Church	18.51643003	73.872293	New Location	2	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
127	Samarth	Maratha Samarat hotel - banerjee chowk	18.52047887	73.872537	New Location	1	3	1	0
128	Deccan	Ajeet Hotel	18.51780789	73.840186	New Location	1	1	0	0
129	Deccan	PYC GYMkhana lane	18.51728841	73.840076	New Location	1	1	0	0
130	Deccan	Boudhani chowk golwikar lab	18.51827109	73.834225	New Location	1	2	0	0
131	Deccan	Padvale Chowk	18.50684098	73.832283	New Location	1	2	0	0
132	Deccan	Swantatra Chowk	18.50994086	73.834064	New Location	2	4	0	0
133	Deccan	SM joshi exit	18.50951037	73.840604	New Location	1	1	0	0
134	Deccan	Central Mall garware college	18.51106473	73.838173	New Location	1	1	0	0
135	Deccan	Garware college and shalla	18.5115944	73.837397	New Location	1	2	1	0
136	Deccan	Shelar mama	18.5138148	73.841564	New Location	1	3	0	0
137	Deccan	Z bridge	18.5178009	73.844035	New Location	2	2	1	0
138	Deccan	Sambhaji Garden	18.52058328	73.846239	Addl Cameras only	0	1	0	0
139	Deccan	Balgandarav chowk	18.52281024	73.848119	Addl Poles and Cameras	4	0	0	4
140	Deccan	Ramee Grand	18.52052097	73.843772	New Location	1	2	1	0
141	Deccan	Darode House	18.52216521	73.844237	New Location	2	4	1	0
142	Deccan	joshiwadewale exit	18.52074447	73.841032	New Location	1	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
143	Deccan	Tukaram paduka chowk	18.52461277	73.841387	New Location	2	0	1	2
144	Deccan	BMCC entry point rupaloli hotel	18.51932698	73.841234	Addl Cameras only	0	1	0	0
145	Deccan	westide mall lane	18.51836234	73.84143	New Location	1	1	1	0
146	Deccan	Gupte Hospital chowk	18.51865927	73.840019	New Location	1	2	0	0
147	Deccan	Ganeshwadi (Joshiwadewale)	18.51946932	73.840829	New Location	1	1	0	0
148	Deccan	BMCC road Kalepath	18.52012529	73.832883	New Location	1	1	0	0
149	Deccan	Ambedkar bhavan entry	18.52444125	73.829729	New Location	1	2	0	0
150	Deccan	german bakery balbarti chowk	18.52025393	73.830488	New Location	1	0	1	0
151	Deccan	Prabhat junction	18.51545583	73.828796	Addl Cameras only	0	1	0	0
152	Deccan	Kanchan gali entry wadeshwer	18.51495265	73.828717	New Location	1	1	0	0
153	Deccan	Kanchan Gali exit hill	18.51571757	73.825241	New Location	1	1	0	0
154	Deccan	Athawale chowk	18.51068427	73.829663	Addl Cameras only	1	3	0	0
155	Deccan	hirvayi garden chowk	18.51481697	73.831514	New Location	1	1	1	0
156	Deccan	Godke chowk	18.51427845	73.833652	Addl Cameras only	0	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
157	Deccan	Rohini bate chowk lane 7	18.5139429	73.836242	New Location	1	4	0	0
158	Deccan	Khanduji baba chowk	18.51460516	73.84235	Addl Poles and Cameras	2	0	0	4
159	Deccan	Woodland hotel	18.50798333	73.834606	New Location	1	1	0	0
160	Deccan	Kalewada	18.50473464	73.833856	New Location	1	1	0	0
161	Deccan	Rrandwane Gaothan	18.50694747	73.834528	New Location	1	3	0	0
162	Deccan	bhide bridge	18.51522193	73.84482	Addl Cameras only	0	2	0	0
163	Deccan	Nadipatra chopati	18.51585619	73.843905	New Location	1	2	0	0
164	Deccan	Nadipatra chopati	18.51585619	73.843905	New Location	1	2	0	0
165	Deccan	Deccan bustop backside	18.51517869	73.84278	New Location	1	1	0	0
166	Deccan	Deccan bustop	18.51615514	73.842844	New Location	1	1	1	0
167	Deccan	Chithle Corner	18.51574828	73.841489	New Location	1	2	0	0
168	Shivaji Nagar	Natawadi underpass	18.53524031	73.848637	New Location	1	1	0	0
169	Shivaji Nagar	Veer Chapekar chowk	18.53135126	73.844584	Addl Cameras only	3	0	0	4
170	Shivaji Nagar	Lalit mahal chowk	18.52824403	73.842876	New Location	1	2	0	0
171	Shivaji Nagar	Modern High school	18.52447992	73.849162	New Location	1	2	0	0
172	Shivaji Nagar	Svarkar bavan joshi wadewale	18.52228283	73.849351	New Location	2	3	1	0
173	Shivaji Nagar	Kude Bridge busstop	18.52201323	73.851433	New Location	1	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
174	Shivaji Nagar	Shivaji Bridge (Corporation Bridge)	18.5231423	73.854425	New Location	2	2	0	0
175	Shivaji Nagar	Mangla takies	18.52354549	73.854282	New Location	2	2	0	0
176	Shivaji Nagar	Ramsir tekdi	18.52482208	73.853709	New Location	1	1	0	0
177	Shivaji Nagar	Shivaji putla	18.52578696	73.853315	New Location	2	1	1	2
178	Shivaji Nagar	Family Court	18.52688813	73.855475	New Location	2	2	1	0
179	Shivaji Nagar	Kamgar Putla Chowk	18.52795012	73.857538	New Location	1	2	1	0
180	Shivaji Nagar	Shivajinagar Court gate 1 and 3	18.52889458	73.854338	Clubbed Location	2	2	1	0
181	Shivaji Nagar	Coep Hostel Gate and sancheti chowk	18.52934727	73.852825	Clubbed Location	5	4	2	2
182	Shivaji Nagar	Central Mall	18.53448852	73.839084	Addl Poles and Cameras	1	2	2	2
183	Shivaji Nagar	Chitranjan Vatika	18.53233659	73.83744	New Location	1	2	1	0
184	Shivaji Nagar	Kedarnath Mandir	18.530926	73.841593	New Location	2	2	1	0
Zone 2									
185	Lashkar	Sharbat wala chowk	18.51466938	73.876941	New Location	1	3	1	0
186	Lashkar	Center streer chowk	18.51395768	73.877994	New Location	1	3	0	0
187	Lashkar	saraf bazar chowk	18.5121361	73.877908	New Location	1	3	0	0
188	Lashkar	bhopale chowk	18.51137352	73.877888	Addl Cameras only	0	0	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
189	Lashkar	Babajaan dargha	18.51105048	73.876393	Addl Cameras only	0	2	0	0
190	Lashkar	Charbaudi chowki	18.51257453	73.876333	New Location	1	2	0	0
191	Lashkar	Saintjones school corner	18.5144822	73.873512	New Location	1	3	0	0
192	Lashkar	Shitla devi chowk	18.51643259	73.874116	New Location	1	3	0	0
193	Lashkar	Bacchu Adda chowk	18.51666308	73.875017	New Location	2	3	1	0
194	Lashkar	jj garden chowk	18.51709641	73.877362	New Location	2	3	0	0
195	Lashkar	Arora tower chowk	18.51755451	73.878975	Addl Poles and Cameras	2	1	1	2
196	Lashkar	mazorine (cover 7 lanes)	18.51649743	73.878874	New Location	4	8	0	0
197	Lashkar	3 topa chowk	18.51782983	73.880632	New Location	1	3	0	0
198	Lashkar	George hotel	18.51749792	73.879553	New Location	1	3	0	0
199	Lashkar	behind victory theatre mangal vihar hotel	18.51544824	73.879706	New Location	1	3	0	0
200	Lashkar	court gali begind jama masjid	18.51359226	73.879453	New Location	1	2	0	0
201	Lashkar	Indira Gandhi chowk	18.51376203	73.879882	New Location	2	2	0	4
202	Lashkar	Mahavir chowk	18.51366347	73.878816	Addl Poles and Cameras	1	2	0	0
203	Lashkar	Estreet road fashion street	18.50911772	73.87983	New Location	1	2	0	0
204	Lashkar	Mohamad rafi chowk	18.51078114	73.878785	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
205	Lashkar	15 August chowk	18.50810319	73.878732	New Location	1	2	0	0
206	Lashkar	Good luck hotel camp	18.50751533	73.877185	New Location	2	3	0	0
207	Lashkar	Tryluck chowk	18.50709474	73.878569	Addl Cameras only	0	1	0	0
208	Lashkar	Y junction singh corner camp	18.50552949	73.878401	New Location	1	2	0	0
209	Lashkar	azam campus front gate	18.5059801	73.876437	New Location	1	2	1	0
210	Lashkar	Poona college chowki	18.50655261	73.8748	New Location	1	3	0	0
211	Lashkar	Barf factory chowk beside poona college	18.50568337	73.874518	New Location	1	2	1	0
212	Lashkar	Solapur bazar chowki	18.50494352	73.882272	New Location	1	1	0	2
213	Lashkar	Deccan Tower	18.50688518	73.879668	New Location	1	0	1	0
214	Lashkar	thane maruti chowk	18.50700345	73.879262	New Location	1	2	0	0
215	Lashkar	1000 oak chowk	18.51065111	73.880125	New Location	2	3	1	0
216	Lashkar	Bishops chowk	18.51046512	73.882564	New Location	2	4	1	0
217	Lashkar	CNP chowk (Saint marry chowk gate 4)	18.51036745	73.884787	New Location	1	2	0	0
218	Lashkar	General beur road	18.51437594	73.883548	New Location	1	2	0	0
219	Lashkar	Gurudwara chowk dargah zawal	18.513515	73.885706	New Location	1	3	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
220	Koregaon Park	Nala garden	18.53586389	73.890526	New Location	1	2	0	0
221	Koregaon Park	Ragvillas society junction	18.53987024	73.891065	New Location	1	3	1	0
222	Koregaon Park	O hotel chowk	18.53960163	73.888254	New Location	1	2	0	0
223	Koregaon Park	Anna Maggi point	18.54223967	73.890249	New Location	1	1	1	0
224	Koregaon Park	Burning ghat	18.54188524	73.887673	New Location	1	2	0	0
225	Koregaon Park	Bund garden Pune metro station	18.5413207	73.884113	New Location	1	2	1	0
226	Koregaon Park	Ruby hall chowk	18.53203759	73.876771	New Location	1	2	1	0
227	Koregaon Park	Narangi baug road	18.53950609	73.879605	New Location	1	2	0	0
228	Koregaon Park	mobaaz chowk	18.53478287	73.87963	Addl Poles and Cameras	1	2	0	0
229	Koregaon Park	Wadia college main entrance road	18.53367536	73.878855	New Location	2	1	0	4
230	Koregaon Park	Circuit house chowk	18.52842854	73.885346	Addl Cameras only	0	1	1	0
231	Koregaon Park	Blue diamond chowk	18.53794835	73.885411	New Location	1	2	1	0
232	Koregaon Park	Atur park conrad hotel lane	18.53580233	73.884802	New Location	1	2	0	0
233	Koregaon Park	Under sadhuwasvani bridge	18.53302241	73.885744	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
234	Koregaon Park	Budrani hospital chowk	18.53307518	73.887539	New Location	1	2	0	0
235	Koregaon Park	Satys gadghe maharaj shala	18.533079	73.889202	New Location	1	1	1	0
236	Koregaon Park	food corporation of India gate	18.53326146	73.891875	New Location	1	3	0	0
237	Koregaon Park	Madarivasti	18.53186847	73.898901	New Location	1	1	0	0
238	Koregaon Park	Publiq hotel chowk	18.53282119	73.899201	New Location	1	2	0	0
239	Koregaon Park	Daily all day hotel lane 7	18.53408034	73.899494	New Location	1	3	0	0
240	Koregaon Park	Lane 5 north main road	18.53933048	73.897314	New Location	1	1	0	0
241	Koregaon Park	Natural Ice cream point	18.53917123	73.900246	New Location	2	3	0	0
242	Koregaon Park	Murphy hotel chowk	18.53653886	73.898611	New Location	2	4	0	0
243	Bund Garden	SBI teasury	18.52205775	73.871154	New Location	1	1	0	0
244	Bund Garden	Rosary school	18.51979264	73.87412	New Location	1	1	0	0
245	Bund Garden	Kolte patil chowk city point	18.5387414	73.875473	New Location	2	3	1	0
246	Bund Garden	Ganpati chowk	18.53498897	73.876121	Addl Poles and Cameras	1	1	1	0
247	Bund Garden	Meru hotel	18.53484934	73.874712	New Location	1	2	0	0
248	Bund Garden	Rajguru chowk	18.53466218	73.87379	New Location	1	1	1	0
249	Bund Garden	Sourabh hall gali	18.53013335	73.874735	New Location	1	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
250	Bund Garden	Pune railways station near outgate	18.52805787	73.874937	New Location	1	2	0	0
251	Bund Garden	Alankar chowk	18.52816055	73.876228	Addl Cameras only	0	1	1	0
252	Bund Garden	GPO bus stop (Hotel ritz)	18.52336467	73.876332	New Location	1	1	0	0
253	Bund Garden	kirade chowk	18.52219604	73.876123	Addl Cameras only	0	1	0	0
254	Bund Garden	BJ medical ground	18.52108113	73.875763	New Location	1	0	1	0
255	Bund Garden	nehru memorial chowk	18.51952587	73.875324	Addl Poles and Cameras	3	0	0	4
256	Bund Garden	Dorabjee chowk	18.51928111	73.876654	New Location	2	3	1	0
257	Bund Garden	Poona club out gate	18.52308838	73.878168	New Location	1	1	1	0
258	Bund Garden	Council hall chowk vidhan bavan	18.52461691	73.87865	New Location	1	1	0	4
259	Bund Garden	Sadhuwaswani chowk	18.52484898	73.876677	New Location	2	5	1	0
260	Bund Garden	central building chowk	18.52434764	73.872276	New Location	1	0	0	3
261	Bund Garden	Ambedkar putla backside	18.52508232	73.871749	New Location	1	1	1	0
262	Bund Garden	Pune Station chowk	18.5280388	73.873321	New Location	2	4	1	4
263	Bund Garden	Tukaram shinde bustop	18.52748178	73.873182	New Location	1	1	1	0
264	Bund Garden	Sasoon Hospital	18.52618386	73.872374	New Location	1	1	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
265	Bharati Vidyapeeth	Sai mandir sachaimata	18.44282947	73.851929	New Location	1	3	0	0
266	Bharati Vidyapeeth	state bank nagr	18.46240069	73.865307	New Location	1	3	1	0
267	Bharati Vidyapeeth	Jama masjid santosh nagar	18.44680163	73.856046	New Location	1	3	0	0
268	Bharati Vidyapeeth	treemurti hotel holiday	18.4211303	73.858697	New Location	1	2	0	0
269	Bharati Vidyapeeth	bhilarwadi phata	18.42236356	73.860225	New Location	1	3	0	0
270	Bharati Vidyapeeth	Ganesh chowk mangnewadi	18.43177189	73.859452	New Location	2	4	0	0
271	Bharati Vidyapeeth	mangnewadi petrol pump	18.43211255	73.858584	New Location	1	3	0	2
272	Bharati Vidyapeeth	Nimbalkarwadi	18.43142868	73.872719	New Location	1	3	0	0
273	Bharati Vidyapeeth	Barat nagar	18.43567305	73.867904	New Location	1	3	0	0
274	Bharati Vidyapeeth	warkhade nagar	18.44040427	73.864036	New Location	1	3	0	0
275	Bharati Vidyapeeth	Guzar wadi phata	18.43812917	73.860538	New Location	2	3	0	0
276	Bharati Vidyapeeth	Ganga ocean society	18.45579327	73.866129	New Location	1	3	0	0
277	Bharati Vidyapeeth	Yashwant vihar	18.44904131	73.870094	New Location	1	2	0	0
278	Bharati Vidyapeeth	vishwagym	18.44928748	73.870513	New Location	1	2	0	2
279	Bharati Vidyapeeth	khan petrol pump katraj kondwa road	18.44843003	73.868388	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
280	Bharati Vidyapeeth	Samarth nagar Rajas society	18.45238838	73.869179	New Location	1	3	0	0
281	Bharati Vidyapeeth	katraj komdwa under bridge	18.4476365	73.861918	New Location	1	2	0	0
282	Bharati Vidyapeeth	Huzur paga shala	18.45297707	73.865221	New Location	1	3	0	0
283	Bharati Vidyapeeth	Katraj chowki	18.4468099	73.858659	New Location	1	0	1	0
284	Bharati Vidyapeeth	Katraj chowk	18.44818641	73.858413	Addl Poles and Cameras	1	0	0	2
285	Bharati Vidyapeeth	Pmpml depot katraj	18.44722527	73.858676	Addl Poles and Cameras	1	1	0	0
286	Bharati Vidyapeeth	Katraj juna bustop	18.44767753	73.85952	New Location	2	1	1	0
287	Bharati Vidyapeeth	Bhagwa chowk	18.44459276	73.8563	New Location	1	3	0	0
288	Bharati Vidyapeeth	Anjani nagar chowk	18.44234539	73.85782	New Location	2	4	0	0
289	Bharati Vidyapeeth	Panicha tanki	18.43943164	73.853368	New Location	2	4	0	0
290	Bharati Vidyapeeth	wonder city	18.44978808	73.853591	New Location	1	1	0	1
291	Bharati Vidyapeeth	Agaha mandir paida datta mandir	18.44920702	73.854122	New Location	1	3	0	0
292	Bharati Vidyapeeth	Dmart service road	18.45213268	73.844286	New Location	1	2	0	0
293	Bharati Vidyapeeth	Silayi chowk	18.45471959	73.836193	New Location	1	2	0	0
294	Bharati Vidyapeeth	Vaswan chowk	18.45527233	73.83594	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
295	Bharati Vidyapeeth	Bairavnath mandir	18.45133059	73.83742	New Location	1	4	0	0
296	Bharati Vidyapeeth	Gaiyemukh chowk	18.44893522	73.838258	New Location	1	3	1	0
297	Bharati Vidyapeeth	ambegaon khurd gram panchayat office	18.44091007	73.840365	New Location	1	2	0	0
298	Bharati Vidyapeeth	swaminarayan mandir both sides of road	18.4432816	73.837127	New Location	2	3	0	0
299	Bharati Vidyapeeth	Dhari pul Jambadwadi	18.43164625	73.839877	New Location	2	2	0	2
300	Bharati Vidyapeeth	Sidhivinayak chowk	18.43309254	73.843892	New Location	1	3	0	0
301	Bharati Vidyapeeth	shani nagar chowk	18.44095168	73.847151	New Location	1	2	1	0
302	Bharati Vidyapeeth	Teloko chowk	18.44455127	73.848476	New Location	2	2	0	0
303	Bharati Vidyapeeth	Datta nagar chowk	18.44976995	73.849704	New Location	2	4	0	0
304	Bharati Vidyapeeth	Datta nagar chowk near highway	18.45044881	73.850109	New Location	1	1	0	0
305	Bharati Vidyapeeth	Rajmata buayari marg	18.45048778	73.851286	New Location	2	4	0	0
306	Bharati Vidyapeeth	Chandarbaugh chowk	18.454039	73.851113	New Location	2	3	1	0
307	Bharati Vidyapeeth	Ramkrushna boys hostel	18.45992573	73.851069	New Location	1	2	0	0
308	Bharati Vidyapeeth	Raje chowk Ambrgaon pathat	18.45922353	73.845981	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
309	Bharati Vidyapeeth	Ganesh chowk ambegaon pathar	18.46286585	73.84593	New Location	1	2	0	0
310	Bharati Vidyapeeth	teen bati	18.46271669	73.843746	New Location	1	3	1	0
311	Bharati Vidyapeeth	Sai sidhi chowk	18.46265202	73.842358	New Location	1	3	0	0
312	Bharati Vidyapeeth	Chintamani chowk	18.45560954	73.839968	New Location	1	3	0	0
313	Bharati Vidyapeeth	Jija mata chowk	18.455979	73.84373	New Location	2	3	1	0
314	Bharati Vidyapeeth	Maharana pratap chowk	18.45697094	73.849073	New Location	2	2	0	0
315	Bharati Vidyapeeth	bhartiya vidyapeeth back market	18.45753022	73.853092	New Location	1	1	0	0
316	Bharati Vidyapeeth	maitri chowk	18.45461846	73.853648	New Location	2	4	0	0
317	Swargate	Vishwa hotel chowk	18.50430537	73.852307	New Location	1	5	1	0
318	Swargate	Gadade patil office	18.49900619	73.850271	New Location	1	2	0	0
319	Swargate	Purohit hospital chowk	18.49744853	73.851028	New Location	2	4	0	0
320	Swargate	Sana Parking	18.49683551	73.857027	New Location	1	2	0	0
321	Swargate	Sai baba mandir	18.49411697	73.858011	New Location	1	2	0	0
322	Swargate	Pachami chowk	18.49248889	73.857728	New Location	2	1	0	4
323	Swargate	City international school	18.49193924	73.858669	New Location	2	3	1	0
324	Swargate	Zambre palace	18.49324639	73.861958	New Location	2	4	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
325	Swargate	minatai thskre vasahat (north south)	18.49238757	73.866477	Addl Poles and Cameras	1	2	0	0
326	Swargate	Cresend high school	18.49264671	73.867759	New Location	1	1	0	0
327	Swargate	Ekta garden	18.48984224	73.871802	New Location	1	2	0	0
328	Swargate	Sailisbury park	18.49479434	73.874932	New Location	1	2	1	0
329	Swargate	Poonawala garden	18.49713566	73.876856	New Location	1	2	0	0
330	Swargate	Radha Swami chowk	18.49641295	73.871572	New Location	1	2	0	0
331	Swargate	Dias plot chowk	18.49649244	73.86933	New Location	2	3	1	0
332	Swargate	Dias Plot canal	18.49814811	73.869487	New Location	1	1	1	0
333	Swargate	Mira society	18.50157872	73.870183	New Location	1	2	0	0
334	Swargate	St. Vincen college behind kumar pacific	18.49948951	73.870363	New Location		2	0	0
335	Swargate	7 luv chowk	18.50143373	73.869319	New Location	1	3	0	0
336	Swargate	Dobi ghat	18.50201339	73.874712	New Location	1	2	0	4
337	Swargate	Swargate pmt bus stop	18.49912097	73.85807	New Location	1	2	1	0
338	Swargate	Swargate (Katraj bustop - Canal)	18.49828296	73.858148	New Location	1	1	1	0
339	Swargate	Dayari bys stop	18.50031957	73.858513	New Location	1	1	1	0
340	Swargate	Natraj hotel entry exit	18.50142151	73.857331	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
341	Swargate	Hirabaug chowk	18.50400432	73.855304	New Location	1	3	1	0
342	Swargate	Sanas Putla	18.50412514	73.85366	New Location	1	2	0	0
343	Swargate	Veer savarkar chowk	18.49968947	73.85388	New Location	2	4	1	0
344	Swargate	Mira mandal chowk	18.49726475	73.854101	New Location	1	1	0	0
345	Swargate	laxmi narayan chowk	18.4967904	73.858021	New Location	1	2	0	0
346	Swargate	Padbander office	18.49796648	73.858334	New Location	1	2	0	0
347	Swargate	Swargate bus stand	18.49926971	73.859651	New Location	4	6	3	0
348	Sahakar Nagar	Khandere junction	18.47671598	73.851046	New Location	1	4	0	0
349	Sahakar Nagar	Vinkar chowk	18.47456728	73.853251	New Location	1	2	0	0
350	Sahakar Nagar	Solapur sideshwer bank	18.47443629	73.855014	New Location	1	2	0	0
351	Sahakar Nagar	Chavan nagar kaman	18.47448746	73.85632	New Location	1	2	1	0
352	Sahakar Nagar	shankar maharaj vasat	18.47180599	73.855859	New Location	1	2	0	0
353	Sahakar Nagar	shankar maharaj math	18.47094716	73.857193	New Location	1	2	0	0
354	Sahakar Nagar	Rajni corner	18.46271669	73.862325	New Location	1	2	0	0
355	Sahakar Nagar	tulza bawani mandir	18.46480798	73.861086	New Location	1	2	1	0
356	Sahakar Nagar	pawar hospital	18.46598495	73.859864	New Location	1	2	0	0
357	Sahakar Nagar	Elora palace	18.46597636	73.857982	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
358	Sahakar Nagar	Chaityana nagar near nagar sevak office	18.46059422	73.857294	New Location	1	2	0	0
359	Sahakar Nagar	mohan nagar Ambe da jada	18.46026411	73.849093	New Location	1	3	0	0
360	Sahakar Nagar	Swarkar chowk	18.46475328	73.848362	New Location	1	3	0	0
361	Sahakar Nagar	Talzai shevadcha Bus Stop	18.46852066	73.846843	New Location	1	3	0	0
362	Sahakar Nagar	Lunkad High school	18.47452308	73.849677	New Location	1	2	0	0
363	Sahakar Nagar	Talzai Mandir	18.47598554	73.844762	New Location	1	3	1	0
364	Sahakar Nagar	Anna bau sathe	18.48597283	73.854	New Location	1	3	1	0
365	Sahakar Nagar	Walvekar lawns	18.48120249	73.857236	New Location	1	2	0	0
366	Sahakar Nagar	Sharda center	18.47915611	73.857095	New Location	1	1	1	0
367	Sahakar Nagar	Padmavati Chowk	18.47624789	73.856394	New Location	2	0	0	4
Zone 3									
368	Dattawadi	E learning chowk	18.4925882	73.854137	New Location	1	4	0	4
369	Dattawadi	Gajanan maharaj chowk	18.49266706	73.84989	New Location	2	4	1	0
370	Dattawadi	999 chowk	18.50090912	73.84214	New Location	1	2	0	0
371	Dattawadi	Mashoba chowk	18.50096667	73.839191	New Location	1	3	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
372	Dattawadi	Asha hotel chowk	18.49622337	73.838444	New Location	1	2	0	2
373	Dattawadi	Nashamaruti chowk	18.49353128	73.835467	New Location	2	4	1	0
374	Dattawadi	Appla chowk jantavashat	18.49393891	73.839459	New Location	1	3	0	0
375	Dattawadi	Panmala chowk	18.49925057	73.843024	New Location	1	1	1	0
376	Dattawadi	Mahalaxmi complex	18.50058958	73.849019	New Location	1	2	1	0
377	Dattawadi	Nilayam bridge	18.49951745	73.850615	New Location	1	2	0	2
378	Dattawadi	NS phadke chowk	18.50438585	73.850667	New Location	3	2	1	4
379	Dattawadi	Gaddaepatil chowk dattwadi	18.49880295	73.850226	New Location	1	1	1	0
380	Alankar	nehru vasat karve road	18.50695958	73.827892	New Location	1	2	1	0
381	Alankar	Abhinav shalla canal road	18.50452048	73.825342	New Location	1	3	0	0
382	Alankar	Nimbalkar chowk	18.50001202	73.822463	New Location	1	2	0	0
383	Alankar	Karishma society canal road kwality hotel	18.50381045	73.821504	New Location	1	2	0	0
384	Alankar	Mrutyanja mandir	18.50281267	73.818363	Addl Poles and Cameras	1	2	1	0
385	Alankar	Rahulnagar junction	18.50078957	73.816169	New Location	1	3	0	0
386	Alankar	Abhishek Udyan	18.49825137	73.815842	New Location	1	2	0	0
387	Alankar	Dhanukar colony lane19 sarasvati school	18.49776555	73.811083	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
388	Alankar	Laxmi nagar Dahanukar	18.49301364	73.811447	New Location	1	2	1	0
389	Alankar	Savahas sabhagrah	18.4941087	73.82044	New Location	1	2	0	0
390	Alankar	Sahavas corner	18.49451188	73.818205	New Location	2	5	1	0
391	Alankar	Girija shankar society K52	18.49498627	73.821978	New Location		1	0	0
392	Alankar	Potnis Parisar	18.49744376	73.822388	New Location	1	2	0	0
393	Alankar	Shivai chowk	18.49151539	73.821679	New Location	1	2	0	0
394	Alankar	Sarovar hotel	18.49019583	73.824565	New Location	1	3	0	0
395	Alankar	Shakti 98	18.49433763	73.825661	New Location	1	2	0	0
396	Alankar	Tol hospital	18.49525213	73.828558	New Location	1	2	0	0
397	Alankar	tathwade Udyan	18.49279786	73.827696	New Location	1	2	1	0
398	Alankar	Rajaram bridge towards Alankar	18.48906799	73.826644	New Location	1	1	0	0
399	Alankar	Rajaram mantri circle (Grija hotel	18.48974681	73.827071	New Location	2	3	0	0
400	Alankar	Amachi hotel	18.49243107	73.82914	New Location	1	1	0	0
401	Alankar	Nav Shayadri chowk	18.49472664	73.830419	New Location	1	3	0	0
402	Alankar	Saket Society	18.49741346	73.832892	New Location	1	2	1	0
403	Alankar	Patwarden baug	18.4991073	73.831134	New Location	1	2	0	0
404	Alankar	CDSS chowk	18.50241587	73.828571	New Location	1	1	0	2

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
405	Alankar	Karnatak high school	18.50158921	73.826784	New Location	2	3	0	0
406	Kothrud	bogle chowk bombay stationary	18.5103737	73.817601	New Location	2	2	1	2
407	Kothrud	Kerewadi chowk	18.50783516	73.823033	Addl Poles and Cameras	4	0	1	4
408	Kothrud	New poona bakery	18.5095606	73.813363	New Location	1	3	0	0
409	Kothrud	Rautwadi end to end (entry and exit)	18.51320921	73.822913	New Location	2	3	0	0
410	Kothrud	Annabau sathe chowk	18.50643053	73.808565	New Location	1	3	1	0
411	Kothrud	Niluphule udyan	18.50536045	73.812864	New Location	1	3	0	0
412	Kothrud	Ideal ground	18.50894731	73.817651	New Location	1	3	0	0
413	Kothrud	Durga chowk cafe	18.51027817	73.815792	New Location	1	1	1	0
414	Kothrud	MIT entry	18.51561935	73.815424	New Location	1	2	1	0
415	Kothrud	New India School ram baug colony	18.51116588	73.813067	New Location	1	3	0	0
416	Kothrud	Shivray Pratishthan	18.50945161	73.811794	New Location	1	2	0	0
417	Kothrud	Jai bhawani nagar entry point	18.50854402	73.81044	New Location	1	1	0	0
418	Kothrud	Shivtirath nagar	18.50831796	73.809233	New Location	1	2	0	0
419	Kothrud	Gujarat colony chowk	18.50748385	73.807645	New Location	2	0	1	4
420	Kothrud	Kiskinda nagar	18.51233123	73.81045	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
421	Kothrud	Pethkar Samrajya	18.51547699	73.807789	New Location	1	3	0	0
422	Kothrud	Motha ganpati chowk	18.51876834	73.80738	New Location	1	2	0	0
423	Kothrud	Dashinmukhi maruti	18.51240385	73.805898	New Location	1	3	0	0
424	Kothrud	Kinara chowk	18.50724063	73.806152	Addl Poles and Cameras	1	2	0	0
425	Kothrud	Suttar hospital	18.5040908	73.807716	Addl Poles and Cameras	1	2	0	0
426	Kothrud	Badhai sweet chowk	18.50204578	73.804967	New Location	1	2	0	0
427	Kothrud	Tejasngar shetreya karyalya	18.50203307	73.809479	New Location	1	2	0	0
428	Kothrud	Belkenagar kaman	18.50202162	73.810495	New Location	2	3	1	0
429	Kothrud	Karve putla	18.50204578	73.815165	Addl Poles and Cameras	2	2	0	4
430	Alankar	Kokan express road	18.49847545	73.813744	Addl Poles and Cameras	1	3	1	0
431	Kothrud	Dhanukar colony	18.49529183	73.813285	Addl Poles and Cameras	1	2	1	2
432	Kothrud	Cummins office	18.49420654	73.808058	New Location	1	2	0	0
433	Kothrud	Gopinath chowk	18.49630309	73.80206	New Location	2	3	1	0
434	Kothrud	Ashish garden	18.50237188	73.802284	New Location	2	4	1	0
435	Kothrud	Shantiban chow	18.50310867	73.797604	New Location	1	2	0	0
436	Kothrud	elavya college chowk	18.50329618	73.796259	New Location	2	2	1	0
437	Kothrud	wholemart chowk	18.50098023	73.795805	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
438	Kothrud	Mahatma chowk	18.4973293	73.794933	New Location	1	3	0	0
439	Kothrud	Dodke construction plaza	18.49774582	73.789121	New Location	2	2	0	1
440	Kothrud	Shrungari math	18.50571962	73.7922	New Location	2	3	0	2
441	Kothrud	Right bhusari (Lohiya jain)	18.50791052	73.788698	Addl Poles and Cameras	1	1	0	2
442	Kothrud	Royal enfield chowk	18.50833068	73.790284	New Location	2	4	1	0
443	Kothrud	Mokate nagar	18.50675705	73.79448	New Location	1	2	0	0
444	Kothrud	Indiranagri chowk	18.50774679	73.797557	New Location	3	2	1	2
445	Warje	Wonder city futur	18.49976784	73.790186	New Location	2	3	0	0
446	Warje	Mahada road slum area	18.49184639	73.78928	New Location	1	2	1	0
447	Uttam Nagar	Kamtevasti	18.46778743	73.784738	New Location	1	0	1	0
448	Uttam Nagar	Navbharat t junction (school)	18.46610961	73.782139	New Location	1	1	0	0
449	Uttam Nagar	Shivane shamsham bhoomi	18.46326496	73.78417	New Location	2	1	0	2
450	Uttam Nagar	Maruti temple shivane	18.46362241	73.782057	New Location	1	3	1	0
451	Uttam Nagar	Ingle Corner	18.46663117	73.777272	New Location	1	2	0	0
452	Uttam Nagar	Achanak chowk	18.46218493	73.773279	New Location	2	3	1	0
453	Uttam Nagar	Kpore gaon entry	18.46051249	73.772364	New Location	2	3	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
454	Uttam Nagar	Khadakvasla t junction	18.44829868	73.762007	New Location	2	3	0	2
455	Uttam Nagar	Anjani managl karyala	18.45057047	73.764745	New Location	1	2	0	0
456	Uttam Nagar	kondhwe dhavde gaothan	18.45605088	73.768817	New Location		3	0	0
457	Uttam Nagar	Bhimnagar	18.45809678	73.770852	New Location	1	1	1	0
458	Uttam Nagar	Deshmukh wadi chowk	18.46677824	73.779116	New Location	1	2	1	0
459	Uttam Nagar	Shinde pul	18.46916474	73.786771	New Location	2	3	1	2
460	Warje	Shivaji Maharaj chowk	18.47477026	73.802501	New Location	1	2	0	0
461	Warje	Hanuman chowk	18.47552419	73.805603	New Location	2	4	0	0
462	Warje	Warje gaothan exit (Donewada exit) near puneri hotel	18.47724575	73.807155	New Location	1	2	0	0
463	Warje	Tapodham kaman	18.48183584	73.807966	New Location	1	2	0	0
464	Warje	Vikas mintra mandal chowk	18.48902093	73.816594	New Location	2	3	1	0
465	Warje	Shahu colony lane 1 cumins college	18.4866126	73.815517	New Location	1	3	1	0
466	Warje	Datte mandap	18.4873204	73.82341	New Location	1	2	0	0
467	Warje	Vandevi chowk	18.49313224	73.815136	New Location	2	3	0	2
468	Warje	karve nagar chowk under bridge	18.48918231	73.814311	Addl Poles and Cameras	1	1	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
469	Warje	Akash nagar	18.48721317	73.807296	New Location	1	3	1	0
470	Sinhgad Road	Sun empire chowk	18.47459644	73.812312	New Location	1	1	0	0
471	Sinhgad Road	Samadhan bhel corner	18.47355096	73.810795	New Location	1	2	0	0
472	Sinhgad Road	Raghunanadan hall	18.47160479	73.811654	New Location	1	2	0	1
473	Sinhgad Road	Raheja city chowk	18.46696909	73.814766	New Location	1	2	0	0
474	Sinhgad Road	Gokul hotel chowk	18.45962488	73.814717	New Location	1	2	0	0
475	Sinhgad Road	Dhayari phata	18.46099605	73.812723	New Location	2	2	1	0
476	Sinhgad Road	dhyari bridge start	18.45974764	73.810304	New Location		3	0	0
477	Sinhgad Road	Nanded city	18.4568577	73.800918	New Location	2	2	0	4
478	Sinhgad Road	Dalvi wala phata	18.4532781	73.795109	New Location	2	3	0	0
479	Sinhgad Road	hanuman mamdri dsk road	18.4470649	73.803991	New Location	1	3	0	0
480	Sinhgad Road	DSK vishwa bus stop	18.44318219	73.798958	New Location	1	2	1	0
481	Sinhgad Road	Umbraya ganpati chowk	18.44726916	73.810242	New Location	2	3	1	0
482	Sinhgad Road	Bhiaravnath mandir dayari	18.44493499	73.809778	New Location	1	1	0	0
483	Sinhgad Road	Samrudhi crystal	18.43329281	73.809398	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
484	Sinhgad Road	Dhayari last bus stop	18.442458	73.809687	New Location	1	2	1	0
485	Sinhgad Road	khadak chowk	18.44262173	73.811225	New Location	1	2	0	0
486	Sinhgad Road	Dhareshwer mandir	18.44087914	73.813639	New Location	1	2	1	0
487	Sinhgad Road	Khandoji shalla	18.44232344	73.814419	New Location	1	2	0	0
488	Sinhgad Road	Benkar vasti chowk	18.44672577	73.81548	New Location	1	1	0	0
489	Sinhgad Road	zeal college road	18.44649938	73.826424	New Location	1	1	0	0
490	Sinhgad Road	Maharaja chowk	18.44456429	73.832209	New Location	1	2	0	0
491	Sinhgad Road	Bhairav nath mandir narhe	18.44913532	73.828899	New Location	1	1	1	0
492	Sinhgad Road	Inglevasti	18.44680131	73.82782	New Location	1	3	0	0
493	Sinhgad Road	Walhekar chowk	18.43909114	73.827993	New Location	1	3	0	0
494	Sinhgad Road	Shree control chowk	18.44477151	73.821586	New Location	2	4	1	0
495	Sinhgad Road	Pari chowk	18.44711999	73.820803	New Location	2	2	1	2
496	Sinhgad Road	Ghandarv hotel	18.45374518	73.820477	New Location	1	3	1	0
497	Sinhgad Road	Manaji nagar	18.4534352	73.824361	New Location	1	2	0	0
498	Sinhgad Road	Thorat misal chowk	18.45756211	73.82488	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
499	Sinhgad Road	Selfie point / Ashirwad hotel	18.45315772	73.828377	New Location	1	2	0	0
500	Sinhgad Road	Swaminarayan behind gali	18.44382467	73.836648	New Location	1	1	0	0
501	Sinhgad Road	bhoomkar nagar	18.44890869	73.833079	New Location	1	3	1	0
502	Sinhgad Road	Pune Bangalore highway	18.45040636	73.831443	New Location	1	2	0	0
503	Sinhgad Road	Katila hotel narhe / shamshan bhumi	18.45451828	73.82762	New Location	1	0	1	0
504	Sinhgad Road	Navale bridge (under bridge) chowk	18.45955217	73.823952	New Location	2	4	1	2
505	Sinhgad Road	Balaji super market NSG	18.46101973	73.825703	New Location	2	3	1	0
506	Sinhgad Road	Navle bridge towards Satara	18.46233157	73.8206	New Location	2	0	0	5
507	Sinhgad Road	Vadgaon bridge pasalkar chowk	18.4661888	73.817018	New Location	2	2	1	4
508	Sinhgad Road	Funtime magli road	18.46810711	73.819173	New Location	1	1	0	2
509	Sinhgad Road	Vadgaon chowki	18.46845167	73.833195	New Location	1	2	0	0
510	Sinhgad Road	Polyhub food court(Sinhgad college)	18.4650256	73.834992	New Location	1	2	1	0
511	Sinhgad Road	Dabhadi chowk (Shivaji maharaj putla)	18.46281623	73.825413	New Location	1	2	0	0
512	Sinhgad Road	Bhairav nath mandir	18.46596269	73.82457	New Location	2	1	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
513	Sinhgad Road	Vadgaon kaman	18.46655941	73.823098	New Location	1	2	1	0
514	Sinhgad Road	Vadgaon	18.46760204	73.821971	New Location	1	2	1	0
515	Sinhgad Road	Bramha hotel chowj	18.47404513	73.82204	New Location	1	2	0	0
516	Sinhgad Road	Goel ganga chowk	18.47000618	73.820349	New Location	2	3	0	0
517	Sinhgad Road	Sinhgad shetrya karyalay	18.47061294	73.825086	New Location	2	0	1	4
518	Sinhgad Road	Tukai nagar	18.4712534	73.826478	New Location	2	4	1	0
519	Sinhgad Road	Samarth nagar	18.47490436	73.826674	New Location	1	2	0	0
520	Sinhgad Road	L 379	18.47532641	73.829891	New Location	1	1	1	0
521	Sinhgad Road	khorade vasti	18.47631416	73.830758	New Location	1	1	0	0
522	Sinhgad Road	vithalwadi canal road	18.48300065	73.82968	New Location	1	1	1	0
523	Sinhgad Road	London bridge	18.47833392	73.831544	New Location	1	2	0	0
524	Sinhgad Road	Rajaram pul chowk	18.48777766	73.829046	Addl Poles and Cameras	1	0	1	0
525	Sinhgad Road	Ranka jewellers chowk	18.4831584	73.827437	New Location	1	3	0	0
526	Sinhgad Road	Hingane chowk	18.48073207	73.825482	New Location	1	1	0	0
527	Sinhgad Road	Santosh hall	18.47839815	73.824049	New Location	2	3	1	2

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
528	Sinhgad Road	Ekta nagri nadi patra	18.47850659	73.82093	New Location	1	2	0	0
529	Sinhgad Road	khade washing center	18.47511201	73.815926	New Location	1	2	1	0
Zone 4									
530	Lonikand	Hingan gaon	18.54080447	74.163544	New Location	1	3	0	0
531	Lonikand	Nahavi sandas	18.60099768	74.154413	New Location	1	3	0	0
532	Lonikand	Sanghvi saandas	18.61738231	74.172984	New Location	2	4	1	0
533	Lonikand	Bhawadi gram panchayat	18.6242995	73.982073	New Location	1	0	1	0
534	Lonikand	Wadgaon shinde	18.62034989	73.947875	New Location	1	0	1	0
535	Lonikand	Phulgaon MIDC	18.65745244	74.011831	New Location	1	1	1	0
536	Lonikand	Tulapur sambhaji	18.66902778	73.994154	New Location	2	1	1	2
537	Vishrantwadi	Kasturba housing chowk	18.57132468	73.879518	New Location	1	3	0	0
538	Vishrantwadi	Navrang plaza	18.57343021	73.884046	Addl Poles and Cameras	1	2	0	0
539	Vishrantwadi	Martobha school bridge	18.57777983	73.885988	New Location	1	2	0	0
540	Vishrantwadi	Sawant petrol pump	18.57353573	73.885323	New Location	1	1	0	0
541	Vishrantwadi	Lane 7 Radha Krishna hotel tingre nagar	18.57431818	73.891634	New Location	1	1	0	0
542	Vishrantwadi	509 chowk or lohegaon entry	18.57546834	73.899331	Addl Poles and Cameras	1	2	0	4

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
543	Vishrantwadi	Lane 13 Kekan chowk	18.57554779	73.89522	New Location	1	2	0	0
544	Vishrantwadi	Poonam medical chowk lane 10	18.57582397	73.893092	New Location	1	2	0	0
545	Vishrantwadi	Shramik nagar chawl	18.58016873	73.889853	New Location	1	2	0	0
546	Vishrantwadi	Vidya nagar last bus stop	18.5797593	73.890413	New Location	1	2	0	0
547	Vishrantwadi	Pragati garden bridge	18.580967	73.893623	New Location	1	2	0	0
548	Vishrantwadi	Vimal classic lane 14 tingre nagar behind MLA house	18.58235419	73.89575	New Location	1	2	0	0
549	Vishrantwadi	Genba sopan school munjaba vasti ganpati chowk	18.59072539	73.895878	New Location	1	2	1	0
550	Vishrantwadi	Jakat naka machi market dhanori road octroi	18.597079	73.907264	New Location	1	1	0	0
551	Vishrantwadi	Vithal mandir chowk dhanori gaon entry	18.59673644	73.896821	New Location	1	2	0	0
552	Vishrantwadi	Gokul bus stop	18.59123861	73.889694	New Location	1	2	0	0
553	Vishrantwadi	Maharana pratap chowk	18.58712203	73.887454	New Location	1	2	0	0
554	Vishrantwadi	Ganpati chowk bairav nagar marathibschool	18.58404799	73.886091	New Location	1	2	0	0
555	Vishrantwadi	Telephone exchange	18.58660086	73.886255	New Location	1	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
556	Vishrantwadi	Ambanagari	18.58100609	73.882004	New Location	1	2	0	0
557	Vishrantwadi	Vishrantiwadi chowk	18.57268494	73.878389	Addl Poles and Cameras	1	2	0	0
558	Vishrantwadi	Maske vasti	18.58365598	73.875026	New Location	2	2	0	4
559	Vishrantwadi	Shivaji chowk (kalas)	18.57901252	73.874555	New Location	1	2	0	0
560	Vishrantwadi	Shanti nagar bridge towards vishrantwadi	18.56872396	73.873873	New Location	1	1	0	1
561	Vishrantwadi	Alandi road junction	18.55221324	73.873199	New Location	2	4	0	0
562	Vishrantwadi	Chandrma chowk	18.54873812	73.871058	Addl Poles and Cameras	2	0	0	4
563	Vishrantwadi	Sant dyaneshwer ghaat or old holkar bridge	18.55372304	73.863918	New Location	1	2	0	2
564	Yerwada	Golf club chowk	18.55263758	73.889605	Addl Cameras only	0	1	0	0
565	Yerwada	parnkoti chowk	18.54498797	73.884663	Addl Cameras only	0	1	0	0
566	Yerwada	Shastri nagar chowk	18.55224248	73.896485	Addl Cameras only	0	1	0	0
567	Yerwada	Cybage chowk	18.5474279	73.897548	New Location	1	2	0	0
568	Yerwada	Diamond chowk	18.55079211	73.887802	New Location	2	3	0	0
569	Yerwada	Chatrapati Shivaji Maharaj chowk	18.55009633	73.886929	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
570	Yerwada	Darbar bakery	18.54743107	73.884161	New Location	1	2	0	0
571	Yerwada	Dharamveer Sambhaji maharaj chowk	18.54821619	73.883689	New Location	1	3	1	0
572	Yerwada	Laxmi nagar raja chowk	18.54932775	73.882078	New Location	2	3	1	0
573	Yerwada	Bagat singh putla	18.54882966	73.878668	New Location	1	3	0	0
574	Yerwada	Sevalal maharaj chowk	18.54660336	73.878237	New Location	2	3	1	0
575	Yerwada	Nurani masjid chowk	18.55110266	73.883289	New Location	1	2	0	0
576	Yerwada	Raat chowk	18.55191891	73.883888	New Location	2	3	1	0
577	Yerwada	Laxmi nagar RK chowk	18.54951465	73.88431	New Location	1	2	0	0
578	Yerwada	sevak chowk	18.54875655	73.884852	New Location	2	3	0	0
579	Yerwada	Shani mandir chowk parnkoti	18.54732681	73.885026	New Location	1	1	0	0
580	Yerwada	babu jagjeevan school	18.54902705	73.89028	New Location	1	2	0	0
581	Yerwada	Mauli chowk	18.55040147	73.892045	New Location	2	4	0	0
582	Yerwada	Wadia bungalow	18.54867041	73.892666	New Location	1	2	0	0
583	Yerwada	Gandhi nagar corner done motors	18.54891199	73.892515	New Location	1	2	0	0
584	Yerwada	Shayadri hospital chowk	18.55417597	73.896818	New Location	1	3	0	0
585	Yerwada	Ishaniya mall chowk	18.55804829	73.894815	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
586	Yerwada	Shanti Rakshak chowk	18.55775016	73.897236	New Location	1	3	0	0
587	Yerwada	Moze school	18.56042987	73.89773	New Location	1	2	1	0
588	Yerwada	Shirke sweets	18.56281205	73.894659	New Location	1	4	0	0
589	Yerwada	Adarsh nagar chowk	18.56641974	73.883422	New Location	1	2	0	0
590	Yerwada	Siddhi vinayak mandir prateek nagar	18.56851388	73.882736	New Location	1	3	0	0
591	Yerwada	Samvak budh vihar	18.57337332	73.88515	New Location	1	2	0	0
592	Yerwada	Commerzone main chowk	18.56399343	73.884986	New Location	2	4	1	0
593	Yerwada	Agrasen high school	18.55522996	73.881141	New Location	1	3	0	0
594	Yerwada	Ambedkar chowk yerwada	18.55288391	73.879654	New Location	2	5	1	0
595	Yerwada	Yerwada fire station	18.55034267	73.878882	New Location	1	2	0	0
596	Yerwada	Sangamwadi parking 1	18.53285711	73.859602	New Location	3	2	1	4
597	Yerwada	Sangamwadi junction	18.54133151	73.870047	New Location	1	3	0	0
598	Yerwada	Road coming from sangamwadi bridge Sadal baba	18.54491931	73.877048	Addl Poles and Cameras	1	2	1	0
599	Yerwada	Tarkeshwer chowk	18.54502865	73.88322	New Location	2	3	1	0
600	Yerwada	Gadital chowk	18.54558047	73.887061	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
601	Yerwada	Gunjan Chowk	18.54585225	73.888798	Addl Cameras only	0	1	0	0
602	Yerwada	Wadgaon sheri junction	18.55864393	73.911573	Addl Poles and Cameras	1	2	0	4
603	Yerwada	Kalyani nagar junction	18.55255366	73.897543	New Location	2	4	1	0
604	Yerwada	St. Arnold schoold	18.55102129	73.912971	New Location	1	2	0	0
605	Yerwada	Bramha sun city chowk	18.54945203	73.91857	New Location	1	3	1	0
606	Yerwada	Vithalanjana Mangal karyala	18.55306223	73.917436	New Location	1	2	0	0
607	Yerwada	Bishop school	18.55234197	73.905987	New Location	1	3	1	0
608	Yerwada	Maarkesh or kwality hotel chowk	18.54860239	73.905887	New Location	1	3	0	0
609	Yerwada	Agha khan bridge	18.54300924	73.904923	New Location	1	3	1	0
610	Chaturshringi	Spicer sanghvi entry point	18.56851325	73.8175	New Location	1	1	0	0
611	Chaturshringi	Manjalkar chowk	18.52719064	73.837786	New Location	1	2	0	0
612	Chaturshringi	Patarkar nagar	18.52389113	73.827079	New Location	1	2	0	0
613	Chaturshringi	Gopal krishan mitra mandal janwadi	18.52926969	73.823497	New Location	1	2	1	0
614	Chaturshringi	Om super market chowk	18.53287142	73.834298	New Location	2	4	0	0
615	Chaturshringi	Cosmos tower and towards bhosle nagar lane	18.54013471	73.832012	New Location	2	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
616	Chaturshringi	SB road juction	18.54080733	73.830585	New Location	1	4	0	0
617	Chaturshringi	ABIL chowk NH 37 dabha	18.53864355	73.834068	New Location	1	2	1	0
618	Chaturshringi	Vidhate vasti	18.56054688	73.798007	New Location	1	3	0	0
619	Chaturshringi	Umaji naik chowk baner pashan link road	18.55177016	73.79791	New Location	1	3	0	0
620	Chaturshringi	Rajwada chowk	18.54567964	73.804284	New Location	1	2	0	0
621	Chaturshringi	Green park chowk	18.551082	73.804895	New Location	1	3	0	0
622	Chaturshringi	abhimanshree pashan	18.54702834	73.817131	New Location	1	2	0	0
623	Chaturshringi	Abhimaan shree chowk pashan	18.54014298	73.817727	New Location	1	2	0	0
624	Chaturshringi	Panchvati junction	18.53919068	73.809395	New Location	1	3	0	0
625	Chaturshringi	HEMRL chowk near bhel	18.52577693	73.781534	New Location	1	1	0	2
626	Chaturshringi	Sutarwadi last bustop	18.53851735	73.781435	New Location	1	3	0	0
627	Chaturshringi	Cosmos bank chowk	18.53710914	73.793964	New Location	1	3	0	0
628	Chaturshringi	Balaji chowk	18.54274096	73.789953	New Location	2	3	1	0
629	Chaturshringi	Saichowk	18.54359602	73.785078	Addl Poles and Cameras	1	1	0	0
630	Chaturshringi	Shivshakti chowk	18.54422095	73.781419	New Location	1	1	1	0
631	Chaturshringi	Sus khind	18.54638594	73.774794	Addl Cameras only	0	0	0	2

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
632	Chaturshringi	Skystories service road sus khind	18.54756044	73.774023	New Location	1	2	0	0
633	Chaturshringi	Nanaware bridge service road	18.55215603	73.772113	New Location	1	2	0	0
634	Chaturshringi	Bitwise underpass or Kia showroom	18.55741038	73.770202	New Location	1	1	1	2
635	Chaturshringi	Kalpatru jade pancard or dhankude udyan	18.55596959	73.783511	New Location	1	3	0	0
636	Chaturshringi	Kalamkar chowk pancard	18.55854857	73.776582	New Location		3	0	0
637	Chaturshringi	Murkute garden or Tapkir chowk	18.55766656	73.772924	New Location	2	2	0	2
638	Chaturshringi	Virbhadra nagar	18.56374743	73.772018	New Location	1	2	0	0
639	Chaturshringi	ganraj mangal karyalya cycar chowk	18.56422131	73.77688	New Location	2	4	1	0
640	Chaturshringi	Cummins high street	18.57066998	73.774808	New Location	2	3	1	0
641	Chaturshringi	Mitcon college	18.57504152	73.76638	New Location	2	2	0	2
642	Chaturshringi	Mamta chowk or sopan baug	18.57369654	73.774726	New Location	1	2	1	0
643	Chaturshringi	Dassra chowk	18.57382144	73.776245	New Location	2	3	1	0
644	Chaturshringi	Balegwadi gaon than	18.57627367	73.779485	New Location	2	2	1	0
645	Chaturshringi	Laxmi mata mandir chowk	18.57371307	73.780747	New Location	1	3	0	0
646	Chaturshringi	DSK gandhkosh	18.56792877	73.786298	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
647	Chaturshringi	Ganesh dutta mandir chowk balewadi	18.56463357	73.782468	New Location	1	2	0	0
648	Chaturshringi	Jupiter chowk	18.56444888	73.787126	New Location	2	4	0	2
649	Chaturshringi	Tamahane chowk aundh baner link road	18.5616421	73.792069	New Location	2	3	1	0
650	Chaturshringi	Westend mall chowk	18.56059578	73.807431	New Location	1	3	1	0
651	Chaturshringi	Seasons Hotel chowk	18.55376182	73.804908	New Location	1	3	0	0
652	Chaturshringi	Gaikwad petrol pump	18.55790431	73.803989	New Location	1	2	0	0
653	Chaturshringi	Ambedkar chowk aundh	18.5602433	73.803135	Addl Cameras only	1	0	1	0
654	Chaturshringi	Medi point junction	18.56548405	73.802338	New Location	1	2	0	0
655	Chaturshringi	Bhale chowk / Nagras chowk	18.56525903	73.807487	New Location	1	2	1	2
656	Chaturshringi	AIIMS chowk	18.56133062	73.80948	New Location	2	4	0	0
657	Chaturshringi	Dmart chowk aundh	18.56694766	73.807562	New Location	1	2	1	0
658	Chaturshringi	Rajiv Gandji bridge Aundh	18.56654382	73.811829	New Location	2	1	0	4
659	Chaturshringi	Kasturbha chowk	18.55608052	73.818036	New Location	2	3	0	0
660	Chaturshringi	Chhatursbungi police station chowk raj bhavan	18.54566566	73.825614	New Location	1	3	0	0
661	Khadki	Symphony hotel chowk	18.54492376	73.838774	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
662	Khadki	Range hill under pass	18.55663993	73.844254	New Location	1	2	0	0
663	Khadki	Khadki church chowk	18.55772123	73.846495	Addl Cameras only		0	1	0
664	Khadki	Khadki railway station	18.56319217	73.841373	New Location	1	3	0	0
665	Khadki	Khadki post office	18.55504879	73.847849	New Location	2	0	0	4
666	Khadki	Poultary chowk egg hatching chowk	18.54913417	73.851116	Addl Cameras only	0	1	0	0
667	Khadki	COEP chowk	18.53029999	73.854561	Addl Cameras only	0	1	0	0
668	Khadki	Patilestate	18.53332536	73.852539	New Location	2	2	0	4
669	Khadki	Patilestate under btidge	18.53232846	73.853453	New Location	1	2	0	0
670	Khadki	Shivaji nagar ST stand	18.54260395	73.851608	New Location	2	2	1	0
671	Khadki	Mula road circle	18.54928547	73.854804	New Location	2	4	0	0
672	Khadki	Holkar bridge	18.55594607	73.861449	New Location	1	2	0	0
673	Khadki	Ammunation factory chowk	18.56260927	73.852372	Addl Cameras only	0	2	0	0
674	Khadki	MSBC chowk	18.56498983	73.850244	Addl Poles and Cameras	1	2	0	0
675	Khadki	Khadki pmt bus depot	18.56640535	73.848531	New Location	2	4	1	0
676	Khadki	Bopodi chowk	18.57181118	73.837361	Addl Poles and Cameras	2	0	0	4

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
677	Khadki	Shahu maharaj chowk	18.57236514	73.833794	New Location	1	3	1	0
678	Khadki	Ambedkar chowk	18.56256901	73.830385	Addl Poles and Cameras	1	2	0	0
679	Khadki	Sai chowk	18.56167388	73.841072	New Location	1	3	0	0
680	Lonikand	Khandve nagar	18.57274205	73.956297	New Location	1	0	1	4
681	Lonikand	Ubale nagar chowk	18.57442687	73.959533	New Location	1	2	1	0
682	Lonikand	Panchsheel tower	18.55407236	73.961424	New Location	1	0	1	0
683	Lonikand	Swari garden	18.57700622	73.96839	New Location	1	2	0	0
684	Lonikand	Wagheshwer chowk	18.58074263	73.978774	New Location	1	3	1	0
685	Lonikand	Bazar chowk back side kalobai nagar	18.57882729	73.98586	New Location	1	2	0	0
686	Lonikand	Awahlwadi fata	18.55660345	73.987753	New Location	1	1	1	0
687	Lonikand	manjri nadi pul	18.52521234	73.989899	New Location	1	0	0	2
688	Lonikand	Shivaji Maharaj chowk manjri khurd	18.52706666	73.99197	New Location		0	1	0
689	Lonikand	kolvadi shivaji maharaj putla	18.52928622	74.034192	New Location	3	1	1	2
690	Lonikand	Rahu road kesanand police chowki	18.5722187	74.021925	New Location	1	1	0	0
691	Lonikand	Shivaji maharaj chowk kesnand gaon	18.57284734	74.02217	New Location	2	2	1	0
692	Lonikand	Magar wasti	18.59461881	74.032692	New Location	1	2	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
693	Lonikand	Pehrne fata nadi pul	18.64310646	74.053464	New Location	2	1	1	4
694	Lonikand	Pimpri Sandas	18.60923134	74.129723	New Location	1	3	1	0
695	Lonikand	Ashtapur fata	18.58654747	74.079511	New Location	2	4	1	4
696	Lonikand	Chatrapati shivaji maharaj chowk	18.63138162	74.063014	New Location	2	4	1	0
697	Lonikand	Pherne fata	18.64127556	74.050691	New Location	1	1	0	0
698	Lonikand	Alandi chowk	18.62699977	74.033793	New Location	2	2	0	4
699	Lonikand	theur fata nagar road	18.62130625	74.026285	New Location	1	2	0	2
700	Lonikand	Katkewadi fata	18.60319786	74.011708	New Location	1	3	0	0
701	Lonikand	Yeshwant nagar chowk	18.59102093	74.002943	New Location	1	2	1	0
702	Lonikand	Bokari phata road	18.58986325	73.997534	New Location	2	4	1	0
703	Lonikand	Keshnand phata	18.581642	73.986669	New Location	1	4	1	0
704	Lonikand	Avlwadi phata	18.58095715	73.984827	New Location	2	3	1	0
705	Lonikand	Shivaji maharaj wagholi wes	18.58227963	73.981972	New Location	2	4	1	0
706	Lonikand	lohegaon chowk	18.58300376	73.979377	New Location	1	3	0	0
707	Lonikand	PMT bus depot lohegaon road	18.58920797	73.956749	New Location	1	0	0	2
708	Viman Nagar (vimantal)	Sant nagar	18.59542265	73.933272	New Location	1	3	0	0
709	Viman Nagar (vimantal)	Rakhpasre vasti	18.60319373	73.927618	New Location	2	4	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
710	Viman Nagar (vimantal)	Pmpml lohegaon bustop	18.59579774	73.924663	New Location	3	3	1	2
711	Viman Nagar (vimantal)	Dhanori Jakatnaka	18.59706565	73.908768	New Location	2	2	1	2
712	Viman Nagar (vimantal)	Ganraj chowk	18.59067137	73.906709	New Location	1	3	0	0
713	Viman Nagar (vimantal)	Kalwad vasti entry	18.58533001	73.905149	New Location	1	1	0	0
714	Viman Nagar (vimantal)	Mahada colony last bus stop	18.57388405	73.912014	New Location	1	2	0	0
715	Viman Nagar (vimantal)	CCD chowk	18.56511087	73.910822	New Location	1	3	0	0
716	Viman Nagar (vimantal)	Nexa showroom	18.56247267	73.906759	Addl Cameras only	0	1	0	0
717	Viman Nagar (vimantal)	Symbosis school lane 22	18.56521273	73.906974	New Location	1	2	0	0
718	Viman Nagar (vimantal)	Airport chowk T point	18.5781225	73.908067	Addl Cameras only	0	2	0	0
719	Viman Nagar (vimantal)	Symbosis college chowk	18.57535552	73.907662	New Location	1	1	1	0
720	Viman Nagar (vimantal)	Dorabjee chowk	18.56839597	73.906783	New Location	2	2	0	2
721	Viman Nagar (vimantal)	Ganga puram chowk	18.56821926	73.910915	New Location	1	3	0	0
722	Viman Nagar (vimantal)	Kailash super market	18.56785408	73.914355	New Location	1	3	0	0
723	Viman Nagar (vimantal)	Pheonix back side entry	18.56205465	73.918066	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
724	Viman Nagar (vimantal)	Shrikrishna chowk	18.56513317	73.918195	Addl Cameras only	0	0	1	0
725	Viman Nagar (vimantal)	Ganpati Mandir chowk	18.56747937	73.918586	New Location	1	2	1	0
726	Chandan Nagar	Stella marris shalla and masjid	18.54609478	73.923823	New Location	2	3	0	0
727	Chandan Nagar	Lonkar madyamic shala vadgaon sheri gaothan	18.54282265	73.923426	New Location	2	4	1	0
728	Chandan Nagar	Post office chowk	18.54374184	73.924468	New Location	1	2	0	0
729	Chandan Nagar	Munurwar society malawadi	18.5438287	73.926939	New Location	1	4	0	0
730	Chandan Nagar	Bank of baroda chowk	18.54489615	73.935867	New Location	2	2	0	0
731	Chandan Nagar	Thite wasti	18.54530155	73.938983	New Location	1	3	0	0
732	Chandan Nagar	EON IT ganga consetela circle	18.55399321	73.950019	New Location	2	4	1	0
733	Chandan Nagar	Rajaram patil nagar	18.55912605	73.949968	New Location	1	1	0	0
734	Chandan Nagar	Bikaner chowk	18.55964798	73.949781	New Location	2	4	1	0
735	Chandan Nagar	Blueberry society	18.56386988	73.950123	New Location	1	3	0	0
736	Chandan Nagar	Aple ghar	18.56794374	73.949237	New Location	1	1	0	0
737	Chandan Nagar	Baji Mandai kharadi entry	18.56870934	73.950233	New Location	1	1	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
738	Chandan Nagar	Kharadi jakat naka	18.56908914	73.950884	New Location	3	1	1	4
739	Chandan Nagar	Samarth nagar chowk	18.56393399	73.953807	New Location	1	2	0	0
740	Chandan Nagar	Gera road jain mandir	18.55969145	73.95512	New Location	1	1	0	0
741	Chandan Nagar	Barclays chowk	18.55357778	73.955901	New Location	2	2	0	0
742	Chandan Nagar	Panchsheel yower circle	18.5512209	73.96026	New Location	2	2	0	4
743	Chandan Nagar	Eon IT phase 2	18.55136362	73.955744	New Location	1	2	0	0
744	Chandan Nagar	Bhulaje pathare chowk	18.54922953	73.95518	New Location	1	2	0	0
745	Chandan Nagar	Khardi chowk back side	18.54833555	73.948658	New Location	3	3	0	0
746	Chandan Nagar	Kharadi police chowki	18.54864753	73.947254	New Location	2	3	1	0
747	Chandan Nagar	WTC chowk	18.55384954	73.947714	New Location	2	4	1	0
748	Chandan Nagar	Shinde dairy chowk	18.55832015	73.946628	New Location	1	2	0	0
749	Chandan Nagar	Cherly society kharadi	18.56060759	73.94564	New Location	1	3	0	0
750	Chandan Nagar	Ahmednagar toll naka (dragha chowk)	18.56484645	73.943703	New Location	1	2	0	0
751	Chandan Nagar	Saraf bazar exit	18.56317342	73.932792	New Location	1	1	0	0
752	Chandan Nagar	Shivraj chowk	18.56208993	73.933936	New Location	1	2	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
753	Chandan Nagar	Saraf bazar chandan nagar entry	18.5621821	73.93268	New Location	1	3	1	0
754	Chandan Nagar	Hanuman mandir chowk	18.56201238	73.932013	New Location	1	2	0	0
755	Chandan Nagar	Machi market chowk/nagpal road	18.55796692	73.931494	New Location	1	3	0	0
756	Chandan Nagar	Durgamatta mandir	18.55774032	73.93803	New Location	2	4	0	0
757	Chandan Nagar	Shankar nagar road kharadi bypass	18.55585341	73.937769	New Location	1	1	0	0
758	Chandan Nagar	Raidison hotel	18.55266936	73.937383	Addl Poles and Cameras	2	0	0	4
759	Chandan Nagar	Reliance mart kharadi	18.55074507	73.937058	New Location	2	3	1	0
760	Chandan Nagar	Zensar chowk	18.55007948	73.939492	New Location	3	4	1	2
761	Chandan Nagar	Kharadi gaon chowk nadipatra road	18.54823908	73.945856	New Location	1	2	0	0
762	Chandan Nagar	Mundhwa bridge	18.53649372	73.933876	New Location	1	2	1	0
763	Chandan Nagar	Khardkar park petrol pump	18.53827823	73.934329	New Location	1	1	0	0
764	Chandan Nagar	Sainath nagar chowk	18.5398305	73.934594	New Location	2	4	1	0
765	Chandan Nagar	Sirvi Bandhu sweek corner chowk	18.5398642	73.931833	New Location	1	2	0	0
766	Chandan Nagar	Jog company near mother terrisa	18.54376036	73.931085	New Location		3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
767	Chandan Nagar	Sundarbai school chowk	18.55737631	73.928139	New Location	1	2	1	0
768	Chandan Nagar	Ganesh nagar chowk	18.55770407	73.926276	New Location	1	1	0	0
769	Chandan Nagar	Anandpark chowk	18.55277234	73.925151	New Location	1	3	0	0
770	Chandan Nagar	Siddhivinayak mandir chowk	18.54676674	73.921334	New Location	2	2	1	0
771	Chandan Nagar	Tempo chowk	18.55128409	73.920957	New Location	1	2	0	0
772	Chandan Nagar	Ramchandra chowk	18.55465942	73.920772	New Location	1	1	0	0
773	Chandan Nagar	Surya cateres chowk sopan nagar	18.55515876	73.918333	New Location	1	2	0	0
774	Chandan Nagar	Sopan nagar road	18.56086241	73.91875	New Location	1	2	0	0
775	Chandan Nagar	In orbit mall corner	18.56169708	73.921102	New Location	1	2	1	0
776	Chandan Nagar	Tata guard room chowk	18.56284605	73.927899	New Location	3	4	2	6
777	Chandan Nagar	Dwarka garden chowk mangal karyala	18.5594986	73.928037	New Location	1	3	1	0
Zone 5									
778	Wanawadi	Jambhulkar chowk	18.49503206	73.900019	New Location	2	4	1	0
779	Wanawadi	Command hospital	18.49641772	73.882004	New Location	1	2	0	0
780	Wanawadi	CDO chowk	18.49957564	73.879786	New Location	2	5	0	1

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
781	Wanawadi	Arjun road	18.50653672	73.88876	New Location	2	1	0	4
782	Wanawadi	Beam center chowk	18.50325048	73.8887	New Location	1	4	0	0
783	Wanawadi	Turf club chowk	18.50644452	73.889831	New Location	2	4	1	0
784	Wanawadi	Gorpadi junction	18.52316725	73.887955	New Location	1	2	0	0
785	Wanawadi	Army public school	18.51533543	73.894469	New Location	2	2	0	2
786	Wanawadi	Empress garden	18.51619572	73.898376	New Location	1	2	0	0
787	Wanawadi	BT kawade nunction highway	18.50649284	73.906949	New Location	1	1	0	0
788	Mundhwa	Guru kripa socity	18.52528959	73.959935	New Location	1	2	0	0
789	Mundhwa	The Orbis School	18.53257292	73.950769	New Location	1	2	1	0
790	Mundhwa	Z corner	18.53586273	73.946703	New Location	1	2	0	0
791	Mundhwa	Renuka mata mandir	18.53481657	73.945968	New Location	1	3	0	0
792	Mundhwa	KCG chowk	18.53342772	73.919929	New Location	1	2	0	0
793	Mundhwa	One lawn chowk	18.53591931	73.910798	New Location	1	3	0	0
794	Mundhwa	Ghorpafi Rly gate	18.52971506	73.902543	New Location	1	2	1	0
795	Mundhwa	Devki palace	18.52362981	73.906249	New Location	1	3	1	0
796	Mundhwa	Vathari mala chowk	18.52355033	73.909601	New Location	1	3	1	0
797	Mundhwa	Ambedkar chowk	18.51967437	73.915704	New Location	2	4	0	0
798	Mundhwa	Tadihutta chowk	18.53227029	73.917245	New Location	2	4	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
799	Mundhwa	Towards Mundhwa gaon	18.53183636	73.932192	New Location	1	1	0	0
800	Mundhwa	Shravastinagar balajinagar gulmohor society t point	18.52477649	73.902978	New Location	1	2	0	0
801	Mundhwa	Sade Satra nali chowk or lonkar chowk	18.52774379	73.949209	New Location	2	4	1	0
802	Mundhwa	Mundhwa gaon pravesh dwar	18.53429142	73.929726	New Location	1	2	0	0
803	Mundhwa	Kumbharwada keshav nagar	18.53479336	73.942585	New Location	1	2	0	0
804	Mundhwa	Kamgar maidan mundhwa	18.53377517	73.932322	New Location	2	4	0	0
805	Mundhwa	Gai raan vasti	18.53695942	73.946274	New Location	1	1	1	0
806	Mundhwa	Bhimnagar corner shahu chowk	18.51973795	73.909677	New Location	1	2	0	0
807	Mundhwa	Basant school-Vikas nagar chowk+ingle hospital	18.5257976	73.899415	New Location	2	4	1	0
808	Mundhwa	Balajinagar church (T point Nr. Sicilia)	18.52086972	73.903373	New Location	1	2	0	0
809	Mundhwa	Daiker gali	18.52795233	73.919511	New Location	1	2	0	0
810	Hadapsar	Gayran vasti	18.53964804	73.948204	New Location	1	2	0	0
811	Hadapsar	hadapasar railwaystation chowk	18.52724341	73.927237	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
812	Mundhwa	passport office mundhwa	18.53101811	73.913762	New Location	1	2	0	0
813	Hadapsar	Mamjri road ganesh mandir	18.5090659	73.958923	New Location	2	4	1	0
814	Hadapsar	Banker kida sankul near shamshan bhoomi	18.49425242	73.956947	New Location	1	2	0	0
815	Hadapsar	karepadal railway crossing	18.48369961	73.939578	New Location	1	3	1	0
816	Hadapsar	Sanjudha complex	18.48218633	73.949253	New Location	2	3	0	0
817	Hadapsar	Bekrai jagat naka shivale chowk	18.4832751	73.953973	New Location	1	4	0	0
818	Hadapsar	Kuthwad godown phursungi	18.47786361	73.965512	New Location	1	2	0	0
819	Hadapsar	Mantri niwas	18.49850382	73.922819	New Location	2	4	0	0
820	Hadapsar	Sayad nagar new english school towards hadpasar	18.49311825	73.928048	New Location	1	2	0	0
821	Hadapsar	Purohit sweets sasane nagar	18.49132715	73.931985	New Location	1	4	1	0
822	Hadapsar	Sasane nagar canal	18.49692136	73.933112	New Location	3	3	1	0
823	Hadapsar	32 number shalla behind vaibhav theatre	18.50037179	73.930962	New Location	2	3	0	0
824	Hadapsar	Amar cottage	18.50663178	73.932177	New Location	1	2	0	0
825	Hadapsar	Ambika super market chowk	18.507997	73.941253	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
826	Mundhwa	Jehangir chowk behind laxmi lawns magarpatta	18.52012517	73.919059	New Location	2	4	1	4
827	Hadapsar	Phursungi Railway over bridge saswad road behind power house	18.47254321	73.959053	New Location	2	3	1	0
828	Hadapsar	Bustop phursungi gaon	18.4735659	73.974922	New Location	2	3	1	0
829	Hadapsar	Ganga nagar rickshaw stand ganga nagar	18.48857577	73.955914	New Location	2	3	1	0
830	Hadapsar	phursungi fata (behind grapes garden majri)	18.49456275	73.98347	New Location	1	1	1	0
831	Hadapsar	Morewasti T point manjri	18.50178379	73.984287	New Location	1	2	0	0
832	Hadapsar	swami samarth mandir bhaparkar mala majri	18.51055986	73.988971	New Location	1	2	0	0
833	Hadapsar	Shivaji chowk manjri gaon	18.52243384	73.984263	New Location	1	2	1	0
834	Hadapsar	Kk gule vidyalaya chowk manjri	18.52126838	73.982565	New Location	1	2	1	0
835	Hadapsar	Gopal pati chowk	18.51221087	73.967343	New Location	1	3	0	0
836	Hadapsar	Chorung building manjri mahadev nagar	18.50451557	73.954193	New Location	1	2	0	0
837	Hadapsar	Sane guruji hospital	18.50388382	73.937701	New Location	1	2	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
838	Hadapsar	Magarpatta south gate	18.50843194	73.929757	New Location	1	0	1	0
839	Hadapsar	Magapatta city main gate	18.51452377	73.932528	New Location	1	2	1	0
840	Lonikalbhor	walava vasti road	18.43756778	73.976661	New Location	1	1	0	0
841	Lonikalbhor	Vithal mandir mastani talav view	18.41592973	74.003714	New Location	1	2	0	0
842	Lonikalbhor	Hotel mountain view	18.4221498	73.983137	New Location	1	2	0	2
843	Lonikalbhor	Wadki nala	18.42682063	73.981556	New Location	1	1	0	0
844	Lonikalbhor	Hotel Vijay executive samor	18.43139306	73.97983	New Location	2	4	1	0
845	Lonikalbhor	Wadki 10th mile	18.44095961	73.974874	New Location	1	1	0	0
846	Lonikalbhor	Wadki gaon...near shambumahadev mandir	18.44015396	73.983857	New Location	1	4	0	0
847	Lonikalbhor	wadkigaon fata....greeting samor	18.44526067	73.972473	New Location	1	2	0	0
848	Lonikalbhor	Datakrupa hardware	18.44899647	73.970385	New Location	2	3	0	0
849	Lonikalbhor	Urulai police chowki	18.4574906	73.966269	New Location	1	3	0	0
850	Lonikalbhor	Autadiwadi fata	18.44329383	73.935735	New Location	1	3	0	0
851	Lonikalbhor	Mantarwadi fata	18.46539503	73.962405	Addl Poles and Cameras	1	2	1	0
852	Lonikalbhor	Savali home	18.45322864	73.958519	New Location	1	4	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
853	Lonikalbhor	Ambedkar chowk urali devachi gaon	18.45613357	73.956188	New Location	2	3	0	0
854	Lonikalbhor	Shyadari chowk urali devachi gaon	18.45573507	73.958443	New Location	1	2	0	0
855	Lonikalbhor	Urali fata	18.45938891	73.965315	New Location	1	1	1	4
856	Lonikalbhor	Mantarwadi kaman	18.46523539	73.961432	New Location	1	1	0	0
857	Lonikalbhor	Khind road katraj bypass road	18.46029242	73.949084	New Location	1	4	0	0
858	Lonikalbhor	Handewadi chowk	18.45550704	73.933016	New Location	2	4	1	0
859	Lonikalbhor	Pandavdand road near chetan gaikwad bungalow	18.48836781	74.006952	New Location	1	2	0	0
860	Lonikalbhor	Raywadi road ambernath mandir	18.47979858	74.027868	New Location	1	2	0	0
861	Lonikalbhor	Shivaji Udyan loni gaon	18.48240192	74.025486	New Location	2	4	0	0
862	Lonikalbhor	Indira nagar chowk	18.48522336	74.020928	New Location	2	4	0	0
863	Lonikalbhor	Rahilkar corner	18.48657152	74.016209	New Location	2	4	0	0
864	Lonikalbhor	Iocl chowk	18.49085942	74.010204	New Location	1	1	0	0
865	Lonikalbhor	kodi fata toll naka	18.49313637	73.993473	New Location	2	3	0	4
866	Lonikalbhor	Phursungi fata near maduban karyalaya	18.4921704	74.00017	New Location	2	3	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
867	Lonikalbhor	Prateek garden mangal karyalaya chowk	18.49101968	74.007843	New Location	1	1	0	0
868	Lonikalbhor	Field guard chowk	18.49041565	74.012107	New Location	1	2	0	0
869	Lonikalbhor	Kadam vasti (Angore wines)	18.48994686	74.015301	New Location	1	1	1	0
870	Lonikalbhor	PDCC bank lane or sambhaji nagar fata	18.4897492	74.016854	New Location	1	1	0	0
871	Lonikalbhor	Kadamvasti chowk	18.48953222	74.018417	New Location	1	2	0	0
872	Lonikalbhor	Loni station chowk	18.48908039	74.021812	New Location	2	2	0	0
873	Lonikalbhor	MIT corner	18.48904319	74.024613	New Location	1	2	1	0
874	Lonikalbhor	MIT college entry gate chowk	18.48996021	74.025476	New Location	1	1	0	0
875	Lonikalbhor	Rajlaxmi magal karyala	18.48872521	74.044911	New Location	1	2	1	0
876	Lonikalbhor	Grand 9 hotel	18.47841246	74.13188	New Location	1	2	0	0
877	Lonikalbhor	Vishwa petrol pump	18.48715316	74.144809	New Location	2	2	0	0
878	Lonikalbhor	irrigation colony chowk	18.4872654	74.140598	New Location	1	1	0	0
879	Lonikalbhor	Talim road chowk	18.48740022	74.134737	New Location	1	1	0	0
880	Lonikalbhor	Elite chowk urali kanchan	18.48752487	74.130297	New Location	1	4	1	0
881	Lonikalbhor	Talawadi chowk	18.48736747	74.133383	New Location	2	2	0	4

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
882	Lonikalbhor	Dost pan center towards railway track	18.4934731	74.133551	New Location	1	1	0	0
883	Lonikalbhor	Rahukar collection towards railway track	18.49349853	74.133032	New Location	1	1	0	0
884	Lonikalbhor	Baif road tambe wasti hotel Jaybhawani	18.49827267	74.140735	New Location	1	1	0	0
885	Lonikalbhor	Dattamandir chowk urali kanchan	18.50439348	74.131938	New Location	2	2	2	0
886	Lonikalbhor	Bodhevasti chowk	18.49589881	74.119273	New Location	2	4	0	0
887	Lonikalbhor	Inamdar vasti near jayneth wines	18.48783679	74.115273	New Location	1	1	0	0
888	Lonikalbhor	Shiva farm chowk near golden lodge	18.48791852	74.106083	Addl Poles and Cameras	2	2	0	0
889	Lonikalbhor	Sortapa wadi fata	18.48794205	74.098927	New Location	2	3	0	0
890	Lonikalbhor	peth gaon fata	18.48824571	74.08935	New Location	1	1	0	0
891	Lonikalbhor	Khorawade fata chowk kunjir wadi	18.48837907	74.077021	New Location	1	1	0	0
892	Lonikalbhor	Naygaon fata chowk	18.48844794	74.070717	New Location	2	2	0	0
893	Lonikalbhor	Kunjirwadi chowk kunjirwadi	18.48846639	74.067401	New Location	2	4	0	0
894	Lonikalbhor	Chintamani Vidyamandir theur	18.52317965	74.043827	New Location	1	2	1	2

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
895	Lonikalbhor	Caramella cake shop	18.52288824	74.045438	New Location	1	2	0	0
896	Lonikalbhor	near PDCC bank theur kakdemala road	18.52230254	74.046643	New Location	1	1	0	0
897	Lonikalbhor	Theur railway bridge or hotel shivrudra	18.49538522	74.050965	New Location	1	2	0	0
898	Lonikalbhor	Theur fata	18.48872967	74.051692	New Location	2	1	1	4
899	Lonikalbhor	Borkarvasti shell petrol pump	18.48877418	74.041889	New Location	1	1	0	0
900	Lonikalbhor	Maliyacha mala (Rajendra petrol pump)	18.48890817	74.035508	New Location	1	2	1	0
901	Lonikalbhor	Shivshakti chowk (joy nest society)	18.48885908	74.029645	New Location	1	1	0	0
902	Lonikalbhor	Lonigaon fata or lonigaon corner	18.48894303	74.027427	New Location	2	4	0	0
903	Wanawadi	Nepiar road junction	18.50295002	73.883072	New Location	1	3	0	0
904	Wanawadi	Racecourse walking plaza wanwadi	18.51511225	73.892048	New Location	1	3	0	0
905	Wanawadi	Mahatma phule sanskritika bhavan	18.49698082	73.900219	New Location	1	2	0	0
906	Wanawadi	Girme school	18.49552458	73.894635	New Location	1	2	0	0
907	Wanawadi	Andha shala ramtekdi	18.50385235	73.916353	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
908	Wanawadi	Sayyadnagar underpass bridge	18.49284162	73.927696	New Location	1	2	1	0
909	Wanawadi	Sayyadnagar Lane 12	18.4872848	73.931273	New Location	1	2	1	0
910	Wanawadi	Ahilyadevi holkar chowk	18.4767341	73.939567	New Location	1	2	1	0
911	Wanawadi	Farate chowk, Kalepadal	18.48275393	73.940312	New Location	2	4	0	0
912	Wanawadi	Kalepadal underpass bridge	18.48634773	73.935299	New Location	1	2	1	0
913	Bibwewadi	JSPM college	18.47321515	73.931808	New Location	1	3	0	0
914	Wanawadi	Shriram chowk, Handewadi road	18.48159297	73.931329	New Location	1	3	0	0
915	Wanawadi	Wadkar mala, sayyed nagar road	18.47979094	73.927569	New Location	2	4	0	0
916	Wanawadi	Mohammad wadi chowk	18.47528055	73.924632	New Location	1	3	0	0
917	Wanawadi	Mountain Hi Hotel Heavan park chowk	18.4811745	73.910181	New Location	1	3	0	0
918	Wanawadi	One plus shopping mall junction	18.48127403	73.903612	New Location	1	3	0	0
919	Wanawadi	Ganga satellite SoC junction	18.48757352	73.89539	New Location	1	2	0	0
920	Wanawadi	Dagad chowk	18.48591782	73.900356	New Location	2	4	0	0
921	Wanawadi	Diamond bakery	18.50491046	73.901025	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
922	Wanawadi	wanawadi bazaar chowki chowk circld	18.49902399	73.895564	New Location	1	2	0	0
923	Wanawadi	Netaji nagar society chowk	18.49108709	73.894757	New Location	1	3	0	0
924	Wanawadi	Mummadevi chowk	18.50650938	73.884296	New Location	2	4	0	4
925	Wanawadi	Fatima nagar chowk	18.5064655	73.902383	New Location	1	3	0	0
926	Wanawadi	BT kawade road & sopan baug under bridge chowk	18.51320312	73.907026	New Location	1	3	0	0
927	Wanawadi	Mahalaxmi temple Dhobarwadi	18.51744197	73.902273	New Location	1	2	0	0
928	Wanawadi	Ghadyal chowk	18.52271932	73.889706	New Location	1	2	0	0
929	Wanawadi	Empress garden road near to Canal	18.50974913	73.897448	New Location	1	3	0	0
930	Wanawadi	BT Kawade canal road	18.51063934	73.907013	New Location	1	3	0	0
931	Wanawadi	Kalubai chowk	18.50651033	73.908473	New Location	1	3	0	0
932	Wanawadi	industrial area underpass	18.49782436	73.922013	New Location	1	4	0	0
933	Wanawadi	Digital hub	18.49901795	73.920796	New Location	1	2	0	0
934	Wanawadi	Sathe Udyan ramtekdi	18.50152817	73.918792	New Location	1	3	1	0
935	Wanawadi	Saibaba mandir,ramtekdi	18.5042733	73.912543	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
936	Bibwewadi	Shivtej nagar	18.46930561	73.872057	New Location	1	2	0	0
937	Bibwewadi	chaitraban zopadpati or savidhan chowk	18.46313775	73.865047	New Location	1	3	1	0
938	Bibwewadi	Dolphin chowk	18.46131995	73.866316	New Location	1	3	0	0
939	Bibwewadi	Upper juna bus stop	18.46107221	73.867494	New Location	1	1	0	0
940	Bibwewadi	Rajiv gandhi nagar chowk	18.46084526	73.869141	New Location	1	1	0	0
941	Bibwewadi	Ramiya nagari water tank	18.46274731	73.87079	New Location	2	2	0	0
942	Bibwewadi	Padmavati nagar	18.46121468	73.872383	New Location	2	2	0	0
943	Bibwewadi	Prasad bibwenagar	18.47153388	73.873742	New Location	1	2	0	0
944	Bibwewadi	Kakdevasti chsndan marble	18.46453226	73.877395	New Location	1	1	0	0
945	Bibwewadi	Near Aai mata mandir	18.47222235	73.874877	New Location	2	3	0	0
946	Bibwewadi	Kothari block	18.47872632	73.857242	New Location	2	2	0	4
947	Bibwewadi	Bibewadi chowk gaon	18.47472978	73.862462	New Location	1	2	0	0
948	Bibwewadi	Papal vasti	18.47272956	73.866416	New Location	1	3	0	0
949	Bibwewadi	Bharat jyoti bus stop	18.47260268	73.863092	New Location	1	2	0	0
950	Bibwewadi	Mahesh society chowk	18.46907601	73.864102	Addl Poles and Cameras	1	2	0	0
951	Bibwewadi	Khumbarwada chowk	18.46902004	73.861756	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
952	Bibwewadi	Anna bhau sathe sabagrah	18.47701489	73.857156	New Location	1	2	0	0
953	Bibwewadi	Amba mata mandir	18.45573348	73.869474	New Location	1	3	0	0
954	Kondhwa	Gausulwara masjid	18.4630045	73.884483	New Location	1	3	0	0
955	Kondhwa	Prince town near punaydham ashram	18.45932753	73.903785	New Location	1	3	0	0
956	Kondhwa	Hanuman chowk	18.46138196	73.879974	New Location	1	3	0	0
957	Kondhwa	Welcome hall chowk	18.46937398	73.891247	New Location	1	3	0	0
958	Kondhwa	Parge nagar	18.46652653	73.891337	New Location	1	3	0	0
959	Kondhwa	Somaji chowk	18.45701388	73.890831	New Location	1	4	1	0
960	Kondhwa	Pratibha pawar school or pawar udyan	18.45594879	73.887462	New Location	1	3	1	0
961	Kondhwa	Shreeji lawns	18.46464992	73.877504	New Location	1	3	0	0
962	Kondhwa	Gokul nagar	18.44962111	73.873959	New Location	2	0	0	4
963	Kondhwa	Kanha hotel	18.45060641	73.879644	New Location	3	6	1	0
964	Kondhwa	Trinity college	18.42409875	73.901907	New Location	1	2	0	0
965	Kondhwa	Pansare nagar	18.43543037	73.890199	New Location	1	2	0	0
966	Kondhwa	Yewalewadi chowk	18.43620455	73.89645	New Location	2	2	0	2
967	Kondhwa	khadi machine chowk towards yewele road	18.45193772	73.891281	New Location	2	4	1	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
968	Kondhwa	Ambekar hotel chowk or padmavati temple	18.44983102	73.910787	New Location	1	4	0	0
969	Kondhwa	Holewasti chowk	18.45879229	73.912105	New Location	1	3	0	0
970	Kondhwa	Ashirwad chowk	18.47522776	73.916264	New Location	1	3	0	0
971	Kondhwa	Kadnagar chowk	18.46277585	73.914225	New Location	1	3	0	0
972	Kondhwa	Hotel betos	18.46338104	73.904308	New Location	1	3	0	0
973	Kondhwa	Dorabjee royal heritage mall	18.46726717	73.905616	New Location	2	3	1	0
974	Kondhwa	Raheja vista chowk or bbc chowk	18.47226751	73.909936	New Location	2	4	1	0
975	Kondhwa	Sainik vihar society	18.47772306	73.908025	New Location	1	2	0	0
976	Kondhwa	Salunkhe vihar opp. healing hands clinic	18.48130869	73.901462	New Location	1	3	1	0
977	Kondhwa	RIMS international school	18.47740697	73.901915	New Location	1	3	0	0
978	Kondhwa	Sai baba mandir	18.47794025	73.89205	New Location	2	2	0	0
979	Kondhwa	Fakri hills	18.48274798	73.891004	New Location	2	3	0	2
980	Market yard	Kamal nursing home	18.48943747	73.861738	New Location	1	1	0	0
981	Market yard	Kalobai mandir	18.4891195	73.866338	New Location	1	2	1	0
982	Market yard	Wakar maha mandal	18.48945432	73.870067	New Location	2	0	0	2
983	Market yard	Market yard pmt depot	18.48390089	73.871385	New Location	1	1	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
984	Market yard	Hotel Khushbu	18.48087274	73.865544	New Location	1	0	1	0
985	Market yard	Vastu nagar	18.48064474	73.86829	New Location	1	2	1	0
986	Market yard	Ganesh Amrutalya	18.48792329	73.857552	New Location	1	1	0	0
987	Market yard	Hotel dev uncle kitchen	18.48016903	73.875454	New Location	1	2	1	0
988	Market yard	Vardhamanpura jain temple chowk	18.47677767	73.875049	New Location	1	2	1	0
989	Market yard	Gangadham T junction	18.47649338	73.874014	New Location	1	2	1	0
990	Market yard	Meghadoot hotel	18.48104349	73.875518	New Location	1	2	1	0
991	Market yard	Raviraj mall	18.48192749	73.876614	New Location	1	2	1	0
992	Market yard	New Era society gangadham chowk	18.48181524	73.872762	New Location	1	2	0	0
993	Market yard	Gangadham chowk	18.48079292	73.872326	Addl Cameras only	0	1	0	0
994	Market yard	Hamalnagar	18.48181365	73.871919	New Location	1	2	0	0
995	Market yard	Bhimalay complex	18.48539157	73.875359	New Location	1	1	1	0
996	Market yard	Sandesh nagar ZP high school	18.48573594	73.874329	New Location	1	1	0	0
997	Market yard	Trupti hotel	18.48579572	73.871444	New Location	1	0	1	0
998	Market yard	Chintamani ganesh mandir	18.48789849	73.873923	New Location	1	2	0	0
999	Market yard	Unity bakery chowk	18.48781009	73.870961	New Location	1	3	0	0

SN	Police Station	Proposed Location Spot	Lat	Long	Category	Poles	Fixed	PTZ	ANPR
1000	Market yard	Gas godown	18.48334379	73.867144	New Location	1	1	0	0
1001	Market yard	Market yard gate 3	18.48770007	73.865733	New Location	2	4	1	0
1002	Market yard	Market yard gate 4	18.48507995	73.864515	New Location	1	1	1	0
1003	Market yard	Swapan purti apartment bayle bazar	18.48268207	73.864024	New Location	1	2	0	0
1004	Market yard	Chandralok hospital chowk	18.48081327	73.861258	New Location	3	3	1	4
1005	Market yard	Uttsav chowk or city pride chowk	18.48933826	73.857499	Addl Poles and Cameras	1	3	0	0
1006	Market yard	Budhvihar chowk prem nagar vasaht	18.48539539	73.863026	New Location	1	2	0	0
1007	Market yard	Magar statue	18.48938818	73.864485	Addl Cameras only	0	1	0	0
	Grand Total					**1238**	**2240**	**328**	**318**

Note: The final count of Poles may differ in the BOQ, the additional poles have kept as buffer

Annexure C: List of Proposed Viewing Centers

SN	Police Station/DCP Offices	Proposed Location Spot	Category
1	Lonikand	Lonikand Police Station	New Viewing Center
2	Lonikalbhor	Lonikalbhor Police Station	New Viewing Center
3	Wagholi	Wagholi Police Station	New Viewing Center
4	Phursungi	Phursungi Police Station	New Viewing Center
5	Baner	Baner Police Station	New Viewing Center
6	Nanded city	Nanded city Police Station	New Viewing Center

SN	Police Station/DCP Offices	Proposed Location Spot	Category
7	Kalepadal	Kalepadal Police Station	New Viewing Center
8	Kharadi	Kharadi Police Station	New Viewing Center
9	DCP Zone 3	DCP Zone 3 office, Kothrud	New Viewing Center
10	DCP Zone 5	DCP Zone 5 office, Hadapsar	New Viewing Center
11	PCSCL	PCSCL command center	New Viewing Center

Signature Not Verified
Digitally signed by SHRIKANT SAHEBRAO PATIL
Date: 2024.02.26 17:59:35 IST
Location: Maharashtra-MH